# Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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# Motivation

- **Economic Problem:** Selecting best options under costly evaluation
  - Determining each option's true quality costs decision-maker time and effort
  - Low-cost screening of all options $\Rightarrow$ high-cost evaluation of surviving options
  - **Examples:** Admissions and hiring, clinical trials, matching platforms
- **Why Graduate Admissions:** Benefits graduate programs and future applicants
  - Graduate programs invest time and money into students' development
  - If programs are undervaluing applicants with certain characteristics, future such applicants may receive formerly-withheld opportunities

# Research Questions

1. Which factors in applications affect admission and success in graduate school?
    - Literature measures success with program completion and job placement ranking
2. Are admissions committees optimizing, or are they making mistakes?
    - Goal is to accept applicants with highest success probabilities who will matriculate
    - Predictors of admission only (success only) may be overweighted (underweighted)
3. If they are making mistakes, how can they make better admissions decisions?
    - In turn, by changing decision rules, how much better of a cohort can they admit?

★ This paper's methods generalize to admissions, hiring, and other settings

# Methodology and Main Results

- Build structural models of admissions committee's decision problem that support:
  - Statistical tests of whether committee's decision rule aligns with desired objective
  - Counterfactual simulations quantifying gains from deviating to model's decision rule
- Estimate model parameters with admissions data from a major research university
  - For many applicants, estimated success probabilities align with those implied by committee's decisions. But committee misweights some applicant characteristics
  - Model increases cohort's average success probability by several percentage points

# Literature Review

- **Graduate Admissions:** Regress acceptance and success on applicant charateristics
  - **Original Literature:** Attiyeh & Attiyeh (1997); Krueger & Wu (2000); Grove & Wu (2007); Athey, Katz, Krueger, Levitt & Poterba (2007)
  - **Study of 2002 Cohort:** Stock, Finegan & Siegfried (2006–2015)
  - **New Predictors/Methods:** Bai, Esche, MacLeod & Shi (2022)
- **Optimal Dynamic Selection Under Costly Evaluation:**
  - **Sequential Search:** McCall (1970); Mortensen (1970); Weitzman (1979)
  - **Rational Inattention:** Sims (2003); Matějka & McKay (2015); Caplin & Dean (2015)

★ **This Paper:** Builds and estimates dynamic selection models of admissions, which test optimality by relating acceptance effects to success effects

# Synthetic Data Protocol (IRB-Approved)

1. **Administrative office receives data from university departments**
    - Algorithmically extracts variables from textual files, such as recommendation letters
2. **Synthetic data (Patki et al. 2016) used for model development**
    - Estimate joint distribution of variables, and simulate synthetic dataset from it
    - Rows do not correspond to real subjects, preventing identity disclosure
    - Cells with $< 5$ comparable observations binned to prevent attribute disclosure
3. **Administrative office executes final estimation code on actual data**
    - Only summary statistics and parameter estimates are returned to researcher

★ **Contribution:** Privacy-protecting framework for researchers studying sensitive data

Synthetic Data Example

# Outline

# Data Description

- **Sample:** Applicants to R1 university's economics PhD program from 2015–2019
- **Dependent Variables:** Measures of admission and success
  - **Admission:** Indicators for all stages of admissions process, including first-round filter
  - **Success:** Completion and placement category/quality for **all** applicants, with academia-equivalent rankings for non-academic placements (Krueger & Wu 2000)
- **Independent Variables:** Objective vs. subjective
  - **Objective:** Demographics and performance measures, such as GRE scores
  - **Subjective:** Textual processing of recommendation letters, application essays, CVs, and supplemental forms using dictionaries to construct variables

Variables | Inference vs. Prediction | Sample Selection

# Reduced-Form Summary

- **Method:** Logistic regression AMEs for acceptance, matriculation, and success
- **Results:** Different independent variables best predict each dependent variable
  - **Accept:** *Committee Score* is best predictor. Conditional on it, other effects (e.g. demographics, *GRE Quantitative Percentile*, *Work Experience*) are smaller
  - **Matriculate:** Predictors of acceptance often negatively predict matriculation. However, best predictor of matriculation is *Open House* attendance
  - **Success:** *PhD Program Rank* is best predictor. Stronger programs place better, and conditional on that, other effects (e.g. *Work Experience*) are smaller

Summary Statistics

# Acceptance, Matriculation, and Success AMEs

| Dependent Variable | Accept | Matriculate | T100 Placement |
|---|---|---|---|
| Winsorized Age | -0.0001 | -0.0223** | 0.0053* |
| Female | 0.0163** | -0.0490 | -0.0032 |
| U.S. | -0.0013 | 0.1472* | 0.0192 |
| International Asian | -0.0214*** | 0.1644** | -0.0127 |
| GRE Quantitative Percentile | 0.0011** | -0.0147** | 0.0006 |
| Undergraduate GPA | 0.0195 | -0.1026 | -0.1096** |
| TOEFL/IELTS | -0.0022 | -0.0631 | 0.0409* |
| Work Experience | 0.0140** | 0.1591** | 0.0358** |
| Committee Score | 0.0788*** | -0.0371 | 0.0007 |
| Advanced Math | 0.0246 | -0.1950** | -0.0287 |
| Letters Terms (%) PC1 | 0.0055 | 0.0671** | 0.0159 |
| CV Coding | -0.0044* | -0.0414** | 0.0125 |
| Essay Economics Terms (%) | 0.0001 | 0.0366** | -0.0104 |
| Essay Positive Terms (%) | -0.0013 | -0.0072 | 0.0123* |
| Open House | | 0.2615*** | |
| Program Rank | | | -0.0011*** |
| Missing Program Rank | | | -0.3965*** |

⏩ Full Logistic Regression AMEs

# Acceptance vs. Success AMEs

| Dependent Variable | Accept | T100 Placement |
|---|---|---|
| Winsorized Age | -0.0001 | 0.0053* |
| Female | 0.0163** | -0.0032 |
| International Asian | -0.0214*** | -0.0127 |
| GRE Quantitative Percentile | 0.0011** | 0.0006 |
| TOEFL/IELTS | -0.0022 | 0.0409* |
| Elite U.S. University | -0.0034 | -0.0192 |
| Work Experience | 0.0140** | 0.0358** |
| #Professor Recommenders | -0.0033 | -0.0069 |
| Missing TOEFL/IELTS | 0.0070 | -0.0008 |
| Committee Score | 0.0788*** | 0.0007 |
| CV Coding | -0.0044* | 0.0125 |
| Essay Economics Terms (%) | 0.0001 | -0.0104 |
| Essay Positive Terms (%) | -0.0013 | 0.0123* |

- Admissions literature regresses acceptance and success on variables in applications, and qualitatively compares the effects
- While such comparisons are of interest, they are insufficient to test optimality

# Insufficiency of Comparing AMEs

- Cannot test optimality by comparing AMEs across regressions. Suppose:
  - Work experience has large effect on success, graduate degree smaller effect
  - 25% of applicants have both, 25% experience only, 25% degree only, 25% neither
  - Committee can accept half of applicants, and other half attend different university
- Optimal committee accepts half with experience and ignores degree
  - **Acceptance Regression:** $\text{AME}_{\text{Experience}} = 1$, $\text{AME}_{\text{Degree}} = 0$
  - **Success Regression:** $1 > \text{AME}_{\text{Experience}} > \text{AME}_{\text{Degree}} > 0$
  - Despite optimality, acceptance and success AMEs not equal across regressions
- **Upshot:** Relationship between AMEs must be interpreted through a model

Wald Tests

# Outline

# Why Structural Models?

- **Reason 1:** Test whether committee's decision rule aligns with desired objective
    - Per literature, committee should accept applicants with highest success probabilities
    - Committees also accept applicants with higher matriculation probabilities
- **Reason 2:** Find counterfactual decision rule in which committee optimizes
    - Quantify gains from deviating from actual to counterfactual decision rule

# Model Framework

- **Portfolio Choice Problem:** Accept subset of applicants with highest sum of success probabilities subject to capacity constraint on number of acceptances

- For each applicant $n \in \{1, \ldots, N\}$, make decision $d_n \in \{\text{Accept}, \text{Reject}\}$ given:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \underbrace{P(y_n = S|x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^{N} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_A,$$

where $x$ = objective variables, $z$ = subjective variables, $y \in \{\text{Success}, \text{Failure}\}$, $N$ = number of applicants, and $N_A$ = maximum number of acceptances

# Model Framework

- **Portfolio Choice Problem:** Accept subset of applicants with highest sum of success probabilities subject to capacity constraint on number of acceptances

$$\max_{(d_1,\dots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \underbrace{P(y_n = S|x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^{N} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_A$$

- BEMS (2022) prove that portfolio choice problem reduces to cutoff rule problem
  - Accept applicant if and only if their success probability exceeds cutoff probability
  - Decisions across applicants now independent, so no need to consider all subsets

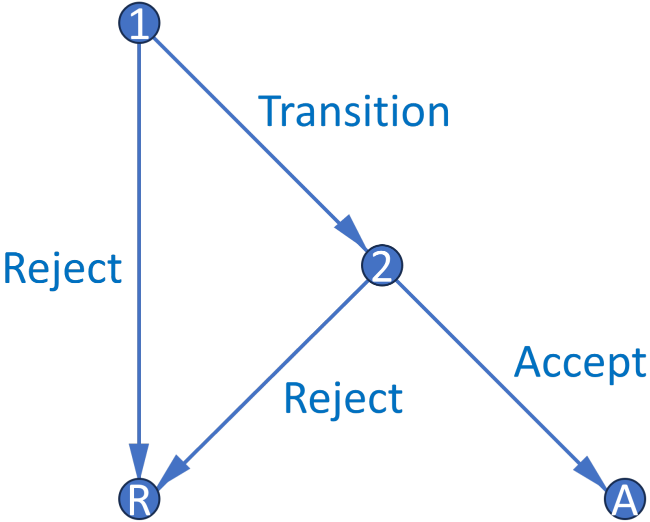
Portfolio Choice to Cutoff Rule

# Cutoff Rule Problem

- Start with two-round cutoff rule problem that accounts for costly evaluation:
  - **Round 1:** See only objective variables (e.g. GRE). Can transition application file to Round 2 to see subjective variables (e.g. recommendation letters), or can reject
  - **Round 2:** If chose to transition application file in Round 1, can accept or reject
  - Assume for now committee transitions and accepts optimal number of files
- Along with the baseline dynamic model, there are additional models
  - Static model better handles high-dimensional subjective variables with missing values
  - Matriculation models account for both success and matriculation probabilities

Models of Admissions

# Decision Graph



# Dynamic Model

- **States:** $r$ = round, $x$ = objective variables, $z$ = subjective variables, $t$ = year, $\overline{k}$ = fixed program rank, $\epsilon_r$ = T1EV preference shock with scale parameter $\sigma$
- **Controls:** $d_1 \in \{\text{Transition}, \text{Reject}\}$, $d_2 \in \{\text{Accept}, \text{Reject}\}$
- **Outcomes:** $y \in \{\text{Success}, \text{Failure}\}$, $y_r^*$ = marginal outcome

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**Round 2:**

$$v_2(x, z, t, \overline{k}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, \overline{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \overline{k}, d_2) = \underbrace{P(y = S|x, z, t, \overline{k})\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S|t)\mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Round 2 Portfolio Choice

# Dynamic Model

**Round 1:**

$$v_1(x, t, \overline{k}, \epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, t, \overline{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \overline{k}, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]\mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Social Surplus}]} + \underbrace{L(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{such that:}$$

$$\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}] = \int_{\widetilde{z}} \sigma \log\left(e^{P(y=S|x,\widetilde{z},t,\overline{k})/\sigma} + e^{P(y_2^*=S|t)/\sigma}\right) dF(\widetilde{z}|x, t)$$

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**Conditional Choice Probabilities:**

$$P(d_r|x,(z),t,\overline{k}) = \frac{\exp(u_r(x,(z),t,\overline{k},d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(x,(z),t,\overline{k},\widetilde{d}_r)/\sigma)}$$

Round 1 Portfolio Choice
Dynamic Parameterization

# Static Model

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- $P(y = S|x, t, \bar{k})$ estimates inconsistent, but fitted values account for $z$'s effect on $y$

Round 1 Portfolio Choice

# Portfolio Choice with Matriculation

- Accept subset of applicants with highest sum of success times matriculation probabilities subject to capacity constraint on expected number of matriculants

- For each applicant $n \in \{1, \ldots, N\}$, make decision $d_n \in \{\text{Accept}, \text{Reject}\}$ given:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \underbrace{P(y_n = S|x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}},$$

$$\text{such that: } \sum_{n=1}^{N} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M,$$

where $(x, z) =$ variables, $y \in \{\text{Success}, \text{Failure}\}$, $m \in \{\text{Matriculate}, \text{Decline}\}$, $N =$ number of applicants, and $N_M =$ maximum number of matriculants

# Portfolio Choice with Matriculation

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$$\text{such that: } \sum_{n=1}^{N} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \le N_M$$

**Theorem 1 (Proof)**

*For large $N$, this problem has a cutoff rule solution on* **success** *probabilities.*

- Likely matriculants more valuable, but also more expensive $\Rightarrow$ effects cancel
- However, matriculation probabilities matter in Round 1 of a two-round model

Cutoff Rule with Matriculation

# Matriculation-Dynamic Model

**Round 2:**

$$v_2(x, z, t, \overline{k}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, \overline{k}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

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**Round 1:**

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$$\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}] = \int_{\widetilde{z}} \sigma \log\left(e^{[P(y=S|x,\widetilde{z},t,\overline{k}) - P(y_2^*=S|t)]/\sigma} + 1\right) P(m = M|x, \widetilde{z}, t) dF(\widetilde{z}|x, t)$$

Dynamic Portfolio Choice with Matriculation

# Outline

# Maximum Likelihood Estimation

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^{N} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}}$$

$$\prod_{n=1}^{N} \underbrace{\left[P_{y_n}(\theta_{S_1}) P_{y_n}(\theta_{S_2})\right]^{\mathbb{1}(y_n=S)} \left[(1 - P_{y_n}(\theta_{S_1}))(1 - P_{y_n}(\theta_{S_2}))\right]^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- In unrestricted likelihood $\mathcal{L}^U$, parameterize CCPs with flexible logistic specifications
- In restricted likelihood $\mathcal{L}^R$, use model CCPs, which depend on structural parameters
- Do success probabilities implied by outcomes align with those implied by CCPs?

Dynamic Likelihood | Matriculation Likelihood | Empirical Considerations

# Cutoff Probability Identification

**Theorem 2 (Proof)**

*The maximum likelihood cutoff probability estimate is such that the number of acceptances $N_A$ equals the expected $N_A$. Also, for any cutoff probability and $N_A$, the likelihood is highest when the committee accepts the best $N_A$ applicants.*

- Identify each year's cutoffs by how selective committee is that year
- While selectivity identifies cutoffs, sorting improves restricted likelihood
  - If committee engages in sorting, Akaike Information Criterion more favorable
  - $\text{AIC} = 2\left[K - \log(\hat{\mathcal{L}}^R)\right]$, such that $K = \dim(\theta^R)$

# Restricted Dynamic Estimates

| | $L(\text{Cutoff})_1$ | $P(\text{Cutoff})_2$ | Sigma |
|---|---|---|---|
| 2015 | 0.3327 | 32.63% | |
| | (0.2495, 0.4437) | (24.06%, 42.54%) | |
| 2016 | 0.3902 | 38.35% | |
| | (0.2622, 0.5807) | (24.57%, 54.29%) | |
| 2017 | 0.4615 | 45.25% | |
| | (0.3928, 0.5423) | (38.04%, 52.68%) | |
| 2018 | 0.3473 | 34.14% | |
| | (0.2828, 0.4263) | (27.30%, 41.70%) | |
| 2019 | 0.4173 | 41.06% | |
| | (0.3429, 0.5078) | (33.39%, 49.19%) | |
| Sigma | | | 0.0018 |
| | | | (0.0014) |
| Observations = 2329 | Parameters = 528 | LL = -13667.0 | AIC = 28390.0 |

Full Dynamic Estimates Full Myopic Estimates

# Restricted Static Estimates

| | $P(\text{Cutoff})_1$ | $P(\text{Cutoff})_2$ | Sigma |
|---|---|---|---|
| 2015 | 29.33% | 36.37% | |
| | (23.18%, 36.34%) | (28.46%, 45.10%) | |
| 2016 | 29.17% | 43.56% | |
| | (22.20%, 37.28%) | (34.01%, 53.62%) | |
| 2017 | 37.04% | 46.49% | |
| | (32.09%, 42.28%) | (37.65%, 55.56%) | |
| 2018 | 38.19% | 34.19% | |
| | (34.05%, 42.50%) | (26.12%, 43.29%) | |
| 2019 | 34.04% | 41.70% | |
| | (28.67%, 39.85%) | (33.20%, 50.72%) | |
| Sigma | | | 0.0056 |
| | | | (0.0035) |
| Observations = 2329 | Parameters = 79 | LL = -2528.4 | AIC = 5214.8 |

Full Static Estimates

# Restricted Matriculation-Dynamic Estimates

| | $L(\text{Cutoff})_1$ | $P(\text{Cutoff})_2$ | Sigma |
|---|---|---|---|
| 2015 | 0.0066 | 31.33% | |
| | (0.0023, 0.0190) | (21.85%, 42.69%) | |
| 2016 | 0.0065 | 41.42% | |
| | (0.0024, 0.0180) | (30.36%, 53.41%) | |
| 2017 | 0.0098 | 45.65% | |
| | (0.0036, 0.0267) | (39.08%, 52.37%) | |
| 2018 | 0.0086 | 31.99% | |
| | (0.0032, 0.0231) | (24.21%, 40.91%) | |
| 2019 | 0.0089 | 39.77% | |
| | (0.0031, 0.0253) | (31.52%, 48.65%) | |
| Sigma | | | 0.0019** |
| | | | (0.0009) |
| Observations = 2329 | Parameters = 567 | LL = -13796.0 | AIC = 28726.0 |

Full Matriculation-Dynamic Estimates

# Outline

# Deviation Gains Test

- Suppose committee adopts model's decision rule. Will average success probability of acceptance/matriculation set increase, and if so, by how much?
  - Solve model with $\sigma = 0$ in CCPs, compute average yearly $P(\text{Success})$ of model's acceptance/matriculation set, and compare to that of actual decision rule's set
  - $\hat{\theta}_S^R$ may be biased to rationalize suboptimal behavior, so plug in $\hat{\theta}_S^U$

★ **Robustness:** Given too few observations for holdout set and infeasibility of large $k$-fold cross-validation due to runtime, implement randomization test

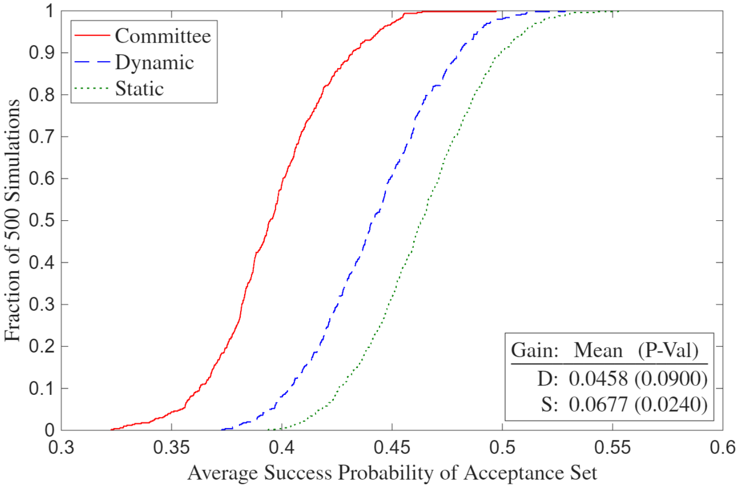
  - Compute average success probabilities with perturbations of $\hat{\theta}_S^U$
  - P-value is fraction of perturbations with committee's $\overline{P(\text{Success})} >$ model's

Deviation Gains Algorithm

# Deviation Gains Interpretation

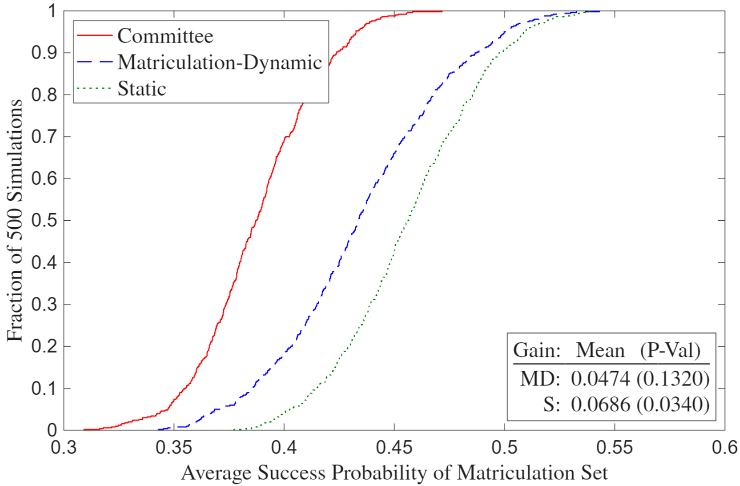
- Given meaningfully large deviation gains, committee may have incorrect beliefs about which types of applicants are most likely to succeed
  - Matriculation rate of model's acceptance set is different from that of actual decision rule's acceptance set, making it necessary to compare matriculation sets
  - In matriculation counterfactuals, compare success probabilities of matriculation sets by using matriculation probabilities to simulate accepted applicants' decisions
- Committee may use success-predicting variables that I don't observe
  - Because the success probability measure used to evaluate gains cannot detect such variables, any edge the committee's private information provides goes unmeasured
  - Comprehensive data, especially committee score in $z$, protects against this limitation

# Deviation Gains



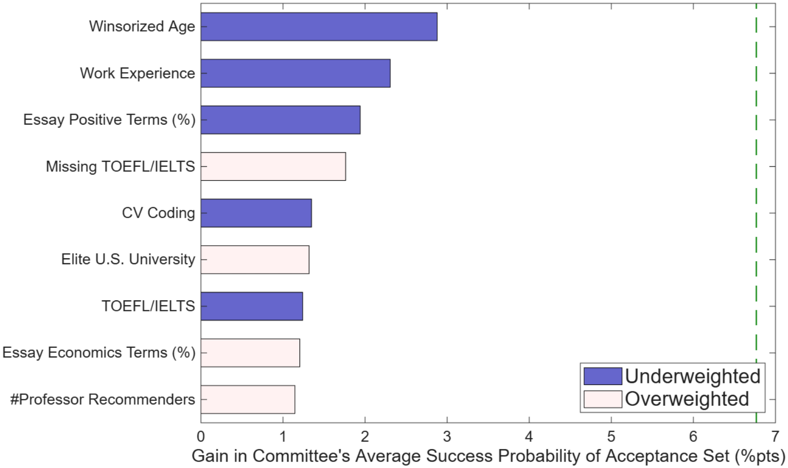

Full Deviation Gains

# Deviation Gains (Matriculation)



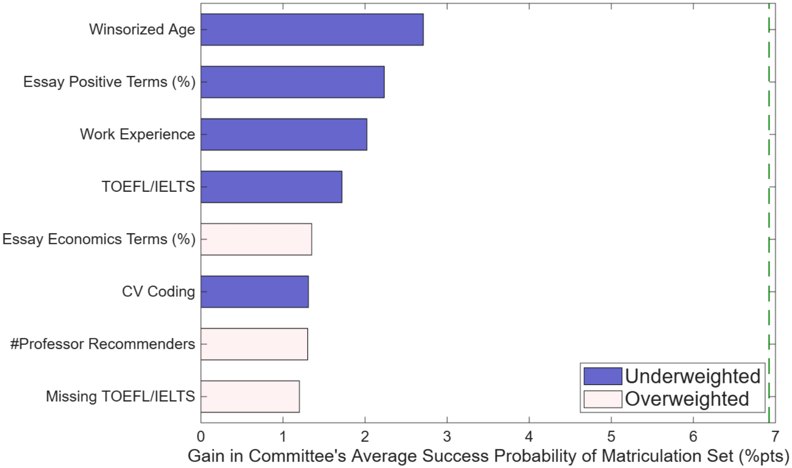

Full Deviation Gains (Matriculation)

# Deviation Gains Attribution



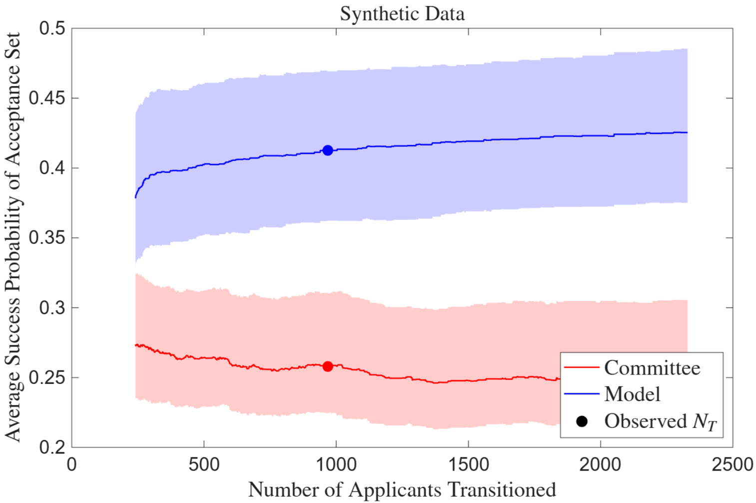

⏩ Full Deviation Gains Attribution ⏩ Deviation Gains Attribution Table

# Deviation Gains Attribution (Matriculation)

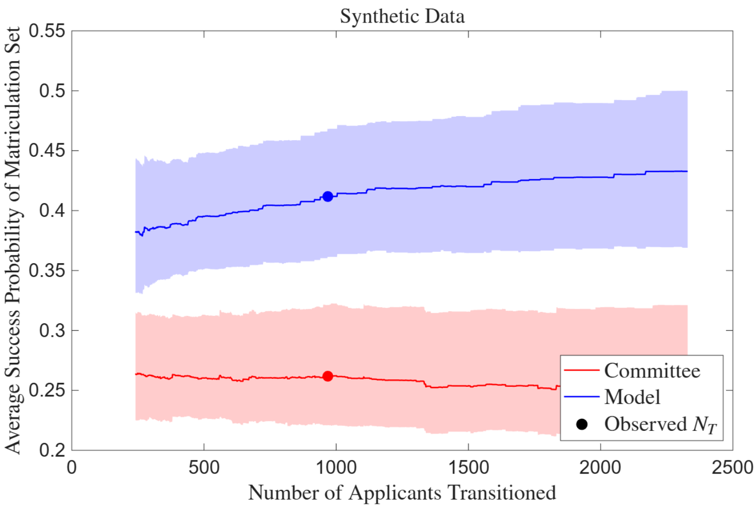



Full Deviation Gains Attribution (Matriculation)
Deviation Gains Attribution Table (Matriculation)

# Benefit of Reading More Files



# Benefit of Reading More Files (Matriculation)



# Conclusion

- **Economic Problem:** Selecting best options under costly evaluation
- **Application:** Graduate admissions using synthetic data protocol to protect privacy
- **Contribution:** Build models to test optimality and quantify deviation gains
    1. For many applicants, estimated success probabilities align with those implied by committee's decisions. But committee misweights some applicant characteristics
    2. Model increases cohort's average success probability by several percentage points
- **Future Research:** Two-sided matching model across multiple universities

▸ Simulated Results ▸ Waitlist Models ▸ Multiyear Models

# Thank You!

# A1: Synthetic Rows $\neq$ Actual Applicants

| Transition | Waitlist | Accept | Matriculate | Success | GRE Quantitative Percentile | Gender | Demographic 1 | Demographic 2 | Advanced Math | Committee Score | Program Rank |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 | 89 | 0 | 0 | 0 | 0 | 5 | 150 |
| 1 | 0 | 1 | 0 | 1 | 96 | 0 | 1 | 0 | 0 | 6 | 23 |
| 0 | 0 | 0 | 0 | 0 | 88 | 0 | 1 | 0 | 1 | . | 150 |
| 0 | 0 | 0 | 0 | 0 | 80 | 1 | 1 | 0 | 1 | . | 150 |
| 0 | 0 | 0 | 0 | 1 | 99 | 0 | 1 | 0 | 0 | . | 29 |
| 0 | 0 | 0 | 0 | 0 | 81 | 0 | 0 | 1 | 1 | . | 23 |
| 1 | 0 | 1 | 1 | 1 | 92 | 0 | 1 | 0 | 0 | 3 | 37 |
| 1 | 1 | 1 | 1 | 1 | 93 | 0 | 0 | 0 | 1 | 3 | 37 |
| 0 | 0 | 0 | 0 | 0 | 83 | 1 | 0 | 1 | 0 | . | 48 |
| 1 | 1 | 0 | 0 | 0 | 87 | 0 | 1 | 0 | 1 | 5 | 34 |

$\Downarrow$

| Transition | Waitlist | Accept | Matriculate | Success | GRE Quantitative Percentile | Gender | Demographic 1 | Demographic 2 | Advanced Math | Committee Score | Program Rank |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 77 | 0 | 0 | 0 | 1 | . | 30 |
| 0 | 0 | 0 | 0 | 1 | 89 | 0 | 0 | 0 | 0 | . | 25 |
| 0 | 0 | 0 | 0 | 0 | 85 | 1 | 0 | 0 | 0 | . | 150 |
| 0 | 0 | 0 | 0 | 0 | 76 | 0 | 1 | 0 | 1 | . | 51 |
| 1 | 1 | 0 | 0 | 0 | 88 | 0 | 1 | 0 | 0 | 8 | 150 |
| 1 | 0 | 1 | 0 | 0 | 88 | 0 | 0 | 0 | 1 | 5 | 150 |
| 1 | 1 | 0 | 0 | 0 | 84 | 0 | 0 | 1 | 0 | 2 | 20 |
| 0 | 0 | 0 | 0 | 0 | 82 | 0 | 1 | 0 | 0 | . | 35 |
| 0 | 0 | 0 | 0 | 0 | 70 | 1 | 0 | 0 | 0 | . | 150 |
| 1 | 1 | 0 | 0 | 1 | 93 | 0 | 1 | 0 | 0 | 5 | 57 |

Synthetic Data Protocol

# A1: Dependent Variables

Table 1a: Dependent Variables

| Admission | Success |
|---|---|
| **$\mathbb{1}$(Transition)** | **$\mathbb{1}$(Complete Program)** |
| $\mathbb{1}$(Accept) | **$\mathbb{1}$(Academic Placement)** |
| **$\mathbb{1}$(Waitlist)** | **$\mathbb{1}$(Government Placement)** |
| **$\mathbb{1}$(Open House)** | **$\mathbb{1}$(Consulting Placement)** |
| $\mathbb{1}$(Matriculate) | **$\mathbb{1}$(Tech Placement)** |
| **$\mathbb{1}$(Attend Program)** | **$\mathbb{1}$(Other Placement)** |
| **Program Rank** | **$\mathbb{1}$(No Placement)** |
| | **Placement Rank** |
| | **$\mathbb{1}$(Top 100 Placement)** |

- Economics-only variables bold, non-economics-only variables italic
- U.S. News PhD program rankings, IDEAS/RePEc placement rankings
  - For non-tenure track, use weighted average of university's and maximum rankings
  - For non-academia, use average of ChatGPT, Claude, and Gemini's rankings

Data Description

# A1: Objective Variables

Table 1b: Objective Independent Variables

| Demographic | Performance |
|---|---|
| **Year** | GRE Verbal Percentile |
| $\mathbb{1}$*(Covid Year)* | GRE Quantitative Percentile |
| Winsorized Age | GRE Analytic Percentile |
| $\mathbb{1}$(Female) | Undergraduate GPA |
| $\mathbb{1}$**(U.S.)** | TOEFL/IELTS |
| $\mathbb{1}$*(U.S. African/Hispanic/Native)* | $\mathbb{1}$(Elite U.S. University) |
| $\mathbb{1}$*(U.S. Asian)* | $\mathbb{1}$(Elite U.S. Liberal Arts College) |
| $\mathbb{1}$*(U.S. Other)* | $\mathbb{1}$(Other U.S. University/College) |
| $\mathbb{1}$(International Asian) | $\mathbb{1}$(Elite Foreign University) |
| $\mathbb{1}$(International Other) | $\mathbb{1}$(Other Foreign University) |
| | $\mathbb{1}$(Graduate Degree) |
| | $\mathbb{1}$(Work Experience) |
| | #Professor Recommenders |
| | #Prolific Recommenders |
| | *Concentration Dummy Variables* |

Data Description

# A1: Subjective Variables

Table 1c: Subjective Independent Variables

| Score/Recommendation Letters | Supplement/CV/Essay |
|---|---|
| **Committee Score** | **#Math Courses** |
| **Letters Length** | **$\mathbb{1}$(Advanced Math)** |
| **Positive Terms (%) (YS 2012)** | **CV Length** |
| **Negative Terms (%) (YS 2012)** | **CV Coding** |
| **Standout Terms (%) (BEMS 2022)** | **CV $\mathbb{1}$(Publication/Working Paper)** |
| **Ability Terms (%) (BEMS 2022)** | **Essay Length** |
| **Research Terms (%) (BEMS 2022)** | **Essay $\mathbb{1}$(Specific Professor)** |
| **Grindstone Terms (%) (BEMS 2022)** | **Essay Economics Terms (%)** |
| **Teaching Terms (%) (BEMS 2022)** | **Essay Research Terms (%)** |
| **Communal Terms (%) (BEMS 2022)** | **Essay Positive Terms (%)** |
| **Agentic Terms (%) (MHM 2009)** | **Essay Negative Terms (%)** |
| | **Field Dummy Variables** |

- Dictionaries from Madera, Hebl & Martin (2009), Young & Soroka (2012), and Bai, Esche, MacLeod & Shi (2022)
- *#Math Courses* and *Advanced Math* objective after 2020

Data Description

# A1: Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)
- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

Data Description

# A1: Academia-Equivalent Rankings (Consulting/Tech/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Econic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

Data Description

# A1: Dictionaries

- **Recommendation Letters:**
  - **Positive/Negative Words:** Young & Soroka (2012)'s Lexicoder Sentiment Dictionary
  - **Standout, Ability, Research, Grindstone, Communal Words:** Bai, Esche, MacLeod & Shi (2022)
  - **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera, Hebl & Martin 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
  - **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop
- **CVs and Essays:**
  - **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape
  - **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
  - **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
  - **Economics:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban

Data Description

# A1: Inference vs. Prediction

- Challenging to obtain consistent estimates due to omitted variables
  - **Example:** Don't observe effort (e.g. hours worked). Can bias coefficients upward if effort correlated with variables and with success irrespective of those variables
  - Existing literature also suffers from this problem, though BEMS (2022) parse recommendation letters for "grindstone terms" (e.g. depend*, diligen*)
- Research questions emphasize prediction over inference
  - Mullainathan & Spiess (2017) distinguish $\hat{\beta}$ vs. $\hat{y}$ problems
  - If GRE predicts success, committee should use it regardless of why

Data Description

# A1: Sample Selection

- Determinants of success conditional on applying vs. matriculating
- BEMS (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples. Consistent with empirical results
- If there aren't some types of outcomes data (e.g. grades, passing comps) for all applicants, or such data are too hard to obtain, use Heckman correction
  - Let $y_n^* = x_n\beta + u_n$, $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$, and $y_n = d_n y_n^*$
  - **Step 1:** Run probit of $d$ on $z$ and compute inverse Mills $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$
  - **Step 2:** Run OLS or heckprobit of $y$ on $(x, \hat{\lambda})$ to get $\hat{\beta}$
  - **Instruments:** Variables that predict matriculation but not the outcome

Data Description

# A1: Wald Tests

- Test if acceptance and/or success AMEs are same across PhD programs
  - **Example:** Does elite undergraduate university predict acceptance and/or success more or less for economics program than for other programs?
  - **Sample:** Applicants to university's economics, government, history, linguistics, and psychology PhD programs from 2020–2023 (when multi-program data is available)
- Can I test if acceptance AMEs equal success AMEs within programs?
  - Predictors of acceptance only (success only) may be overweighted (underweighted), provided committee wants to select based on a "success" model
  - **Problem:** Not a test if admissions decisions are optimal
  - Need structural model of admissions to conclude about optimality

Insufficiency of Comparing AMEs

# A1: Logistic Wald Test Across Programs

Assume $(Y^E \perp\!\!\!\perp Y^{NE})|X$. Let $P(Y_n^E = 1|x_n^E;\theta^E) = \frac{\exp(x_{n,0}^E\theta_0^E + x_{n,1}^E\theta_1^E)}{1+\exp(x_{n,0}^E\theta_0^E + x_{n,1}^E\theta_1^E)}$, and $P(Y_n^{NE} = 1|x_n^{NE};\theta^{NE}) = \frac{\exp(x_{n,0}^{NE}\theta_0^{NE} + x_{n,1}^{NE}\theta_1^{NE})}{1+\exp(x_{n,0}^{NE}\theta_0^{NE} + x_{n,1}^{NE}\theta_1^{NE})}$. Finally, let $N = N_E + N_{NE}$. Then:

$$\mathcal{L}_N^U(y|x;\theta) = \prod_{n=1}^{N_E} P(Y_n^E = 1|x_n^E;\theta^E)^{Y_n^E}\left[1 - P(Y_n^E = 1|x_n^E;\theta^E)\right]^{1-Y_n^E}$$
$$\prod_{n=1}^{N_{NE}} P(Y_n^{NE} = 1|x_n^{NE};\theta^{NE})^{Y_n^{NE}}\left[1 - P(Y_n^{NE} = 1|x_n^{NE};\theta^{NE})\right]^{1-Y_n^{NE}}.$$

- $Y \in \{\mathbb{1}(\text{Accept}), \mathbb{1}(\text{Attend Program}), \mathbb{1}(\text{Complete Program}), \mathbb{1}(\text{Good Placement})\}$
- **Test:** Let $g(\hat{\theta}) = \frac{1}{N_E}\sum_{n=1}^{N_E}\hat{P}_n^E(1-\hat{P}_n^E)\hat{\theta}_1^E - \frac{1}{N_{NE}}\sum_{n=1}^{N_{NE}}\hat{P}_n^{NE}(1-\hat{P}_n^{NE})\hat{\theta}_1^{NE}$. Then: $W = g(\hat{\theta})'[\nabla_\theta g(\hat{\theta})\hat{V}(\hat{\theta})\nabla_\theta g(\hat{\theta})']^{-1}g(\hat{\theta}) \sim \chi_Q^2$. Compare AMEs per Mood (2010)

Insufficiency of Comparing AMEs

# A1: Linear Wald Test Across Programs

Assume $\left(Y^E \perp\!\!\!\perp Y^{NE}\right)\big|X$, i.e. $\left(\text{Economics} \perp\!\!\!\perp \text{Non-Economics}\right)\big|\text{Predictors}$.

Let $f\left(y_n^E \middle| x_n^E; \beta^E, \sigma_E^2\right) = \frac{1}{\sqrt{2\pi\sigma_E^2}} \exp\left[\frac{-1}{2\sigma_E^2}\left(y_n^E - x_{n,0}^E\beta_0^E - x_{n,1}^E\beta_1^E\right)^2\right]$, and

$f\left(y_n^{NE} \middle| x_n^{NE}; \beta^{NE}, \sigma_S^2\right) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}} \exp\left[\frac{-1}{2\sigma_{NE}^2}\left(y_n^{NE} - x_{n,0}^{NE}\beta_0^{NE} - x_{n,1}^{NE}\beta_1^{NE}\right)^2\right]$. Then:

$$\mathcal{L}_N^U\left(y \middle| x; \beta, \sigma^2\right) = \prod_{n=1}^{N_E} f\left(y_n^E \middle| x_n^E; \beta^E, \sigma_E^2\right) \prod_{n=1}^{N_{NE}} f\left(y_n^{NE} \middle| x_n^{NE}; \beta^{NE}, \sigma_{NE}^2\right).$$

- $Y \in \{\mathbb{1}(\text{Standardized Committee Score}), \mathbb{1}(\text{Standardized Placement Rank})\}$
- **Test:** Let $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$. (Can also do likelihood-ratio test.) Then $W = g(\hat{\beta}_1)'[\nabla_{\beta,\sigma^2} g(\hat{\beta}_1)\hat{V}(\hat{\beta}, \hat{\sigma}^2)\nabla_{\beta,\sigma^2} g(\hat{\beta}_1)']^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$
- **Assumption:** Homoskedasticity (i.e. $\forall n,\ \sigma_n^2 = \sigma^2$)

Insufficiency of Comparing AMEs

# A1: Additional Questions

- Are there different determinants of different types of success? (AKKLP 2007)
  - Need different skills at different stages in the program, such as grades vs. placement
  - For some programs, initial placement matters less than for economics
- Different determinants for international applicants? (KW 2000)
  - **Admission:** May have different preparation and/or interests
  - **Completion:** Depending on country, may have fewer outside options
  - **Placement:** May return to native country after graduation
- Different determinants over time? (2015 to 2023)
  - COVID-19 and other macro factors; changes in professions (e.g. more empirics)
  - For economics, increased importance of elite predocs and Fed RAships

▸▸ Insufficiency of Comparing AMEs

# A2: Portfolio Choice to Cutoff Rule (BEMS 2022)

**Portfolio Choice Problem:**

$$\max_{(d_1,\dots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n, z_n)\mathbb{1}(d_n = A), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A$$

$$\mathcal{L}(d_1,\dots,d_N;\lambda) = \sum_{n=1}^{N} \left[P(y_n = S|x_n, z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\right] + \lambda(N_A - N)$$

**Cutoff Rule Solution:**

$$\forall n \in \{1,\dots,N\},\ d_n(x_n, z_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S|x_n, z_n) < \lambda^* \end{cases}$$

**Upshot:** $\lambda^*$ is cutoff probability. Denote it hereafter as $P(y^* = S)$

- $P(y^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$

Model Framework

# A2.1: Portfolio Choice to Cutoff Rule with Shocks

**Portfolio Choice Problem:**

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \Big\{ \big[P(y_n = S|x_n, z_n) + \epsilon_n(A)\big] \mathbb{1}(d_n = A) + \epsilon_n(R)\mathbb{1}(d_n = R) \Big\},$$

$$\text{such that: } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A. \quad \text{Then: } \mathcal{L}(d_1, \ldots, d_N; \lambda) =$$

$$\sum_{n=1}^{N} \Big\{ \big[P(y_n = S|x_n, z_n) + \epsilon_n(A)\big] \mathbb{1}(d_n = A) + \big[\lambda + \epsilon_n(R)\big] \mathbb{1}(d_n = R) \Big\} + \lambda(N_A - N)$$

**Cutoff Rule Solution:**

$$\forall n \in \{1, \ldots, N\}, d_n(x_n, z_n, \epsilon_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) + \epsilon_n(A) > \lambda^* + \epsilon_n(R) \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) + \epsilon_n(A) = \lambda^* + \epsilon_n(R) \\ R & \text{if } P(y_n = S|x_n, z_n) + \epsilon_n(A) < \lambda^* + \epsilon_n(R) \end{cases}$$

**Theorem 3 (Proof)**

*For large $N$, there is a cutoff rule solution with logistic CCPs.*

Model Framework

# A2: Models of Admissions

1. **Dynamic Model:** Committee ranks applicants based on conditional expectation in Round 1 about those who survive to Round 2, reads files of those above Cutoff 1, and rejects the rest. Favors those in Round 1 with objective variables positively correlated with subjective variables that predict success. Then re-ranks survivors based on all variables, accepts survivors above Cutoff 2, and rejects the rest
   1. **Static Model:** In Round 1, rank applicants based only on objective variables
   2. **Myopic Model:** Like dynamic, but distribution of $z$ not conditional on $x$
2. **Matriculation-Dynamic Model:** In Round 1, rank applicants based on combination of success and matriculation probabilities (largely success probabilities)
3. **Waitlist-Dynamic/Hybrid Models:** In Round 1, rank applicants based on success probabilities. After Round 1, accept or waitlist applicants in Round 2. After acceptances matriculate or decline, accept or reject waitlisted applicants in Round 3. Favor likely matriculants in Round 2 to increase selectivity in Round 3

Cutoff Rule Problem

# A2: Description of Models

- **Round 2:** If accept applicant, get success probability $P(y = S|x, z, t, \overline{k})$, where Success $= \mathbb{1}(\text{Complete Program})$ or $\mathbb{1}(\text{Good Placement})$. If reject applicant, get $P(y_2^* = S|t) = P(\text{Success})$ of year $t$'s marginal accepted applicant

- **Round 1:** If reject applicant, get $L/P(y_1^* = S|t) = L/P(\text{Success})$ of year $t$'s marginal transitioned applicant. Can be less or greater than $P(y_2^* = S|t)$ depending on value placed on committee's time vs. admitting a successful cohort
  - **Dynamic:** If transition applicant, get $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}] =$ expected value of Round 2, where $z$ is integrated out conditional on $(x, t)$ based on $F(z|x, t)$
  - **Static:** If transition applicant, get $P(y = S|x, t, \overline{k})$, or $P(\text{Success})$ unconditional on $z$. It may differ from $P(y = S|x, z, t, \overline{k})$, though it still accounts for $z$'s effect on $y$
  - **Myopic:** In $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]$, replace $F(z|x, t)$ with $F(z|t)$

Cutoff Rule Problem

# A2: Portfolio Choice to Cutoff Rule, Round 2

**Portfolio Choice Problem:**

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|x_n, z_n)\mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A$$

$$\mathcal{L}(d_{2,1},\ldots,d_{2,N_T};\lambda_2) = \sum_{n=1}^{N_T} \left[P(y_n = S|x_n, z_n)\mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)\right] + \lambda_2(N_A - N_T)$$

**Cutoff Rule Solution:**

$$\forall n \in \{1,\ldots,N_T\},\ d_{2,n}(x_n, z_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|x_n, z_n) < \lambda_2^* \end{cases}$$

**Upshot:** $\lambda_2^*$ is cutoff probability. Denote it hereafter as $P(y_2^* = S)$

- $P(y_2^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$

Dynamic Model

# A2: Portfolio Choice to Cutoff Rule, Round 1 (Dynamic)

**Portfolio Choice Problem:**

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n;\overline{\lambda_2^*})\mathbb{1}(d_{1,n}=T), \text{ such that: } \sum_{n=1}^{N}\mathbb{1}(d_{1,n}=T)\leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n)=\int_{\widetilde{z}_n}\max\{P(y_n=S|x_n,\widetilde{z}_n),\overline{\lambda_2^*}\}dF_{z|x}(\widetilde{z}_n|x_n)$$

$$\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1)=\sum_{n=1}^{N}\left[EV_{2,n}(x_n)\mathbb{1}(d_{1,n}=T)+\lambda_1\mathbb{1}(d_{1,n}=R)\right]+\lambda_1(N_T-N)$$

**Cutoff Rule Solution:**

$$\forall n\in\{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*)=\begin{cases} T & \text{if } EV_{2,n}(x_n)>\lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n)=\lambda_1^* \\ R & \text{if } EV_{2,n}(x_n)<\lambda_1^* \end{cases}$$

**Upshot:** $\lambda_1^*$ is cutoff level. Denote it hereafter as $L(y_1^*=S)\in\left[EV_{2,N_T+1},EV_{2,N_T}\right]$

▸▸ Dynamic Model

# A2: Dynamic Parameterization 1

**Unrestricted CCPs:**

$$P_{d_1} = P\left(d_1 = T \middle| x, t; \theta_T\right) = \frac{\exp(f_T(x,t)\theta_T)}{1 + \exp(f_T(x,t)\theta_T)}$$

$$P_{d_2} = P\left(d_2 = A \middle| x, z, t; \theta_A\right) = \frac{\exp(f_A(x,z,t)\theta_A)}{1 + \exp(f_A(x,z,t)\theta_A)}$$

**Success Probability:**

$$P_y = P\left(y = S \middle| x, z, t, k; \theta_S\right) = \frac{\exp(f_S(x,z,t,k)\theta_S)}{1 + \exp(f_S(x,z,t,k)\theta_S)}$$

**Cutoff Level/Probability:**

$$L_{y_1^*} = L\left(y_1^* = S \middle| t; \theta_{S_1^*}\right) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_2^*} = P\left(y_2^* = S \middle| t; \theta_{S_2^*}\right) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1 + \exp(f_{S_2^*}(t)\theta_{S_2^*})}$$

Dynamic Model

# A2: Dynamic Parameterization 2

$$\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}] = \int_{\widetilde{z}} \sigma \log \left( \sum_{\widetilde{d}_2 \in \{A,R\}} e^{u_2(x,\widetilde{z},t,\overline{k},\widetilde{d}_2)/\sigma} \right) dF(\widetilde{z}|x, t)$$

$$= \int_{\widetilde{z}_1} \cdots \int_{\widetilde{z}_J} \left[ \cdot \right] dF_1(\widetilde{z}_1|\widetilde{z}_2, \ldots, \widetilde{z}_J, x, t) \cdots dF_J(\widetilde{z}_J|x, t)$$

- **Continuous:** $f_{z|x,t}(z_j|x, t; \beta_{F_{z|x,t,j}}, \sigma^2_{F_{z|x,t,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,t,j}}}} \exp\left[\frac{-1}{2\sigma^2_{F_{z|x,t,j}}}(z_j - g(x, t)\beta_{F_{z|x,t,j}})^2\right]$
- **Binary:** $P(z_j = 1|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}{1+\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}$
- **Multinomial:** $P(z_j = \ell|x, t; \theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta^{\ell}_{F_{z|x,t,j}}\right)}{1+\sum_{i=1}^{L_j-1} \exp\left(g(x,t)\theta^{i}_{F_{z|x,t,j}}\right)}, \ \ell = 1, \ldots, L_j - 1$

Dynamic Model

# A2: Portfolio Choice to Cutoff Rule, Round 1 (Static)

**Portfolio Choice Problem:**

$$\max_{(d_{1,1},\dots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n)\mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T$$

$$\mathcal{L}(d_{1,1},\dots,d_{1,N};\lambda_1) = \sum_{n=1}^{N} \left[P(y_n = S|x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1\mathbb{1}(d_{1,n} = R)\right] + \lambda_1(N_T - N)$$

**Cutoff Rule Solution:**

$$\forall n \in \{1,\dots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S|x_n) > \lambda_1^* \\ \{T,R\} & \text{if } P(y_n = S|x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S|x_n) < \lambda_1^* \end{cases}$$

**Upshot:** $\lambda_1^*$ is cutoff probability. Denote it hereafter as $P(y_1^* = S)$

- $P(y_1^* = S) \in \left[P(y_{N_T+1} = S|x_{N_T+1}), P(y_{N_T} = S|x_{N_T})\right]$

Static Model

# A2: Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_F\right)}_{\text{Density of } z^c|z^{\neg c}, x, t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}}$$

$$\prod_{n=N_T+1}^{N} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{f\left(z_n^{\neg c} | x_n, t_n; \theta_F\right)}_{\text{Density of } z^{\neg c}|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Success often unobservable for applicants from fewer than six years ago

Maximum Likelihood Estimation

# A2: Heckman (1979)-Corrected Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{\Phi(x_n, t_n; \theta_H)}_{\text{Selection}} \underbrace{f\left(z_n \middle| x_n, t_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z|x,t} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}}_{\text{Conditional choice probabilities}}$$

$$\underbrace{(1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probability}} \prod_{n=N_T+1}^{N} \underbrace{(1 - \Phi(x_n, t_n; \theta_H))}_{\text{Selection}} \underbrace{f\left(z_n^{\neg c} \middle| x_n, t_n, \frac{-\phi(x_n, t_n; \theta_H)}{1 - \Phi(x_n, t_n; \theta_H)}; \theta_F\right)}_{\text{Density of } z^{\neg c}|x,t}$$

$$\underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- *Committee Score* unobservable for applicants who didn't survive Round 1

Maximum Likelihood Estimation

# A2: Empirical Considerations

- **Selection Effects:** Track success for all applicants, including non-matriculants
  - In selected samples, success parameters are biased down (e.g. quantitative GRE)
- **Potential Outcomes:** Track which graduate program all applicants attended, and add *Program Rank* as a control variable in success blocks of likelihood
  - In decision blocks containing success, fix *Program Rank* at this university's rank
  - If a variable only predicts success by helping applicant get into elite graduate program, then committee should not use that variable to rank applicants
- **Unobservables:** I may not observe all information that committee observes
  - **Example:** Recommendation letter has nuances that algorithms can't detect
  - Include subjective committee score in $z$ as a proxy for this information

Maximum Likelihood Estimation

# A2: Cutoff Rule with Matriculation

**Portfolio Choice Problem:**

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_n = A),$$

$$\text{such that: } \sum_{n=1}^{N} P(m_n = M|w_n)\mathbb{1}(d_n = A) \leq N_M$$

$$\mathcal{L}(d_1,\ldots,d_N;\lambda) = \sum_{n=1}^{N} \left[P(y_n = S|w_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\right]P(m_n = M|w_n) + \lambda(N_M - \overline{N_M})$$

**Cutoff Rule Solution:**

$$\forall n \in \{1,\ldots,N\},\ d_n(w_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda^* \\ \{A,R\} & \text{if } P(y_n = S|w_n) = \lambda^* \\ R & \text{if } P(y_n = S|w_n) < \lambda^* \end{cases}$$

▸▸ Portfolio Choice with Matriculation

# A2: Cutoff Rule with Matriculation (Large $N$)

**Portfolio Choice Problem:**

$$\max_{\delta \in \{\delta | \delta : \mathbb{R}^K \to [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w) P(m = M|w) \delta(w) f(w) dw,$$

$$\text{such that: } \int_{w \in \mathbb{R}^K} P(m = M|w) \delta(w) f(w) dw \leq N_M$$

$$\mathcal{L}(\delta; \lambda) = \int_{w \in \mathbb{R}^K} \left[ P(y = S|w)\delta(w) + \lambda(1 - \delta(w)) \right] P(m = M|w) f(w) dw + \lambda(N_M - \overline{N_M})$$

**Cutoff Rule Solution:**

$$\forall w \in \mathbb{R}^K,\ \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0,1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^* \end{cases}$$

**Upshot:** Point-identified cutoff probability $\lambda^*$ is with respect to success probabilities

Portfolio Choice with Matriculation

# A2: Cutoff Rule with Matriculation, Round 2

**Portfolio Choice Problem:**

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A),$$

$$\text{such that: } \sum_{n=1}^{N_T} P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A) \leq N_M$$

$$\mathcal{L}(d_{2,1},\ldots,d_{2,N_T};\lambda_2)=\sum_{n=1}^{N_T}\left[P(y_n=S|w_n)\mathbb{1}(d_{2,n}=A)+\lambda_2\mathbb{1}(d_{2,n}=R)\right]P(m_n=M|w_n)+\lambda_2(N_M-\overline{N_M})$$

**Cutoff Rule Solution:**

$$\forall n \in \{1,\ldots,N_T\},\ d_{2,n}(w_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S|w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|w_n) < \lambda_2^* \end{cases}$$

Matriculation-Dynamic Model

# A2: Cutoff Rule with Matriculation, Round 1

**Portfolio Choice Problem:**

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n;\overline{\lambda_2^*})\mathbb{1}(d_{1,n}=T), \text{ such that: } \sum_{n=1}^{N}\mathbb{1}(d_{1,n}=T)\leq N_T, \text{ where:}$$

$$EV_{2,n}(x_n)=\int_{\widetilde{z}_n}\max\{P(y_n=S|x_n,\widetilde{z}_n)-\overline{\lambda_2^*},0\}P(m_n=M|x_n,\widetilde{z}_n)dF_{z|x}(\widetilde{z}_n|x_n)$$

$$\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1)=\sum_{n=1}^{N}\left[EV_{2,n}(x_n)\mathbb{1}(d_{1,n}=T)+\lambda_1\mathbb{1}(d_{1,n}=R)\right]+\lambda_1(N_T-N)$$

**Cutoff Rule Solution:**

$$\forall n\in\{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*)=\begin{cases} T & \text{if } EV_{2,n}(x_n)>\lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n)=\lambda_1^* \\ R & \text{if } EV_{2,n}(x_n)<\lambda_1^* \end{cases}$$

Matriculation-Dynamic Model

a

# A2: Matriculation-Dynamic Likelihood

**Matriculation Probability:**

$$P_m = P\left(m = M \middle| x, z, t; \theta_M\right) = \frac{\exp(f_M(x,z,t)\theta_M)}{1 + \exp(f_M(x,z,t)\theta_M)}$$

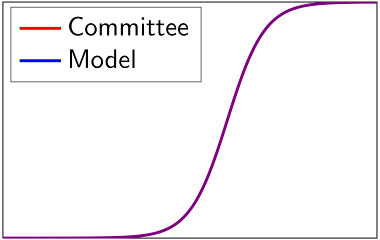
$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{f(z_n|x_n, t_n; \theta_F)}_{\text{Density of } z|x,t} \underbrace{P_{d_{1,n}}(\theta_T)\left[P_{d_{2,n}}(\theta_A)P_{m_n}(\theta_M)^{\mathbb{1}(m_n=M)}(1-P_{m_n}(\theta_M))^{\mathbb{1}(m_n=D)}\right]^{\mathbb{1}(d_{2,n}=A)}}_{\text{Conditional choice and matriculation probabilities}}$$

$$\underbrace{(1-P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional Choice Probability}} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Outcome probabilities}} \prod_{n=N_T+1}^{N} \underbrace{f(z_n|x_n, t_n; \theta_F)}_{\text{Density of } z|x,t} \underbrace{(1-P_{d_{1,n}}(\theta_T))}_{\text{Conditional Choice Probability}}$$

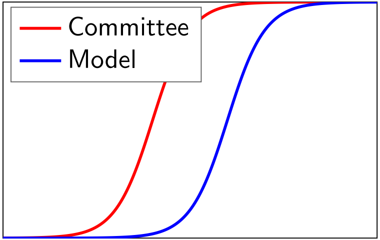
Maximum Likelihood Estimation

# A2: How Deviation Gains Test Works

1. Write Bellman equations for committee's optimization problem
2. Estimate parameters. Rationality assumption distorts estimates
3. Re-estimate parameters without rationality, and use to solve model
4. Compare selections from model's decision rule to actual decision rule
5. To prevent overfitting, use perturbations of estimates for comparison




Optimal Committee




Suboptimal Committee

Deviation Gains Test

# A2: Deviation Gains Algorithm 1

**Dynamic/Static/Myopic:**

1. Rank each $t$'s applicants by $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]$ with $\sigma = 0$ for dynamic and myopic, or $P(y = S|x, t, \overline{k})$ for static. Transition top $N_T$, and reject the rest
2. Rank surviving applicants by $P(y = S|x, z, t, \overline{k})$, and accept top $N_A$ survivors. Compute $\overline{P(y = S|x, z, t, \overline{k})}$ of acceptance set, and compare to committee's set

**Matriculation-Dynamic/Static:**

1. Rank each $t$'s applicants by $\mathbb{E}[v_2(x, z, t, \overline{k}, \epsilon_2)|x, t, \overline{k}]$ with $\sigma = 0$ for dynamic, or $P(y = S|x, t, \overline{k})$ for static (best analogue). Transition top $N_T$, and reject the rest
2. Rank surviving applicants by $P(y = S|x, z, t, \overline{k})$ and accept them in descending order, each matriculating with probability $P(m = M|x, z, t)$, until $N_M$ matriculate. Compute $\overline{P(y = S|x, z, t, \overline{k})}$ of matriculation set, and compare to committee's set

Deviation Gains Test

# A2: Deviation Gains Algorithm 2

**Waitlist-Dynamic/Hybrid:**

1. Rank each $t$'s applicants by $\mathbb{E}[v_2(x, z, t, \overline{k}, \ddot{m}_2, \epsilon_2)|x, t, \overline{k}]$ with $\sigma = 0$ for dynamic, or $P(y = S|x, t, \overline{k})$ for hybrid. Transition top $N_T$, and reject the rest
2. Rank surviving applicants, with $\sigma = 0$ and $\beta = 1$, by $P(y = S|x, z, t, \overline{k}) - \beta\mathbb{E}[v_3(x, z, t, \overline{k}, \overline{m_{2A}}, \epsilon_3)|x, z, t, \overline{k}, \ddot{m}_2]$. Accept top $N_{2A}$, each matriculating with probability $P(m_2 = M|x, z, t)$, and waitlist the rest
3. Rank waitlisted applicants by $P(y = S|x, z, t, \overline{k})$ and accept them in descending order, each matriculating with probability $P(m_3 = M|x, z, t)$, until $N_M$ matriculate. Compute $\overline{P(y = S|x, z, t, \overline{k})}$ of matriculation set, and compare to committee's set

Deviation Gains Test

# A2: Randomization Test

## Theorem 4 (Proof)

*Let $d^R$ be the model's decision rule, and $d^U$ the actual decision rule. By optimality:*

$$\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U\right)\mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\}$$
$$\geq \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \hat{\theta}_S^U\right)\mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\}$$

- But for perturbation $\widetilde{\theta_{S,t}^U}$ of $\hat{\theta}_S^U$'s asymptotic distribution, can have:

$$\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \widetilde{\theta_{S,t}^U}\right)\mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}_U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\}$$
$$< \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n, t_n, \bar{k}_n; \widetilde{\theta_{S,t}^U}\right)\mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}_U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\}$$

- For matriculation and waitlist models, compare matriculants, not acceptances

Deviation Gains Test

# A2: Summary of Simulated Results

- **Simulated Results:** Estimation results are well behaved
  - **Dynamic/Static:** Deviation gains significantly positive for simulated dataset where committee misweights applicant characteristics, but not for optimal simulated dataset
  - **Matriculation-Dynamic/Static:** Same as above
  - **Waitlist-Dynamic/Hybrid:** Same as above, and matriculation estimates as expected
- Example of cutoff probability reversal if committee transitions few files in Round 1:
  - Three applicants: A, B, and C. Committee transitions two, accepts one
  - A is best and C is worst overall, but A is worst in objective variables
  - Committee rejects A in Round 1 and accepts B, who is worse than A
- By assuming $h$ hours to read application file and $w$ hourly wage, and calculating model's $\overline{P(\text{Success})}$ induced by all possible $N_T$, can put dollar value on success
  - Alternatively, what if committee uses AI to read every applicant's file?

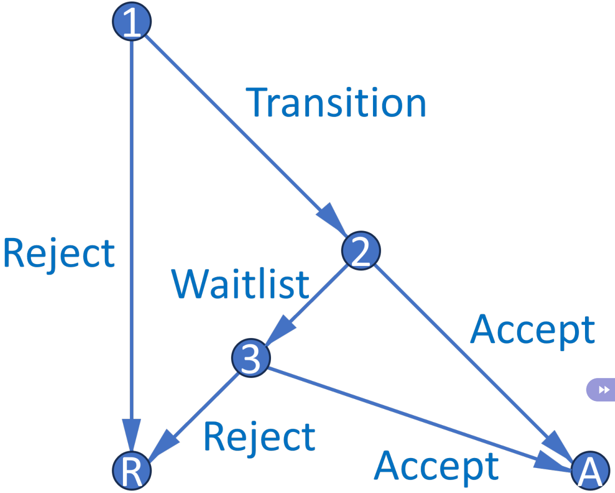
Conclusion

# A2: Extensions

- **Extension 1:** Uncover additional variables that predict matriculation but not failure
  - **Example:** Applicants with work experience more likely to succeed and matriculate
  - Can help committee find "people that we can afford" (*Moneyball*)
- **Extension 2:** Relax assumption that committee maximizes $P(\text{Success})$
  - Instead maximize combination of $P(\text{Success})$ and other objectives
  - **Example:** Committee may want diverse cohort (e.g. across fields): $P(\text{Success})^{\alpha} \times \text{Diversity}^{1-\alpha}$, where $\alpha \in [0, 1]$ can be estimated
- **Extension 3:** Waitlist and multiyear models (see Appendix 3)
  - Three-round models, and two-round models that repeat over infinite horizon
  - States contain info about past rounds/years, and committee values future rounds/years (memory includes matriculation probability and diversity)

Conclusion

# A3.1: Waitlist Decision Graph




- Three-round model features waitlist and accounts for applicants' decisions
- Dynamic mechanism incentivizes giving early acceptances to likely matriculants
- In practice, need several years of matriculation data for identification

Conclusion

# A3.1: Waitlist Model, Round 3

- Respecify Round 3 cutoff $P(y_3^* = S|t)$, where $t =$ year, as $P(y_3^* = S|\overline{m_{2A}})$
  - $\overline{m_{2A}} =$ average early accepted applicant's matriculation probability
  - $P(y_3^* = S|\overline{m_{2A}})$ increases in $\overline{m_{2A}}$ because larger $\overline{m_{2A}}$ means fewer openings

$$v_3(x, z, t, \overline{k}, \overline{m_{2A}}, \epsilon_3) = \max_{d_3 \in \{A,R\}} u_3(x, z, t, \overline{k}, \overline{m_{2A}}, d_3) + \epsilon_3(d_3), \text{ such that:}$$

$$u_3(x, z, t, \overline{k}, \overline{m_{2A}}, d_3) = \underbrace{P(y = S|x, z, t, \overline{k})\mathbb{1}(d_3 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_3^* = S|\overline{m_{2A}})\mathbb{1}(d_3 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Conclusion

# A3.1: Waitlist Model, Round 2

- What information available at start of Round 2 determines $\overline{m_{2A}}$?
  - $\ddot{m}_2$ is defined for all transitioned applicants as $\overline{m_2}$ with $m_2$ excluded, where $m_2 =$ applicant's own matriculation probability. Within a year, high $m_2$ implies low $\ddot{m}_2$
  - $\ddot{m}_2$ appears when applicant is waitlisted $\Rightarrow$ represents $\overline{m_2}$ of applicants committee *can* accept (vs. $\overline{m_{2A}}$, which represents $\overline{m_2}$ of applicants committee *does* accept)

---

$$v_2(x,z,t,\overline{k},\ddot{m}_2,\epsilon_2) = \max_{d_2 \in \{A,W\}} u_2(x,z,t,\overline{k},\ddot{m}_2,d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x,z,t,\overline{k},\ddot{m}_2,d_2) = \underbrace{P(y=S|x,z,t,\overline{k})\mathbb{1}(d_2=A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{\beta\mathbb{E}[v_3(x,z,t,\overline{k},\overline{m_{2A}},\epsilon_3)|x,z,t,\overline{k},\ddot{m}_2]\mathbb{1}(d_2=W)}_{\text{If waitlist, get discounted } \mathbb{E}[\text{Social Surplus}]}$$

$$\mathbb{E}[v_3(x,z,t,\overline{k},\overline{m_{2A}},\epsilon_3)|x,z,t,\overline{k},\ddot{m}_2] = \int_{\widetilde{\overline{m_{2A}}}} \sigma \log\left(\sum_{\tilde{d}_3 \in \{A,R\}} e^{u_3(x,z,t,\overline{k},\widetilde{\overline{m_{2A}}},\tilde{d}_3)/\sigma}\right) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)$$

Conclusion

# A3.1: Waitlist Dynamics

## Theorem 5 (Proof)

*Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

- Compared to low $m_2$ applicant, high $m_2$ applicant has low $\ddot{m}_2$, and therefore low $\overline{F}_{\overline{m_{2A}}|\ddot{m}_2}$, $\mathbb{E}[P(\text{Cutoff})_3]$, and $\mathbb{E}[v_3]$. This increases relative payoff of accepting them
- Why not accept low $m_2$ applicant and fill spot with waitlisted high $m_2$ applicant?
  1. $m_2$ declines between Rounds 2 and 3 and has farther to fall for high $m_2$ applicant
  2. In general, matriculation probabilities can change unpredictably between rounds

★ **Upshot:** Accepting applicants who want to matriculate benefits the program

Conclusion

# A3.1: Waitlist Model, Round 1

**Round 1:**

$$v_1(x,t,\overline{k},\epsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,\overline{k},d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x,t,\overline{k},d_1) = \underbrace{\mathbb{E}[v_2(x,z,t,\overline{k},\ddot{m}_2,\epsilon_2)|x,t,\overline{k}]\mathbb{1}(d_1=T)}_{\text{If transition, get } \mathbb{E}[\text{Social Surplus}]} + \underbrace{L(y_1^*=S|t)\mathbb{1}(d_1=R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{such that:}$$

$$\mathbb{E}[v_2(x,z,t,\overline{k},\ddot{m}_2,\epsilon_2)|x,t,\overline{k}] = \int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2\in\{A,W\}} e^{u_2(x,\widetilde{z},t,\overline{k},\ddot{m}_2(x,\widetilde{z},t),\widetilde{d}_2)/\sigma}\right) dF_{z|x,t}(\widetilde{z}|x,t)$$

**Conditional Choice Probabilities:**

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states},d_r)/\sigma)}{\sum_{\widetilde{d}_r\in D_r}\exp(u_r(\text{states},\widetilde{d}_r)/\sigma)}$$

Conclusion

# A3.1: Waitlist Model, Round 1

**Round 1 (Hybrid):**

$$v_1(x, t, \overline{k}, \epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, t, \overline{k}, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, \overline{k}, d_1) = \underbrace{P(y = S|x, t, \overline{k})\mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

**Conditional Choice Probabilities:**

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states}, d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(\text{states}, \widetilde{d}_r)/\sigma)}$$

Conclusion

# A3.1: Waitlist-Dynamic Parameterization 1

**Unrestricted CCPs:**

$$P_{d_1} = P\big(d_1 = T \big| x, t; \theta_T\big) = \frac{\exp(f_T(x,t)\theta_T)}{1 + \exp(f_T(x,t)\theta_T)}$$

$$P_{d_2} = P\big(d_2 = A \big| x, z, \ddot{m}_2; \theta_{A_2}\big) = \frac{\exp(f_{A_2}(x,z,\ddot{m}_2)\theta_{A_2})}{1 + \exp(f_{A_2}(x,z,\ddot{m}_2)\theta_{A_2})}$$

$$P_{d_3} = P\big(d_3 = A \big| x, z, \overline{m_{2A}}; \theta_{A_3}\big) = \frac{\exp(f_{A_3}(x,z,\overline{m_{2A}})\theta_{A_3})}{1 + \exp(f_{A_3}(x,z,\overline{m_{2A}})\theta_{A_3})}$$

**Success/Matriculation Probabilities:**

$$P_y = P\big(y = S \big| x, z, t, k; \theta_S\big) = \frac{\exp(f_S(x,z,t,k)\theta_S)}{1+\exp(f_S(x,z,t,k)\theta_S)}, P_{m_2} = P\big(m_2 = M \big| x, z, t; \theta_{M_2}\big) = \frac{\exp(f_{M_2}(x,z,t)\theta_{M_2})}{1+\exp(f_{M_2}(x,z,t)\theta_{M_2})}$$

**Cutoff Level/Probability:**

$$L_{y_1^*} = L\big(y_1^* = S \big| t; \theta_{S_1^*}\big) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_3^*} = P\big(y_3^* = S \big| \overline{m_{2A}}; \theta_{S_3^*}\big) = \frac{\exp(f_{S_3^*}(\overline{m_{2A}})\theta_{S_3^*})}{1 + \exp(f_{S_3^*}(\overline{m_{2A}})\theta_{S_3^*})}$$

Conclusion

# A3.1: Waitlist-Dynamic Parameterization 2

$$\mathbb{E}[v_2(x,z,t,\overline{k},\ddot{m}_2,\epsilon_2)|x,t,\overline{k}] = \int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2\in\{A,R\}} e^{u_2(x,\widetilde{z},t,\overline{k},\ddot{m}_2(x,\widetilde{z},t),\widetilde{d}_2)/\sigma}\right) dF_{z|x,t}(\widetilde{z}|x,t)$$

$$= \int_{\widetilde{z}_1}\cdots\int_{\widetilde{z_{J_z}}}\Big[\cdot\Big] dF_{z|x,t,1}(\widetilde{z}_1|\widetilde{z}_2,\ldots,\widetilde{z_{J_z}},x,t)\cdots dF_{z|x,t,J_z}(\widetilde{z_{J_z}}|x,t)$$

$$\mathbb{E}[v_3(x,z,t,\overline{k},\overline{m_{2A}},\epsilon_3)|x,z,t,\overline{k},\ddot{m}_2] = \int_{\widetilde{\overline{m_{2A}}}} \sigma \log\left(\sum_{\widetilde{d}_3\in\{A,R\}} e^{u_3(x,z,t,\overline{k},\widetilde{\overline{m_{2A}}},\widetilde{d}_3)/\sigma}\right) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)$$

- **Continuous:** $f_{z|x,t}(z_j|x,t;\beta_{F_{z|x,t,j}},\sigma^2_{F_{z|x,t,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,t,j}}}} \exp\left[\frac{-1}{2\sigma^2_{F_{z|x,t,j}}}(z_j - g(x,t)\beta_{F_{z|x,t,j}})^2\right]$
- **Binary:** $P(z_j = 1|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}{1+\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}$
- **Multinomial:** $P(z_j = \ell|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta^{\ell}_{F_{z|x,t,j}}\right)}{1+\sum_{i=1}^{L_j-1}\exp\left(g(x,t)\theta^{i}_{F_{z|x,t,j}}\right)}, \quad \ell = 1,\ldots,L_j-1$

Conclusion

# A3.1: Waitlist-Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_{F_z}\right)}_{\text{Density of } z^c|z^{\neg c}, x, t} \underbrace{f\left(\overline{m_{2A,n}} | \ddot{m_{2,n}}; \theta_{F_m}\right)}_{\text{Density of } \overline{m_{2A}}|\ddot{m}_2} \underbrace{P_{d_{1,n}}(\theta_T)}_{\text{Conditional choice probability}}$$

$$\underbrace{P_{d_{2,n}}(\theta_{A_2})^{\mathbb{1}(d_{2,n}=A)}(1 - P_{d_{2,n}}(\theta_{A_2}))^{\mathbb{1}(d_{2,n}=W)}}_{\text{Conditional choice probabilities}} \prod_{n=1}^{N_W} \underbrace{P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}(1 - P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)}}_{\text{Conditional choice probabilities}}$$

$$\prod_{n=N_T+1}^{N} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{f\left(z_n^{\neg c} | x_n, t_n; \theta_{F_z}\right)}_{\text{Density of } z^{\neg c}|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Estimate $\theta_{M_2}$ with 1st-stage likelihood to get $m_{2,n}$ = matriculation probability $P_{m_{2,n}}$: $\mathcal{L}_{N_T-N_W}^{M_2}(\theta_{M_2}) = \prod_{n=N_W+1}^{N_T} P_{m_{2,n}}(\theta_{M_2})^{\mathbb{1}(m_{2,n}=M)}(1 - P_{m_{2,n}}(\theta_{M_2}))^{\mathbb{1}(m_{2,n}=D)}$
- Use Murphy-Topel SEs, and for $\mathcal{L}_R$, use $(\theta_F^U, \theta_S^U)$ as initial guess of $(\theta_F^R, \theta_S^R)$

Conclusion

# A3.1: Waitlist Notes

- **Estimation:** Estimate $f_{\overline{m_{2A}}|\ddot{m}_2}$ on all transitioned applicants; slope parameter's expected sign is positive from variation in $\overline{m_2}$ across years (e.g. due to Covid)
    - **Assumption:** Effect of $\ddot{m}_2$ on $\overline{m_{2A}}$ is stationary across years, and therefore can be used to forecast changes in $\overline{m_{2A}}$ based on changes in $\ddot{m}_2$ from changes in $d_2$
- **Dynamics:** What if committee doesn't think dynamically?
    - If $\ddot{m}_2$ increasing increases $\overline{F}_{\overline{m_{2A}}|\ddot{m}_2}$, which increases $\mathbb{E}[P(\text{Cutoff})_3]$, then it should
    - Don't need success data for all years to identify these two effects
- **Over/Undershooting:** Waitlist helps avoid both in reality, not just in expectation

Conclusion

# A3.1: Symmetric Waitlist Model 1

**Round 3:**

$$v_3(x, z, t, \overline{k}, \overline{m_{2A}}, \overline{m_3}, \epsilon_3) = \max_{d_3 \in \{A,R\}} u_3(x, z, t, \overline{k}, \overline{m_{2A}}, \overline{m_3}, d_3) + \epsilon_3(d_3), \text{ such that:}$$

$$u_3(x, z, t, \overline{k}, \overline{m_{2A}}, \overline{m_3}, d_3) = P(y = S|x, z, t, \overline{k})\mathbb{1}(d_3 = A) + P(y_3^* = S|\overline{m_{2A}}, \overline{m_3})\mathbb{1}(d_3 = R)$$

**Round 2:**

$$v_2(x, z, t, \overline{k}, \ddot{m}_2, m_3, \epsilon_2) = \max_{d_2 \in \{A,W\}} u_2(x, z, t, \overline{k}, \ddot{m}_2, m_3, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, \overline{k}, \ddot{m}_2, m_3, d_2) =$$
$$P(y = S|x, z, t, \overline{k})\mathbb{1}(d_2 = A) + \beta\mathbb{E}[v_3(x, z, t, \overline{k}, \overline{m_{2A}}, \overline{m_3}, \epsilon_3)|x, z, t, \overline{k}, \ddot{m}_2, m_3]\mathbb{1}(d_2 = W), \text{ such that:}$$

$$\mathbb{E}[v_3(x, z, t, \overline{k}, \overline{m_{2A}}, \widetilde{\overline{m_3}}, \epsilon_3)|x, z, t, \overline{k}, \ddot{m}_2, m_3] =$$
$$\int_{\widetilde{\overline{m_{2A}}}} \int_{\widetilde{\overline{m_3}}} \sigma \log\left(\sum_{\widetilde{d}_3 \in \{A,R\}} e^{u_3(x,z,t,\overline{k},\widetilde{\overline{m_{2A}}},\widetilde{\overline{m_3}},\widetilde{d}_3)/\sigma}\right) dF_{\overline{m_3}|m_3}(\widetilde{\overline{m_3}}|m_3)\, dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)$$

Conclusion

# A3.1: Symmetric Waitlist Model 2

**Round 1:**

$$v_1(x,t,\overline{k}) = \max_{d_1 \in \{T,R\}} u_1(x,t,\overline{k},d_1) + \epsilon_1(d_1), \text{ such that:}$$

- **Hybrid:** $u_1(x,t,\overline{k},d_1) = P(y = S|x,t,\overline{k})\mathbb{1}(d_1 = T) + P(y_1^* = S|t)\mathbb{1}(d_1 = R)$
- **Dynamic:**
  $u_1(x,t,\overline{k},d_1) = \mathbb{E}[v_2(x,z,t,\overline{k},\ddot{m}_2,m_3,\epsilon_2)|x,t,\overline{k}]\mathbb{1}(d_1 = T) + L(y_1^* = S|t)\mathbb{1}(d_1 = R)$, such that:

$$\mathbb{E}[v_2(x,z,t,\overline{k}\ddot{m}_2,m_3,\epsilon_2)|x,t,\overline{k}] = \int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2 \in \{A,R\}} e^{u_2(x,\widetilde{z},t,\overline{k},\ddot{m}_2(x,\widetilde{z},t),m_3(x,\widetilde{z},t),\widetilde{d}_2)/\sigma}\right) dF_{z|x,t}(\widetilde{z}|x,t)$$

**Conditional Choice Probabilities:**

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states},d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(\text{states},\widetilde{d}_r)/\sigma)}$$

Conclusion

# A3.1: Symmetric Waitlist-Dynamic Parameterization 1

**Unrestricted CCPs:**

$$P_{d_1} = P\big(d_1 = T \big| x, t; \theta_T\big) = \frac{\exp(f_T(x,t)\theta_T)}{1+\exp(f_T(x,t)\theta_T)}$$

$$P_{d_2} = P\big(d_2 = A \big| x, z, \ddot{m}_2, m_3; \theta_{A_2}\big) = \frac{\exp(f_{A_2}(x,z,\ddot{m}_2,m_3)\theta_{A_2})}{1+\exp(f_{A_2}(x,z,\ddot{m}_2,m_3)\theta_{A_2})}$$

$$P_{d_3} = P\big(d_3 = A \big| x, z, \overline{m_{2A}}, \overline{m_3}; \theta_{A_3}\big) = \frac{\exp(f_{A_3}(x,z,\overline{m_{2A}},\overline{m_3})\theta_{A_3})}{1+\exp(f_{A_3}(x,z,\overline{m_{2A}},\overline{m_3})\theta_{A_3})}$$

**Success/Matriculation Probabilities:**

$$P_y = P\big(y = S \big| x, z, t, k; \theta_S\big) = \frac{\exp(f_S(x,z,t,k)\theta_S)}{1+\exp(f_S(x,z,t,k)\theta_S)}, \quad P_{m_3} = P\big(m_3 = M \big| x, z, t; \theta_{M_3}\big) = \frac{\exp(f_{M_3}(x,z,t)\theta_{M_3})}{1+\exp(f_{M_3}(x,z,t)\theta_{M_3})}$$

**Cutoff Levels/Probabilities:**

$$L_{y_1^*} = L\big(y_1^* = S \big| t; \theta_{S_1^*}\big) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_3^*} = P\big(y_3^* = S \big| \overline{m_{2A}}, \overline{m_3}; \theta_{S_3^*}\big) = \frac{\exp(f_{S_3^*}(\overline{m_{2A}},\overline{m_3})\theta_{S_3^*})}{1+\exp(f_{S_3^*}(\overline{m_{2A}},\overline{m_3})\theta_{S_3^*})}$$

Conclusion

# A3.1: Symmetric Waitlist-Dynamic Parameterization 2

★ Assume $(\overline{m_{2A}}, \overline{m_3})$ are conditionally independent:

$$\mathbb{E}[v_2(x,z,t,\overline{k},\ddot{m}_2,m_3,\epsilon_2)|x,t,\overline{k}] = \int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2\in\{A,R\}} e^{u_2(x,\widetilde{z},t,\overline{k}\ddot{m}_2(x,\widetilde{z},t),m_3(x,\widetilde{z},t),\widetilde{d}_2)/\sigma}\right) dF_{z|x,t}(\widetilde{z}|x,t)$$

$$= \int_{\widetilde{z}_1}\cdots\int_{\widetilde{z_{J_z}}}\Big[\cdot\Big] dF_{z|x,t,1}(\widetilde{z}_1|\widetilde{z}_2,\ldots,\widetilde{z_{J_z}},x,t)\cdots dF_{z|x,t,J_z}(\widetilde{z_{J_z}}|x,t)$$

$$\mathbb{E}[v_3(x,z,t,\overline{k},\overline{m_{2A}},\overline{m_3},\epsilon_3)|x,z,t,\overline{k},\ddot{m}_2,m_3] = \int_{\widetilde{\overline{m_{2A}}}}\int_{\widetilde{\overline{m_3}}} \sigma \log\left(\sum_{\widetilde{d}_3\in\{A,R\}} e^{u_3(x,z,t,\overline{k},\widetilde{\overline{m_{2A}}},\widetilde{\overline{m_3}},\widetilde{d}_3)/\sigma}\right) dF_{\overline{m_3}|m_3}(\widetilde{\overline{m_3}}|m_3)\, dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)$$

- **Continuous:** $f_{z|x,t}(z_j|x,t;\beta_{F_{z|x,t,j}},\sigma^2_{F_{z|x,t,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,t,j}}}} \exp\left[\frac{-1}{2\sigma^2_{F_{z|x,t,j}}}(z_j - g(x,t)\beta_{F_{z|x,t,j}})^2\right]$
- **Binary:** $P(z_j=1|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}{1+\exp\left(g(x,t)\theta_{F_{z|x,t,j}}\right)}$
- **Multinomial:** $P(z_j=\ell|x,t;\theta_{F_{z|x,t,j}}) = \frac{\exp\left(g(x,t)\theta^{\ell}_{F_{z|x,t,j}}\right)}{1+\sum_{i=1}^{L_j-1}\exp\left(g(x,t)\theta^{i}_{F_{z|x,t,j}}\right)},\ \ell = 1,\ldots,L_j-1$

Conclusion

# A3.1: Symmetric Waitlist-Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_{F_z}\right)}_{\text{Density of } z^c|z^{\neg c},x,t} \underbrace{f\left(\overline{m_{2A,n}} | \ddot{m}_{2,n}; \theta_{F_{m_2}}\right)}_{\text{Density of } \overline{m_{2A}}|\ddot{m}_2} \underbrace{f\left(\overline{m_{3,n}} | m_{3,n}; \theta_{F_{m_3}}\right)}_{\text{Density of } \overline{m_3}|m_3} \underbrace{P_{d_{1,n}}(\theta_T)}_{\text{Conditional choice probability}}$$

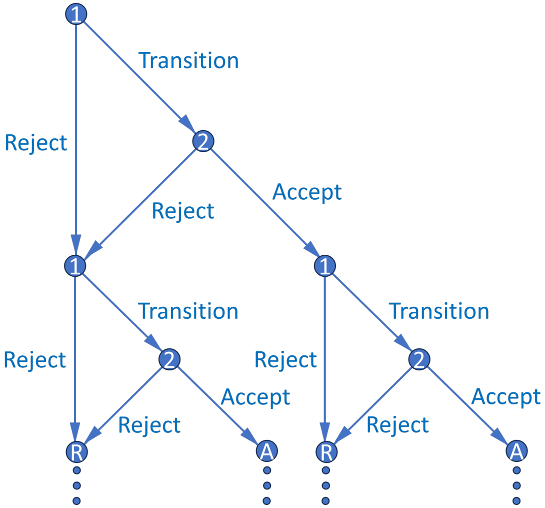
$$\underbrace{P_{d_{2,n}}(\theta_{A_2})^{\mathbb{1}(d_{2,n}=A)}(1 - P_{d_{2,n}}(\theta_{A_2}))^{\mathbb{1}(d_{2,n}=W)}}_{\text{Conditional choice probabilities}} \prod_{n=1}^{N_W} \underbrace{P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}(1 - P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)}}_{\text{Conditional choice probabilities}}$$

$$\prod_{n=N_T+1}^{N} \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}} \prod_{n=1}^{N} \underbrace{f\left(z_n^{\neg c} | x_n, t_n; \theta_{F_z}\right)}_{\text{Density of } z^{\neg c}|x,t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Estimate $\theta_{M_3}$ with 1st-stage likelihood to get $m_{3,n}$ = matriculation probability $P_{m_{3,n}}$: $\mathcal{L}_{N_W}^{M_3}(\theta_{M_3}) = \prod_{n=1}^{N_W} P_{m_{3,n}}(\theta_{M_3})^{\mathbb{1}(d_{3,n}=A, m_{3,n}=M)}(1 - P_{m_{3,n}}(\theta_{M_3}))^{\mathbb{1}(d_{3,n}=A, m_{3,n}=D)}$
- Use Murphy-Topel SEs, and for $\mathcal{L}_R$, use $(\theta_F^U, \theta_S^U)$ as initial guess of $(\theta_F^R, \theta_S^R)$

Conclusion

# A3.2: Multiyear Decision Graph




Conclusion

# A3.2: Multiyear Memory (Matriculation)

- Suppose committee maximizes discounted “lifetime” cohort success
- Respecify $P(y_2^* = S|t)$, where $t =$ year, as $L(y_2^* = S|c_{-1})$
  - $c_{-1} =$ previous year's cohort size. $L(\text{Cutoff}_2)$ increases in $c_{-1}$ because committee must keep total number of students in program relatively constant across years
- What information available at start of Round 2 determines $c$?
  - **If and only if $\boldsymbol{d_2 = A}$:** $m =$ applicant's matriculation probability
  - For simplicity, suppose matriculation probabilities are exogenous, and success probabilities depend only on applicant characteristics (i.e. not program rank)

Conclusion

# A3.2: Multiyear Dynamics (Matriculation)

**Theorem 6 (Proof)**

*Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.*

- With memory, optimal dynamic decision may not equal optimal myopic one
  - High $m$ means high $\overline{F}_{c|m}$, which means high $\mathbb{E}[L(\text{Cutoff}_2)']$ and future EV
  - Accepting likely matriculants above this year's margin means that next year in expectation, committee won't have to accept as many marginal applicants
- Harder to estimate since multiyear problem has no closed-form solution
  - Principal component analysis can help reduce dimensionality

Conclusion

# A3.2: Multiyear Model (Matriculation), Round 2

$$v_2(x, z, t, m, c_{-1}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, m, c_{-1}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$\begin{aligned} u_2(x, z, t, m, c_{-1}, d_2) = &\{P(y = S|x, z, t) + \beta\mathbb{E}[v_1(x', t', c, {\epsilon_1}')|m]\}\mathbb{1}(d_2 = A) \\ &+ \{L(y_2^* = S|c_{-1}) + \beta\mathbb{E}[v_1(x', t', c, {\epsilon_1}')]\}\mathbb{1}(d_2 = R), \text{ such that:} \end{aligned}$$

$$\mathbb{E}[v_1(x', t', c, {\epsilon_1}')|(m)] = \int_c \int_{t'} \int_{x'} \sigma \log\left( \sum_{{d_1}' \in \{T,R\}} e^{u_1(x',t',c,{d_1}')/\sigma} \right) dF_x(x')dF_t(t')dF_{c|(m)}(c|(m))$$

**Conditional Choice Probabilities:**

$$P(d_2|x, z, t, m, c_{-1}) = \frac{\exp(u_2(x, z, t, m, c_{-1}, d_2)/\sigma)}{\sum_{\widetilde{d_2} \in \{A,R\}} \exp(u_2(x, z, t, m, c_{-1}, \widetilde{d_2})/\sigma)}$$

Conclusion

# A3.2: Multiyear Model (Matriculation), Round 1

$$v_1(x,t,c_{-1},\epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x,t,c_{-1},d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$\begin{aligned} u_1(x,t,c_{-1},d_1) = {} & \mathbb{E}[v_2(x,z,t,m,c_{-1},\epsilon_2)|x,t,c_{-1}]\mathbb{1}(d_1 = T) \\ & + \{L(y_1^* = S|t) + \beta\mathbb{E}[v_1(x',t',c,\epsilon_1{}')]\}\mathbb{1}(d_1 = R), \text{ such that:} \end{aligned}$$

$$\begin{aligned} & \mathbb{E}[v_2(x,z,t,m,c_{-1},\epsilon_2)|x,t,c_{-1}] = \\ & \int_{\widetilde{m}}\int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2 \in \{A,R\}} e^{u_2(x,\widetilde{z},t,\widetilde{m},c_{-1},\widetilde{d}_2)/\sigma}\right) dF_{z|x,t}(\widetilde{z}|x,t)\, dF_{m|x,t}(\widetilde{m}|x,t) \end{aligned}$$

**Conditional Choice Probabilities:**

$$P(d_1|x,t,c_{-1}) = \frac{\exp(u_1(x,t,c_{-1},d_1)/\sigma)}{\sum_{\widetilde{d}_1 \in \{T,R\}} \exp(u_1(x,t,c_{-1},\widetilde{d}_1)/\sigma)}$$

Conclusion

# A3.2: Multiyear (Matriculation) Parameterization 1

**Unrestricted CCPs:**

$$P_{d_1} = P\left(d_1 = T \middle| x, t, c_{-1}; \theta_T\right) = \frac{\exp(f_T(x, t, c_{-1})\theta_T)}{1 + \exp(f_T(x, t, c_{-1})\theta_T)}$$

$$P_{d_2} = P\left(d_2 = A \middle| x, z, m, c_{-1}; \theta_A\right) = \frac{\exp(f_A(x, z, m, c_{-1})\theta_A)}{1 + \exp(f_A(x, z, m, c_{-1})\theta_A)}$$

**Success Probability:**

$$P_y = P\left(y = S \middle| x, z, t; \theta_S\right) = \frac{\exp(f_S(x, z, t)\theta_S)}{1 + \exp(f_S(x, z, t)\theta_S)}$$

**Cutoff Levels:**

$$L_{y_1^*} = L\left(y_1^* = S \middle| t; \theta_{S_1^*}\right) = f_{S_1^*}(t)\theta_{S_1^*}$$

$$L_{y_2^*} = L\left(y_2^* = S \middle| c_{-1}; \theta_{S_2^*}\right) = f_{S_2^*}(c_{-1})\theta_{S_2^*}$$

Conclusion

# A3.2: Multiyear (Matriculation) Parameterization 2

- Assume $(x, t)$ are independent, and next year is independent of current year:

$$\mathbb{E}[v_1(x', t', c, \epsilon_1{}')|(m)] = \int_c \int_{t'} \int_{x'} \sigma \log\left( \sum_{d_1{}' \in \{T,R\}} e^{u_1(x',t',c,d_1{}')/\sigma} \right) dF_x(x') dF_t(t') dF_{c|(m)}(c|(m)) =$$

$$\int_c \int_{t_1{}'} \cdots \int_{t_{J_t}{}'} \int_{x_1{}'} \cdots \int_{x_{J_x}{}'} \left[\cdot\right] dF_{x,1}(x_1{}'|x_2{}', \ldots, x_{J_x}{}') \cdots dF_{x,J_x}(x_{J_x}{}') dF_{t,1}(t_1{}'|t_2{}', \ldots, t_{J_t}{}') \cdots dF_{t,J_t}(t_{J_t}{}') dF_{c|(m)}(c|(m))$$

- Assume $(z, m)$ are conditionally independent:

$$\mathbb{E}[v_2(x, z, t, m, c_{-1}, \epsilon_2)|x, t, c_{-1}] = \int_{\widetilde{m}} \int_{\widetilde{z}} \sigma \log\left( \sum_{\widetilde{d}_2 \in \{A,R\}} e^{u_2(x,\widetilde{z},t,\widetilde{m},c_{-1},\widetilde{d}_2)/\sigma} \right) dF_{z|x,t}(\widetilde{z}|x, t) dF_{m|x}(\widetilde{m}|x, t) =$$

$$\int_{\widetilde{m}} \int_{\widetilde{z}_1} \cdots \int_{\widetilde{z_{J_z}}} \left[\,\cdot\,\right] dF_{z|x,t,1}(\widetilde{z}_1|\widetilde{z}_2, \ldots, \widetilde{z_{J_z}}, x, t) \cdots dF_{z|x,t,J_z}(\widetilde{z_{J_z}}|x, t) dF_{m|x,t}(\widetilde{m}|x, t)$$

Conclusion

# A3.2: Multiyear (Matriculation) Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_A} \underbrace{f\left(c_n | m_n; \theta_{F_c}\right)}_{\text{Density of } c|m} \prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_{F_z}\right)}_{\text{Density of } z^c|z^{\neg c}, x, t} \underbrace{f\left(m_n | x_n, t_n; \theta_{F_m}\right)}_{\text{Density of } m|x, t}$$

$$\underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^{N} \underbrace{\left(1 - P_{d_{1,n}}(\theta_T)\right)}_{\text{Conditional choice probability}}$$

$$\prod_{n=1}^{N} \underbrace{f\left(x_n; \theta_{F_x}\right) f\left(t_n; \theta_{F_t}\right) f\left(z_n^{\neg c} | x_n, t_n; \theta_{F_z}\right)}_{\text{Densities of } x,\ t, \text{ and } z^{\neg c}|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Estimate $f_(c|m)$ on $N_A$ accepted applicants only
- As with waitlist-dynamic model, for $\mathcal{L}_R$, use $(\theta_F^U, \theta_S^U)$ as initial guess of $(\theta_F^R, \theta_S^R)$

Conclusion

# A3.2: Multiyear Memory and Dynamics (Diversity)

- Let additional state $a =$ any diversity variable affecting success
  - For example, $a = \mathbb{1}(\text{TheoryField})$; theory is less common than applied
  - Let $\bar{a}_{-1} =$ percentage of last year's acceptance set that's theory
  - If $|a - \bar{a}_{-1}|$ is small, $P(\text{Success})$ may decrease because future advisor may be too busy (effect may be unidentifiable if program never gets too imbalanced)

**Theorem 7 (Proof)**

*Consider any two applicants, one with $a = 1$ and one with $a = 0$, but with equal success probabilities and therefore different values of $x$ or $z$. Suppose $P(a' = 1) = p > 0$. Then for all sufficiently small $p$, the $a = 1$ applicant has a strictly greater Round 2 acceptance probability.*

- Optimal dynamic decision may not equal optimal myopic one. May accept diverse applicant with marginally lower $P(\text{Success})$ to increase $P(\text{Success})$ of next year's pool
  - Relative to $a = 0$, $a = 1 \Rightarrow$ higher $\overline{F}_{\bar{a}|a} \Rightarrow$ higher $\mathbb{E}[P(\text{Success})']$ and future EV

Conclusion

# A3.2: Multiyear Model (Diversity), Round 2

$$v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, a, \bar{a}_{-1}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$\begin{aligned} u_2(x, z, t, a, \bar{a}_{-1}, d_2) = &\{P(y = S|x, z, t, |a - \bar{a}_{-1}|) + \beta\mathbb{E}[v_1(x', t', \bar{a}, \epsilon_1')|a]\}\mathbb{1}(d_2 = A) \\ &+ \{L(y_2^* = S|t) + \beta\mathbb{E}[v_1(x', t', \bar{a}, \epsilon_1')]\}\mathbb{1}(d_2 = R), \text{ such that:} \end{aligned}$$

$$\mathbb{E}[v_1(x', t', \bar{a}, \epsilon_1')|a] = \int_{\widetilde{\bar{a}}} \int_{\widetilde{t'}} \int_{\widetilde{x'}} \sigma \log \left( \sum_{d_1' \in \{T,R\}} e^{u_1(\widetilde{x'}, \widetilde{\bar{a}}, \widetilde{t'}, d_1')/\sigma} \right) dF_{x'}(\widetilde{x'}) dF_{t'}(\widetilde{t'}) dF_{\bar{a}|a}(\widetilde{\bar{a}}|a)$$

**Conditional Choice Probabilities:**

$$P(d_2|x, z, t, a, \bar{a}_{-1}) = \frac{\exp(u_2(x, z, t, a, \bar{a}_{-1}, d_2)/\sigma)}{\sum_{\widetilde{d}_2 \in \{A,R\}} \exp(u_2(x, z, t, a, \bar{a}_{-1}, \widetilde{d}_2)/\sigma)}$$

Conclusion

# A3.2: Multiyear Model (Diversity), Round 1

$$v_1(x,t,\bar{a}_{-1},\epsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,\bar{a}_{-1},d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$\begin{aligned} u_1(x,t,\bar{a}_{-1},d_1) = {} & \mathbb{E}[v_2(x,z,t,a,\bar{a}_{-1},\epsilon_2)|x,t,\bar{a}_{-1}]\mathbb{1}(d_1 = T) \\ & + \{L(y_1^* = S|t) + \beta\mathbb{E}[v_1(x',t',\bar{a},\epsilon_1')]\}\mathbb{1}(d_1 = R), \text{ such that:} \end{aligned}$$

$$\mathbb{E}[v_2(x,z,t,a,\bar{a}_{-1},\epsilon_2)|x,t,\bar{a}_{-1}] = \int_{\tilde{a}}\int_{\tilde{z}} \sigma \log\left(\sum_{\tilde{d}_2\in A,R} e^{u_2(x,\tilde{z},t,\tilde{a},\bar{a}_{-1},\tilde{d}_2)/\sigma}\right) dF_{z|x,t}(\tilde{z}|x,t)dF_a(\tilde{a})$$

$$\mathbb{E}[v_1(x',t',\bar{a},\epsilon_1')] = \int_{\tilde{\bar{a}}}\int_{\tilde{t'}}\int_{\tilde{x'}} \sigma \log\left(\sum_{d_1'\in\{T,R\}} e^{u_1(\tilde{x'},\tilde{t'},\tilde{\bar{a}},d_1')/\sigma}\right) dF_{x'}(\tilde{x'})dF_{t'}(\tilde{t'})dF_{\bar{a}}(\tilde{\bar{a}})$$

**Conditional Choice Probabilities:**

$$P(d_1|x,t,\bar{a}_{-1}) = \frac{\exp(u_1(x,t,\bar{a}_{-1},d_1)/\sigma)}{\sum_{\tilde{d}_1\in\{T,R\}}\exp(u_1(x,t,\bar{a}_{-1},\tilde{d}_1)/\sigma)}$$

Conclusion

# A3.2: Multiyear (Diversity) Parameterization 1

**Unrestricted CCPs:**

$$P_{d_1} = P\big(d_1 = T \big| x, t, \bar{a}_{-1}; \theta_T\big) = \frac{\exp(f_T(x, t, \bar{a}_{-1})\theta_T)}{1 + \exp(f_T(x, t, \bar{a}_{-1})\theta_T)}$$

$$P_{d_2} = P\big(d_2 = A \big| x, z, t, a, \bar{a}_{-1}; \theta_A\big) = \frac{\exp(f_A(x, z, t, a, \bar{a}_{-1})\theta_A)}{1 + \exp(f_A(x, z, t, a, \bar{a}_{-1})\theta_A)}$$

**Success Probability:**

$$P_y = P\big(y = S \big| x, z, t, |a - \bar{a}_{-1}|; \theta_S\big) = \frac{\exp(f_S(x, z, t, |a - \bar{a}_{-1}|)\theta_S)}{1 + \exp(f_S(x, z, t, |a - \bar{a}_{-1}|)\theta_S)}$$

**Cutoff Levels:**

$$L_{y_1^*} = L\big(y_1^* = S \big| t; \theta_{S_1^*}\big) = f_{S_1^*}(t)\theta_{S_1^*}$$
$$L_{y_2^*} = L\big(y_2^* = S \big| t; \theta_{S_2^*}\big) = f_{S_2^*}(t)\theta_{S_2^*}$$

Conclusion

# A3.2: Multiyear (Diversity) Parameterization 2

- Assume $(x, t, \bar{a})$ mutually independent, such that $\bar{a}|a$:

$$\mathbb{E}[v_1(x', t', \bar{a}, \epsilon_1')|a] = \int_{\widetilde{\bar{a}}} \int_{\widetilde{t'}} \int_{\widetilde{x'}} \sigma \log \left( \sum_{d_1' \in \{T,R\}} e^{u_1(\widetilde{x'}, \widetilde{t'}, \widetilde{\bar{a}}, d_1')/\sigma} \right) dF_{x'}(\widetilde{x'}) dF_{t'}(\widetilde{t'}) dF_{\bar{a}|a}(\widetilde{\bar{a}}|a) =$$

$$\int_{\widetilde{\bar{a}}} \int_{\widetilde{t_1'}} \cdots \int_{\widetilde{t_{J_t}'}} \int_{\widetilde{x_1'}} \cdots \int_{\widetilde{x_{J_x}'}} \left[ \cdot \right] F_{x,1}({x_1}'|{x_2}', \ldots, {x_{J_x}}') \cdots dF_{x,J_x}({x_{J_x}}') dF_{t,1}({t_1}'|{t_2}', \ldots, {t_{J_t}}') \cdots dF_{t,J_t}({t_{J_t}}') dF_{\bar{a}|a}(\widetilde{\bar{a}}|a)$$

- Assume $(z, a)$ independent, such that $z|x$:

$$\mathbb{E}[v_2(x, z, t, a, \bar{a}_{-1}, \epsilon_2)|x, t, \bar{a}_{-1}] = \int_{\widetilde{a}} \int_{\widetilde{z}} \sigma \log \left( \sum_{\widetilde{d_2} \in A,R} e^{u_2(x, \widetilde{z}, t, \widetilde{a}, \bar{a}_{-1}, \widetilde{d_2})/\sigma} \right) dF_{z|x,t}(\widetilde{z}|x, t) dF_a(\widetilde{a})$$

$$= \int_{\widetilde{a}} \int_{\widetilde{z_1}} \cdots \int_{\widetilde{z_J}} \left[ \cdot \right] dF_1(\widetilde{z_1}|\widetilde{z_2}, \ldots \widetilde{z_J}, x, t) \cdots dF_J(\widetilde{z_J}|x, t) dF_a(\widetilde{a})$$

Conclusion

# A3.2: Multiyear (Diversity) Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_A} \underbrace{f\left(\bar{a}_n | a_n; \theta_{F_{\bar{a}}}\right)}_{\text{Density of } \bar{a}|a} \prod_{n=1}^{N_T} \underbrace{f\left(z_n^c | z_n^{\neg c}, x_n, t_n; \theta_{F_z}\right)}_{\text{Density of } z^c|z^{\neg c}, x, t} \underbrace{f\left(a_n; \theta_{F_a}\right)}_{\text{Density of } a}$$

$$\underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)}(1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \prod_{n=N_T+1}^{N} \underbrace{\left(1 - P_{d_{1,n}}(\theta_T)\right)}_{\text{Conditional choice probability}}$$

$$\prod_{n=1}^{N} \underbrace{f\left(x_n; \theta_{F_x}\right) f\left(t_n; \theta_{F_t}\right) f\left(z_n^{\neg c} | x_n, t_n; \theta_{F_z}\right)}_{\text{Densities of } x,\ t, \text{ and } z^{\neg c}|x, t} \underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}}$$

- Estimate $f(\bar{a}|a)$ on $N_A$ accepted applicants only
- As with waitlist-dynamic model, for $\mathcal{L}_R$, use $(\theta_F^U, \theta_S^U)$ as initial guess of $(\theta_F^R, \theta_S^R)$

Conclusion

# A4: Economics Department Summary Statistics

Table 2: Economics Department Summary Statistics

(a) Objective Variables

| | Full Sample Mean | Full Sample SD | Transition = 1 Mean | Transition = 1 SD | Accept = 1 Mean | Accept = 1 SD | Matriculate = 1 Mean | Matriculate = 1 SD |
|---|---|---|---|---|---|---|---|---|
| Transition | 0.4204 | 0.4937 | 1.0000 | 0.0000 | 1.0000 | 0.0000 | 1.0000 | 0.0000 |
| Waitlist | 0.3324 | 0.4712 | 0.7686 | 0.4220 | 0.0743 | 0.2628 | 0.0896 | 0.2877 |
| Accept | 0.1103 | 0.3134 | 0.2625 | 0.4402 | 1.0000 | 0.0000 | 1.0000 | 0.0000 |
| Matriculate | 0.0339 | 0.1811 | 0.0807 | 0.2725 | 0.3074 | 0.4623 | 1.0000 | 0.0000 |
| Open House | 0.0353 | 0.1847 | 0.0817 | 0.2741 | 0.3267 | 0.4702 | 0.5522 | 0.5010 |
| Attend Program | 0.6024 | 0.4895 | 0.6752 | 0.4685 | 0.7704 | 0.4214 | 1.0000 | 0.0000 |
| Complete Program | 0.5578 | 0.4968 | 0.6210 | 0.4854 | 0.6654 | 0.4728 | 0.6709 | 0.4729 |
| Top 100 Placement | 0.1971 | 0.3979 | 0.2472 | 0.4316 | 0.3074 | 0.4623 | 0.3418 | 0.4773 |
| Program Rank | 88.9133 | 57.4823 | 78.5844 | 57.1361 | 63.3828 | 54.1845 | 37.0000 | 0.0000 |
| Missing Program Rank | 0.4199 | 0.4937 | 0.3422 | 0.4747 | 0.2490 | 0.4333 | 0.0000 | 0.0000 |
| Placement Rank | 127.7505 | 34.9306 | 123.0222 | 37.2570 | 117.2696 | 40.5746 | 117.4430 | 37.5728 |
| Missing Placement Rank | 0.4444 | 0.4970 | 0.3830 | 0.4864 | 0.3346 | 0.4728 | 0.3291 | 0.4729 |
| 2016 | 0.2052 | 0.4040 | 0.1696 | 0.3754 | 0.1790 | 0.3841 | 0.2025 | 0.4045 |
| 2017 | 0.2014 | 0.4011 | 0.1788 | 0.3833 | 0.1829 | 0.3873 | 0.1772 | 0.3843 |
| 2018 | 0.1928 | 0.3946 | 0.2686 | 0.4435 | 0.2062 | 0.4054 | 0.2532 | 0.4376 |
| 2019 | 0.2027 | 0.4021 | 0.2084 | 0.4064 | 0.2179 | 0.4136 | 0.2152 | 0.4136 |
| Winsorized Age | 25.9910 | 3.1479 | 26.2635 | 3.1823 | 26.0817 | 3.0652 | 26.3291 | 3.2806 |
| Female | 0.3912 | 0.4881 | 0.4147 | 0.4929 | 0.4397 | 0.4973 | 0.4177 | 0.4963 |
| U.S. | 0.2357 | 0.4245 | 0.2380 | 0.4261 | 0.2996 | 0.4590 | 0.3038 | 0.4628 |
| International Asian | 0.4809 | 0.4997 | 0.5230 | 0.4997 | 0.3424 | 0.4754 | 0.4304 | 0.4983 |
| GRE Verbal Percentile | 68.0129 | 22.5869 | 76.8421 | 18.3703 | 79.5703 | 18.3047 | 76.9744 | 19.4041 |
| GRE Quantitative Percentile | 88.4550 | 11.6330 | 93.1362 | 5.4135 | 93.5391 | 4.4614 | 92.3077 | 5.7554 |
| GRE Analytic Percentile | 50.0613 | 27.5744 | 56.3653 | 26.5466 | 60.3398 | 26.0880 | 57.4615 | 26.6481 |
| Undergraduate GPA | 3.6307 | 0.3026 | 3.6645 | 0.3003 | 3.7227 | 0.2610 | 3.7068 | 0.2880 |
| TOEFL/IELTS | 7.4376 | 0.5587 | 7.5641 | 0.5834 | 7.7011 | 0.6667 | 7.4464 | 0.6137 |
| Elite U.S. University | 0.1267 | 0.3327 | 0.1736 | 0.3790 | 0.2257 | 0.4188 | 0.1772 | 0.3843 |
| Elite U.S. Liberal Arts College | 0.0296 | 0.1696 | 0.0347 | 0.1832 | 0.0428 | 0.2028 | 0.0633 | 0.2450 |
| Elite Foreign University | 0.0386 | 0.1928 | 0.0490 | 0.2160 | 0.0350 | 0.1842 | 0.0253 | 0.1581 |
| Other Foreign University | 0.4049 | 0.4910 | 0.4188 | 0.4936 | 0.3619 | 0.4815 | 0.4430 | 0.4999 |
| Missing Undergraduate Institution | 0.2203 | 0.4145 | 0.1910 | 0.3933 | 0.1984 | 0.3996 | 0.1266 | 0.3346 |
| Graduate Degree | 0.5067 | 0.5001 | 0.5730 | 0.4949 | 0.5136 | 0.5008 | 0.5190 | 0.5028 |
| Work Experience | 0.5247 | 0.4995 | 0.5894 | 0.4922 | 0.6770 | 0.4685 | 0.7468 | 0.4376 |
| #Professor Recommenders | 2.6230 | 0.7371 | 2.6302 | 0.7165 | 2.5992 | 0.7118 | 2.5696 | 0.7104 |
| #Prolific Recommenders | 1.2602 | 1.0686 | 1.3912 | 1.0840 | 1.4825 | 1.0424 | 1.3038 | 1.0664 |
| Missing GRE | 0.0335 | 0.1800 | 0.0102 | 0.1006 | 0.0039 | 0.0624 | 0.0127 | 0.1125 |
| Missing Undergraduate GPA | 0.7853 | 0.4107 | 0.7487 | 0.4340 | 0.6770 | 0.4685 | 0.6835 | 0.4681 |
| Missing TOEFL/IELTS | 0.6076 | 0.4884 | 0.5935 | 0.4914 | 0.6615 | 0.4741 | 0.6456 | 0.4814 |
| Missing Winsorized Age | 0.0009 | 0.0293 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Observations | 2329 | | 979 | | 257 | | 79 | |

(b) Subjective Variables

| | Full Sample Mean | Full Sample SD | Transition = 1 Mean | Transition = 1 SD | Accept = 1 Mean | Accept = 1 SD | Matriculate = 1 Mean | Matriculate = 1 SD |
|---|---|---|---|---|---|---|---|---|
| Committee Score | 6.4070 | 1.8701 | 6.4070 | 1.8701 | 8.5003 | 0.8151 | 8.4854 | 0.7157 |
| #Math Courses | 5.7637 | 1.8154 | 5.8173 | 1.7904 | 5.6992 | 1.8640 | 5.6585 | 1.6219 |
| Advanced Math | 0.4860 | 0.5004 | 0.4909 | 0.5007 | 0.5680 | 0.4973 | 0.4651 | 0.5047 |
| Letters Length/100 | 8.3474 | 3.5749 | 8.5333 | 3.8263 | 10.0514 | 4.2864 | 9.7088 | 3.4876 |
| Letters Positive Terms (%) | 11.9770 | 2.2993 | 11.9976 | 2.2077 | 12.2167 | 1.9255 | 12.5110 | 2.0904 |
| Letters Negative Terms (%) | 5.7286 | 1.3427 | 5.7216 | 1.3318 | 5.7446 | 1.2313 | 5.8938 | 1.2871 |
| Letters Standout Terms (%) | 0.6241 | 0.4020 | 0.6241 | 0.4029 | 0.7151 | 0.4509 | 0.7077 | 0.5109 |
| Letters Ability Terms (%) | 1.3326 | 0.6239 | 1.3457 | 0.5964 | 1.3796 | 0.5620 | 1.4334 | 0.6037 |
| Letters Research Terms (%) | 3.4837 | 1.3319 | 3.5208 | 1.3423 | 3.8548 | 1.1984 | 3.8395 | 1.2121 |
| Letters Grindstone Terms (%) | 1.8189 | 0.7734 | 1.8674 | 0.7808 | 1.9975 | 0.7588 | 2.1600 | 0.8093 |
| Letters Teaching Terms (%) | 1.7379 | 0.7710 | 1.7489 | 0.7759 | 1.7275 | 0.7536 | 1.8976 | 0.7577 |
| Letters Communal Terms (%) | 0.5946 | 0.3890 | 0.6019 | 0.3954 | 0.5494 | 0.3234 | 0.5630 | 0.3875 |
| Letters Agentic Terms (%) | 0.4768 | 0.3150 | 0.4842 | 0.3229 | 0.5039 | 0.3222 | 0.5597 | 0.3314 |
| Letters Terms (%) PC1 | -0.0579 | 1.6183 | 0.0000 | 1.5768 | 0.2176 | 1.3817 | 0.5697 | 1.5386 |
| Letters Terms (%) PC2 | -0.0271 | 1.1177 | 0.0000 | 1.1446 | 0.1002 | 1.1774 | 0.1699 | 1.2823 |
| CV Length/100 | 4.3532 | 2.4237 | 4.3982 | 2.4758 | 4.9641 | 2.7175 | 4.8261 | 2.3473 |
| CV Coding | 2.8321 | 2.1135 | 2.7931 | 2.1000 | 2.8618 | 2.1129 | 2.5122 | 1.6450 |
| CV Publication/Working Paper | 0.2518 | 0.4346 | 0.2571 | 0.4377 | 0.3659 | 0.4836 | 0.3902 | 0.4939 |
| Essay Length/100 | 4.5099 | 1.5741 | 4.5173 | 1.5805 | 4.7488 | 1.6373 | 4.9816 | 1.7932 |
| Essay Specific Professor | 0.4822 | 0.5003 | 0.5219 | 0.5003 | 0.5691 | 0.4972 | 0.6512 | 0.4822 |
| Essay Economics Terms (%) | 9.7284 | 2.6219 | 9.7216 | 2.6635 | 9.5303 | 2.7324 | 9.8941 | 3.2608 |
| Essay Research Terms (%) | 5.7975 | 2.0501 | 5.7402 | 2.0168 | 5.7313 | 1.8797 | 5.5231 | 1.8432 |
| Essay Positive Terms (%) | 11.8293 | 2.6518 | 11.7324 | 2.6650 | 11.4734 | 2.4592 | 11.3169 | 2.4455 |
| Essay Negative Terms (%) | 6.2844 | 1.9056 | 6.3233 | 1.9712 | 6.2889 | 2.0860 | 6.2651 | 1.9436 |
| Missing Committee Score | 0.5844 | 0.4929 | 0.0112 | 0.1055 | 0.0195 | 0.1384 | 0.0633 | 0.2450 |
| Missing Application Pdf | 0.8154 | 0.3881 | 0.6650 | 0.4722 | 0.5136 | 0.5008 | 0.4557 | 0.5012 |
| Missing #Math Courses | 0.8201 | 0.3842 | 0.6701 | 0.4704 | 0.5214 | 0.5005 | 0.4810 | 0.5028 |
| Missing Letters | 0.8175 | 0.3863 | 0.6691 | 0.4708 | 0.5136 | 0.5008 | 0.4557 | 0.5012 |
| Missing CV | 0.8210 | 0.3835 | 0.6742 | 0.4689 | 0.5214 | 0.5005 | 0.4810 | 0.5028 |
| Missing Essay | 0.8192 | 0.3849 | 0.6731 | 0.4693 | 0.5214 | 0.5005 | 0.4557 | 0.5012 |
| Observations | 2329 | | 979 | | 257 | | 79 | |

Reduced-Form Summary

# A4: Economics Department Correlation Matrix

## Table 3: Economics Department Correlation Matrix

| | Trans | Wait | Acc | Mat | OH | Att | Com | T100 | ProR | MProR | PlaR | MPlaR | 2016 | 2017 | 2018 | 2019 | Age |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Transition | 1 | | | | | | | | | | | | | | | | |
| Waitlist | 0.8083 | 1 | | | | | | | | | | | | | | | |
| Accept | 0.4136 | -0.1908 | 1 | | | | | | | | | | | | | | |
| Matriculate | 0.2200 | -0.0994 | 0.5320 | 1 | | | | | | | | | | | | | |
| Open House | 0.2192 | -0.1104 | 0.5496 | 0.5400 | 1 | | | | | | | | | | | | |
| Attend Program | 0.1266 | 0.0767 | 0.1209 | 0.1522 | 0.0833 | 1 | | | | | | | | | | | |
| Complete Program | 0.1085 | 0.0889 | 0.0763 | 0.0427 | 0.0154 | 0.9124 | 1 | | | | | | | | | | |
| Top 100 Placement | 0.1073 | 0.0778 | 0.0977 | 0.0682 | 0.0498 | 0.4025 | 0.4412 | 1 | | | | | | | | | |
| Program Rank | -0.1531 | -0.0909 | -0.1565 | -0.1693 | -0.0999 | -0.8635 | -0.7942 | -0.4106 | 1 | | | | | | | | |
| Missing Program Rank | -0.1341 | -0.0860 | -0.1219 | -0.1594 | -0.0856 | -0.9548 | -0.8749 | -0.4018 | 0.9044 | 1 | | | | | | | |
| Placement Rank | -0.1153 | -0.0802 | -0.1057 | -0.0553 | -0.0666 | -0.5176 | -0.5673 | -0.9004 | 0.4999 | 0.5032 | 1 | | | | | | |
| Missing Placement Rank | -0.1051 | -0.0835 | -0.0778 | -0.0435 | -0.0164 | -0.9084 | -0.9957 | -0.4431 | 0.7924 | 0.8708 | 0.5698 | 1 | | | | | |
| 2016 | -0.0752 | -0.0883 | -0.0229 | -0.0013 | -0.0126 | -0.0194 | 0.0073 | -0.0005 | 0.0077 | 0.0178 | -0.0069 | -0.0073 | 1 | | | | |
| 2017 | -0.0480 | -0.0732 | -0.0162 | -0.0113 | 0.0162 | 0.0076 | 0.0138 | 0.0096 | -0.0025 | -0.0064 | -0.0040 | -0.0138 | -0.2552 | 1 | | | |
| 2018 | 0.1637 | 0.1695 | 0.0120 | 0.0287 | 0.0009 | 0.0101 | 0.0078 | 0.0096 | -0.0118 | -0.0233 | -0.0076 | -0.0034 | -0.2483 | -0.2454 | 1 | | |
| 2019 | 0.0121 | -0.0050 | 0.0133 | 0.0058 | -0.0045 | 0.0145 | -0.0565 | -0.0081 | 0.0123 | 0.0017 | 0.0166 | 0.0543 | -0.2562 | -0.2532 | -0.2464 | 1 | |
| Winsorized Age | 0.0738 | 0.0157 | 0.0102 | 0.0201 | 0.0122 | -0.0260 | -0.0179 | 0.0398 | 0.0474 | 0.0304 | -0.0148 | 0.0229 | 0.0102 | -0.0091 | -0.0128 | 0.0160 | 1 |
| Female | 0.0411 | 0.0255 | 0.0350 | 0.0102 | 0.0328 | -0.0068 | -0.0179 | -0.0189 | 0.0074 | 0.0168 | 0.0007 | 0.0198 | 0.0066 | -0.0163 | 0.0142 | 0.0358 | -0.1468 |
| U.S. | 0.0046 | -0.0393 | 0.0530 | 0.0301 | 0.0318 | -0.1069 | -0.1165 | -0.0310 | 0.0900 | 0.1075 | 0.0538 | 0.1141 | 0.0083 | 0.0263 | -0.0073 | -0.0409 | -0.1537 |
| International Asian | 0.0717 | 0.1739 | -0.0976 | -0.0189 | -0.0210 | 0.1744 | 0.1736 | 0.0416 | -0.1725 | -0.1816 | -0.0646 | -0.1707 | -0.0210 | -0.0162 | -0.0107 | 0.0407 | -0.0407 |
| GRE Verbal Percentile | 0.3349 | 0.2346 | 0.1825 | 0.0746 | 0.1228 | 0.0524 | 0.0343 | 0.0479 | -0.0681 | -0.0536 | -0.0548 | -0.0358 | -0.0272 | -0.0158 | 0.0260 | 0.0343 | -0.1056 |
| GRE Quantitative Percentile | 0.3448 | 0.2791 | 0.1559 | 0.0623 | 0.0605 | 0.1957 | 0.1930 | 0.0980 | -0.2102 | -0.1965 | -0.1287 | -0.1899 | 0.0061 | 0.0095 | 0.0116 | -0.0255 | -0.0622 |
| GRE Analytic Percentile | 0.1960 | 0.1198 | 0.1330 | 0.0505 | 0.0710 | -0.0468 | -0.0700 | -0.0083 | 0.0359 | 0.0504 | 0.0058 | 0.0695 | -0.0542 | 0.0113 | 0.0537 | 0.0379 | -0.1533 |
| Undergraduate GPA | 0.0517 | -0.0095 | 0.0747 | 0.0322 | 0.0440 | 0.0329 | 0.0179 | -0.0166 | -0.0382 | -0.0336 | 0.0056 | -0.0179 | -0.0272 | -0.0058 | 0.0260 | 0.0109 | -0.0853 |
| TOEFL/IELTS | 0.1251 | 0.1060 | 0.0898 | 0.0017 | 0.0413 | -0.0200 | -0.0127 | 0.0342 | -0.0053 | 0.0226 | -0.0431 | 0.0108 | -0.0018 | 0.0047 | 0.0176 | -0.0092 | -0.1177 |
| Elite U.S. University | 0.1203 | 0.0775 | 0.1048 | 0.0285 | 0.0769 | -0.0151 | -0.0196 | -0.0167 | -0.0171 | 0.0108 | 0.0124 | 0.0179 | -0.0305 | -0.0271 | 0.0070 | 0.0392 | -0.0769 |
| Elite U.S. Liberal Arts College | 0.0256 | -0.0020 | 0.0274 | 0.0372 | 0.0205 | -0.0081 | -0.0280 | -0.0038 | -0.0103 | 0.0104 | -0.0012 | 0.0272 | 0.0178 | 0.0133 | -0.0405 | -0.0125 | -0.0639 |
| Elite Foreign University | 0.0459 | 0.0691 | -0.0066 | -0.0130 | -0.0062 | 0.0218 | 0.0305 | 0.0071 | -0.0445 | -0.0261 | 0.0027 | -0.0314 | 0.0084 | 0.0215 | -0.0246 | -0.0290 | -0.0129 |
| Other Foreign University | 0.0241 | 0.0431 | -0.0309 | 0.0146 | -0.0327 | 0.0571 | 0.0564 | 0.0311 | -0.0260 | -0.0496 | -0.0454 | -0.0512 | -0.0228 | -0.0085 | -0.0772 | 0.1586 | 0.1709 |
| Missing Undergraduate Institution | -0.0601 | -0.0583 | -0.0185 | -0.0424 | -0.0215 | 0.0169 | 0.0435 | 0.0180 | -0.0124 | -0.0198 | -0.0227 | -0.0458 | 0.0300 | -0.0008 | 0.1946 | -0.2576 | -0.0011 |
| Graduate Degree | 0.1131 | 0.1254 | 0.0049 | 0.0046 | 0.0248 | 0.0529 | 0.0706 | 0.0226 | -0.0399 | -0.0548 | -0.0328 | -0.0698 | 0.0443 | 0.1379 | -0.1556 | -0.0729 | 0.3704 |
| Work Experience | 0.1103 | 0.0043 | 0.1074 | 0.0834 | 0.0484 | -0.0723 | -0.0789 | 0.0371 | 0.0722 | 0.0851 | -0.0148 | 0.0795 | -0.0017 | -0.0002 | -0.0056 | 0.0093 | 0.4141 |
| #Professor Recommenders | 0.0083 | 0.0374 | -0.0114 | -0.0136 | -0.0257 | 0.0606 | 0.0466 | -0.0073 | -0.0472 | -0.0547 | 0.0069 | -0.0514 | -0.0791 | 0.0186 | 0.0329 | 0.0390 | -0.2398 |
| #Prolific Recommenders | 0.1044 | 0.0794 | 0.0733 | 0.0076 | 0.0374 | 0.1042 | 0.1084 | 0.0632 | -0.1156 | -0.1054 | -0.0567 | -0.1070 | -0.0262 | 0.0330 | 0.0430 | 0.0062 | 0.0200 |
| Missing GRE | -0.1102 | -0.1318 | -0.0579 | -0.0217 | -0.0358 | -0.0731 | -0.0649 | -0.0322 | 0.0715 | 0.0786 | 0.0303 | 0.0641 | 0.0118 | -0.0042 | -0.0123 | 0.0071 | 0.0718 |

| | LPC1 | LPC2 | CVN | CVCod | CVPap | EssN | EssProf | EssEcon | EssRes | EssPos | EssNeg | MScore | MPDF | MNMath | ML | MCV | MEss |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Letters Terms (%) PC1 | 1 | | | | | | | | | | | | | | | | |
| Letters Terms (%) PC2 | -0.0204 | 1 | | | | | | | | | | | | | | | |
| CV Length/100 | 0.0820 | 0.0721 | 1 | | | | | | | | | | | | | | |
| CV Coding | 0.0147 | 0.1218 | 0.2274 | 1 | | | | | | | | | | | | | |
| CV Publication/Working Paper | 0.1158 | 0.0865 | 0.2482 | 0.1413 | 1 | | | | | | | | | | | | |
| Essay Length/100 | -0.0332 | 0.0460 | 0.0706 | 0.1168 | 0.0772 | 1 | | | | | | | | | | | |
| Essay Specific Professor | -0.0084 | 0.0360 | 0.0795 | 0.1038 | 0.2923 | 0.0701 | 1 | | | | | | | | | | |
| Essay Economics Terms (%) | [illegible] | 0.1110 | 0.0016 | 0.0578 | 0.0391 | -0.0593 | 0.0049 | 1 | | | | | | | | | |
| Essay Research Terms (%) | [illegible] | 0.0886 | 0.0185 | 0.1414 | 0.0621 | 0.0651 | -0.0097 | 0.1593 | 1 | | | | | | | | |
| Essay Positive Terms (%) | [illegible] | -0.0494 | -0.0413 | -0.0248 | -0.0291 | -0.1956 | -0.0535 | -0.0689 | -0.0250 | 1 | | | | | | | |
| Essay Negative Terms (%) | [illegible] | -0.0083 | 0.0279 | 0.0566 | 0.0102 | 0.0716 | 0.0666 | -0.1217 | -0.1206 | -0.0714 | 1 | | | | | | |
| Missing Committee Score | [illegible] | -0.0054 | -0.0070 | 0.0098 | -0.1443 | 0.0100 | -0.2336 | -0.0045 | 0.0027 | 0.0134 | -0.0106 | 1 | | | | | |
| Missing Application Pdf | [illegible] | 0.0218 | 0.0165 | -0.0164 | -0.4566 | 0.0042 | -0.6494 | -0.0023 | -0.0250 | -0.0327 | 0.0183 | 0.3105 | 1 | | | | |
| Missing #Math Courses | [illegible] | 0.0164 | 0.0076 | -0.0239 | -0.4531 | 0.0007 | -0.6399 | 0.0044 | -0.0279 | -0.0345 | 0.0283 | 0.3149 | 0.9843 | 1 | | | |
| Missing Letters | [illegible] | 0.0219 | 0.0199 | -0.0202 | -0.4599 | 0.0066 | -0.6462 | 0.0039 | -0.0208 | -0.0261 | 0.0113 | 0.3076 | 0.9929 | 0.9769 | 1 | | |
| Missing CV | [illegible] | 0.0187 | 0.0168 | -0.0168 | -0.4653 | 0.0026 | -0.6299 | -0.0129 | -0.0385 | -0.0266 | 0.0242 | 0.3061 | 0.9814 | 0.9650 | 0.9740 | 1 | |
| Missing Essay | 0.0253 | 0.0200 | 0.0134 | -0.0246 | -0.4464 | 0.0042 | -0.6578 | -0.0023 | -0.0253 | -0.0331 | 0.0185 | 0.3034 | 0.9871 | 0.9710 | 0.9798 | 0.9709 | 1 |

Observations: 2329

▸▸ Reduced-Form Summary

# A4: Economics Department Rec Letters Principal Components

Table 4: Economics Department Rec Letters Principal Components

| | PC1 | PC2 |
|---|---|---|
| Positive | 0.5044 | -0.1769 |
| Negative | 0.1479 | 0.3759 |
| Standout | 0.2115 | -0.5082 |
| Ability | 0.4667 | -0.0177 |
| Research | 0.1214 | 0.6418 |
| Grindstone | 0.3879 | 0.3075 |
| Teaching | 0.3708 | 0.0743 |
| Communal | 0.1284 | -0.1983 |
| Agentic | 0.3768 | -0.1324 |
| Proportion of Variance | 0.2763 | 0.1456 |

Reduced-Form Summary

# A4: Economics Department Logistic Regression Results

Table 5a: Economics Department Logistic Regression Results

| Dependent Variable | Accept | Accept | Matriculate | T100 Placement | T100 Placement | T100 Placement | Placement Rank | Placement Rank | Placement Rank |
|---|---|---|---|---|---|---|---|---|---|
| 2016 | -0.3826 | -0.5049 | -0.2577 | 0.0230 | 0.0301 | 0.0885 | 1.8038 | 1.7607 | -0.6618 |
| | (0.2365) | (0.4703) | (0.6853) | (0.1708) | (0.1711) | (0.1887) | (4.2583) | (4.2504) | (3.6266) |
| 2017 | -0.0954 | -1.3247*** | 0.1357 | 0.0829 | 0.0711 | 0.1460 | 2.7746 | 3.0235 | 0.7473 |
| | (0.2564) | (0.4881) | (0.6732) | (0.1707) | (0.1721) | (0.1934) | (4.2295) | (4.2188) | (3.7157) |
| 2018 | -0.2206 | -3.7609*** | 0.9476 | 0.0624 | -0.0392 | 0.0142 | 2.7265 | 5.2685 | 4.9770 |
| | (0.2648) | (0.6518) | (0.6200) | (0.1761) | (0.1807) | (0.2011) | (4.4414) | (4.5011) | (3.8786) |
| 2019 | -0.1059 | -3.1670*** | 0.5909 | 0.0594 | 0.0144 | 0.0366 | 5.7085 | 6.6750 | 5.2241 |
| | (0.2653) | (0.7114) | (0.6855) | (0.1821) | (0.1830) | (0.2052) | (4.6400) | (4.6266) | (4.1424) |
| Winsorized Age | -0.0218 | -0.0080 | -0.1567** | 0.0240 | 0.0242 | 0.0420* | 0.3596 | 0.3632 | -0.1277 |
| | (0.0342) | (0.0592) | (0.0778) | (0.0207) | (0.0205) | (0.0236) | (0.5457) | (0.5410) | (0.4839) |
| Female | 0.3759** | 0.8784** | -0.3445 | -0.0880 | -0.1026 | -0.0256 | 1.8085 | 2.0921 | -1.6375 |
| | (0.1657) | (0.3813) | (0.3794) | (0.1126) | (0.1130) | (0.1252) | (2.8776) | (2.8750) | (2.5117) |
| U.S. | -0.4982* | -0.0681 | 1.0348* | 0.1167 | 0.1396 | 0.1520 | -0.6372 | -0.9759 | -0.7059 |
| | (0.2624) | (0.4538) | (0.6098) | (0.1903) | (0.1909) | (0.2236) | (4.7990) | (4.7913) | (4.3464) |
| International Asian | -1.1569*** | -1.1507*** | 1.1559** | 0.1695 | 0.1949 | -0.1005 | -7.8212** | -8.2008** | 3.3007 |
| | (0.2129) | (0.4146) | (0.4904) | (0.1379) | (0.1395) | (0.1546) | (3.5528) | (3.5708) | (3.1577) |
| GRE Verbal Percentile | 0.0264*** | 0.0079 | -0.0107 | 0.0049 | 0.0030 | 0.0007 | -0.1958** | -0.1470* | -0.0330 |
| | (0.0054) | (0.0130) | (0.0118) | (0.0031) | (0.0032) | (0.0036) | (0.0787) | (0.0806) | (0.0735) |
| GRE Quantitative Percentile | 0.1329*** | 0.0589** | -0.1033** | 0.0245*** | 0.0184** | 0.0047 | -0.8498*** | -0.7196*** | -0.2428* |
| | (0.0202) | (0.0275) | (0.0464) | (0.0077) | (0.0075) | (0.0084) | (0.1567) | (0.1578) | (0.1460) |
| GRE Analytic Percentile | 0.0076** | 0.0094 | -0.0119 | -0.0020 | -0.0027 | -0.0021 | 0.0656 | 0.0821 | 0.0160 |
| | (0.0039) | (0.0085) | (0.0095) | (0.0025) | (0.0025) | (0.0029) | (0.0669) | (0.0671) | (0.0618) |
| Undergraduate GPA | 1.2726* | 1.0509 | -0.7209 | -0.3976 | -0.4722 | -0.8687* | 4.8040 | 6.9311 | 17.2597** |
| | (0.6975) | (1.2453) | (1.2760) | (0.3912) | (0.3848) | (0.4442) | (10.1461) | (10.0615) | (7.7507) |
| TOEFL/IELTS | 0.4236 | -0.1165 | -0.4434 | 0.1778 | 0.1437 | 0.3240* | -2.2504 | -1.2960 | -7.6837* |
| | (0.2924) | (0.5976) | (0.5525) | (0.1571) | (0.1559) | (0.1757) | (4.3852) | (4.3431) | (3.9285) |
| Elite U.S. University | 0.7328*** | -0.1822 | -0.4980 | -0.0159 | -0.0853 | -0.1523 | -2.3790 | -0.3552 | -0.3450 |
| | (0.2774) | (0.5918) | (0.5799) | (0.2133) | (0.2133) | (0.2344) | (5.2934) | (5.2869) | (4.4943) |
| Elite U.S. Liberal Arts College | 0.0167 | 0.5446 | 0.9907 | 0.0731 | 0.0803 | -0.1010 | -3.9470 | -3.8047 | -3.4006 |
| | (0.4568) | (0.6896) | (0.8447) | (0.3370) | (0.3361) | (0.3936) | (8.8571) | (8.7916) | (8.2332) |
| Elite Foreign University | 0.1244 | -0.5507 | -0.0673 | 0.1210 | 0.0751 | 0.0969 | -2.7705 | -1.5414 | -2.2142 |
| | (0.5319) | (0.9725) | (1.1224) | (0.3239) | (0.3259) | (0.3526) | (7.4748) | (7.4679) | (6.1380) |
| Other Foreign University | 0.5323* | 0.6328 | 0.7373 | 0.1483 | 0.1091 | 0.1587 | -6.7584 | -5.7186 | -9.1260** |
| | (0.3022) | (0.6505) | (0.6347) | (0.1964) | (0.1970) | (0.2216) | (4.9679) | (4.9620) | (4.3486) |
| Missing Undergraduate Institution | 0.3260 | 0.2482 | -0.6851 | 0.2292 | 0.2082 | 0.1770 | -7.9862 | -7.5382 | -8.1758* |
| | (0.3111) | (0.6016) | (0.7260) | (0.2017) | (0.2022) | (0.2345) | (5.1133) | (5.1129) | (4.5085) |
| Graduate Degree | 0.1726 | -0.2058 | -0.2954 | -0.0406 | -0.0921 | -0.1835 | -3.2803 | -2.0242 | 2.6215 |
| | (0.1788) | (0.3845) | (0.5017) | (0.1231) | (0.1273) | (0.1370) | (3.1662) | (3.2087) | (2.7452) |
| Work Experience | 0.7785*** | 0.7519** | 1.1183** | 0.2298* | 0.1772 | 0.2839** | -3.5942 | -2.2559 | -6.4380** |
| | (0.1915) | (0.3400) | (0.5291) | (0.1253) | (0.1270) | (0.1422) | (3.2005) | (3.2301) | (2.9093) |
| #Professor Recommenders | -0.0377 | -0.1797 | 0.3587 | -0.0472 | -0.0513 | -0.0547 | 0.8765 | 0.9972 | 0.7371 |
| | (0.1079) | (0.2434) | (0.3115) | (0.0749) | (0.0754) | (0.0862) | (2.0492) | (2.0495) | (1.7160) |
| #Prolific Recommenders | 0.2208*** | 0.2081 | -0.2993 | 0.1011* | 0.0899* | 0.0600 | -1.4623 | -1.1967 | 0.1726 |
| | (0.0811) | (0.1721) | (0.2060) | (0.0528) | (0.0533) | (0.0585) | (1.3680) | (1.3670) | (1.1704) |
| Missing GRE | -1.5399 | 1.0141 | | -0.4515 | -0.3258 | 0.0174 | 11.5202 | 8.5434 | -3.7106 |
| | (0.9821) | (1.1356) | | (0.3510) | (0.3544) | (0.3797) | (8.6385) | (8.6602) | (6.1802) |
| Missing Undergraduate GPA | -0.3432 | -0.2798 | 0.0798 | -0.1010 | -0.0639 | -0.0513 | 2.4296 | 1.5256 | -0.8710 |
| | (0.2210) | (0.4413) | (0.4943) | (0.1565) | (0.1585) | (0.1756) | (3.9808) | (3.9842) | (3.3572) |
| Missing TOEFL/IELTS | 0.4719** | 0.3743 | -0.0734 | -0.1261 | -0.1447 | -0.0065 | 4.0029 | 4.3290 | -2.3792 |
| | (0.2062) | (0.4223) | (0.5891) | (0.1267) | (0.1282) | (0.1443) | (3.3209) | (3.3244) | (2.9657) |
| Committee Score | | 4.2419*** | -0.2608 | | 0.0943** | 0.0059 | | -2.5088** | 0.6884 |
| | | (0.4534) | (0.2250) | | (0.0456) | (0.0488) | | (1.1860) | (1.0132) |
| #Math Courses | -0.0949 | -0.2706 | 0.0755 | 0.0626 | 0.0662 | 0.0008 | -2.3828 | -2.4550 | 0.9116 |
| | (0.0773) | (0.2829) | (0.1680) | (0.0759) | (0.0766) | (0.0885) | (1.8033) | (1.8149) | (1.5082) |
| Advanced Math | 0.6333** | 1.3229 | -1.3706** | -0.1932 | -0.2078 | -0.2275 | 3.0400 | 3.4570 | 1.9957 |
| | (0.3036) | (0.9203) | (0.5621) | (0.2605) | (0.2639) | (0.2949) | (6.4446) | (6.4377) | (5.5819) |
| Letters Length/100 | 0.1358*** | 0.1084 | -0.0910 | 0.0067 | -0.0018 | -0.0150 | -0.2593 | -0.0078 | 0.7206 |
| | (0.0518) | (0.1424) | (0.0712) | (0.0335) | (0.0339) | (0.0388) | (0.8254) | (0.8325) | (0.7084) |
| Letters Terms (%) PC1 | 0.1507 | 0.2939 | 0.4716** | 0.0362 | 0.0251 | 0.1260 | 1.6082 | 1.8374 | -0.5612 |
| | (0.0923) | (0.2912) | (0.2051) | (0.0732) | (0.0748) | (0.0915) | (2.0561) | (2.0540) | (1.8186) |
| Letters Terms (%) PC2 | -0.0913 | -0.4980 | 0.2724 | -0.1226 | -0.1199 | -0.1420 | 0.7118 | 0.5616 | 0.1155 |
| | (0.1464) | (0.4143) | (0.2604) | (0.1162) | (0.1177) | (0.1272) | (2.7992) | (2.8081) | (2.3450) |
| CV Length/100 | 0.0361 | 0.0559 | -0.1986 | 0.0301 | 0.0274 | 0.0352 | -1.0456 | -1.0120 | -0.8208 |
| | (0.0617) | (0.1186) | (0.1337) | (0.0562) | (0.0563) | (0.0593) | (1.3224) | (1.3130) | (1.0252) |
| CV Coding | -0.0439 | -0.2369* | -0.2911** | 0.0716 | 0.0710 | 0.0992 | -1.9601 | -1.8941 | -2.0142 |
| | (0.0765) | (0.1381) | (0.1390) | (0.0532) | (0.0532) | (0.0654) | (1.5459) | (1.5358) | (1.3384) |
| CV Publication/Working Paper | 0.6162* | 0.8309 | -0.1668 | 0.2940 | 0.2834 | 0.1310 | -11.5166 | -10.8849 | -4.6204 |
| | (0.3421) | (0.9326) | (0.5756) | (0.2998) | (0.3036) | (0.3383) | (7.2873) | (7.3131) | (6.7001) |
| Essay Length/100 | 0.1747* | 0.2225 | 0.1776 | -0.0008 | -0.0008 | -0.0711 | -0.4758 | -0.5351 | 1.8425 |
| | (0.0933) | (0.3920) | (0.1536) | (0.0751) | (0.0764) | (0.0821) | (1.8648) | (1.8721) | (1.6240) |
| Essay Specific Professor | 0.3842 | 0.4170 | 0.6845 | 0.2144 | 0.1774 | 0.2872 | -6.8688 | -5.9104 | -7.5240 |
| | (0.2908) | (0.7014) | (0.5119) | (0.2407) | (0.2410) | (0.2789) | (6.0223) | (5.9982) | (5.2718) |
| Essay Economics Terms (%) | -0.0169 | 0.0070 | 0.2576** | -0.1126** | -0.1150** | -0.0822 | 3.6540*** | 3.7277*** | 2.1702* |
| | (0.0547) | (0.0866) | (0.1159) | (0.0533) | (0.0531) | (0.0618) | (1.2540) | (1.2496) | (1.1179) |
| Essay Research Terms (%) | -0.0018 | -0.0459 | -0.1868 | 0.0697 | 0.0642 | 0.0150 | -3.7084** | -3.6795** | -1.0575 |
| | (0.0736) | (0.1690) | (0.1451) | (0.0633) | (0.0639) | (0.0674) | (1.6220) | (1.6209) | (1.3903) |
| Essay Positive Terms (%) | -0.0202 | -0.0700 | -0.0504 | 0.0724 | 0.0772* | 0.0977* | -0.9256 | -0.9929 | -0.7266 |
| | (0.0512) | (0.1044) | (0.1331) | (0.0453) | (0.0458) | (0.0542) | (1.1634) | (1.1496) | (0.9752) |
| Essay Negative Terms (%) | -0.0151 | -0.0886 | 0.0019 | -0.0386 | -0.0454 | -0.0472 | 0.8164 | 0.9436 | 0.9507 |
| | (0.0770) | (0.1557) | (0.1071) | (0.0663) | (0.0663) | (0.0748) | (1.6317) | (1.6259) | (1.3985) |
| Missing Committee Score | | -0.5168 | | | -0.3458** | -0.2199 | | 9.2234*** | 2.8589 |
| | | (0.6383) | | | (0.1397) | (0.1519) | | (3.4582) | (3.0258) |
| Missing Application Pdf | -1.0107*** | 0.1926 | -0.7407 | 0.1461 | 0.2686 | 0.2416 | -5.1611 | -8.4497 | -6.8690 |
| | (0.3188) | (0.8088) | (0.6084) | (0.2442) | (0.2479) | (0.2763) | (6.0328) | (6.1404) | (5.4325) |
| Open House | | | 1.8384*** | | | | | | |
| | | | (0.4115) | | | | | | |
| Program Rank | | | | | | -0.0087*** | | | 0.1813*** |
| | | | | | | (0.0021) | | | (0.0404) |
| Missing Program Rank | | | | | | -3.1432*** | | | 89.4721*** |
| | | | | | | (0.4077) | | | (5.6947) |
| Sample | All | All | Accept = 1 | All | All | All | All | All | All |
| Observations | 2329 | 2329 | 257 | 2329 | 2329 | 2329 | 2329 | 2329 | 2329 |
| Pseudo-$R^2$ | 0.2957 | 0.8108 | 0.2877 | 0.0308 | 0.0353 | 0.2430 | 0.0089 | 0.0097 | 0.1065 |
| CV Pseudo-$R^2$ | 0.2047 | 0.7160 | -0.2841 | -0.0107 | -0.0069 | 0.1999 | 0.0014 | 0.0022 | 0.0992 |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

Acceptance, Matriculation, and Success AMEs

# A4: Economics Department Logistic Regression AMEs

Table 5b: Economics Department Logistic Regression AMEs

| Dependent Variable | Accept | Accept | Matriculate | T100 Placement | T100 Placement | T100 Placement | Placement Rank | Placement Rank | Placement Rank |
|---|---|---|---|---|---|---|---|---|---|
| 2016 | -0.0274 | -0.0094 | -0.0367 | 0.0035 | 0.0046 | 0.0112 | 0.8694 | 0.8485 | -0.3038 |
| | (0.0171) | (0.0085) | (0.0976) | (0.0262) | (0.0261) | (0.0238) | (2.0523) | (2.0482) | (1.6650) |
| 2017 | -0.0068 | -0.0246*** | 0.0193 | 0.0127 | 0.0108 | 0.0184 | 1.3374 | 1.4571 | 0.3431 |
| | (0.0184) | (0.0090) | (0.0958) | (0.0262) | (0.0263) | (0.0244) | (2.0386) | (2.0331) | (1.7054) |
| 2018 | -0.0158 | -0.0699*** | 0.1348 | 0.0096 | -0.0060 | 0.0018 | 1.3142 | 2.5391 | 2.2847 |
| | (0.0189) | (0.0086) | (0.0857) | (0.0270) | (0.0276) | (0.0254) | (2.1405) | (2.1686) | (1.7791) |
| 2019 | -0.0076 | -0.0589*** | 0.0841 | 0.0091 | 0.0022 | 0.0046 | 2.7516 | 3.2169 | 2.3982 |
| | (0.0190) | (0.0109) | (0.0970) | (0.0279) | (0.0279) | (0.0259) | (2.2355) | (2.2284) | (1.8977) |
| Winsorized Age | -0.0016 | -0.0001 | -0.0223** | 0.0037 | 0.0037 | 0.0053* | 0.1734 | 0.1751 | -0.0586 |
| | (0.0024) | (0.0011) | (0.0108) | (0.0032) | (0.0031) | (0.0030) | (0.2629) | (0.2606) | (0.2222) |
| Female | 0.0269** | 0.0163** | -0.0490 | -0.0135 | -0.0157 | -0.0032 | 0.8717 | 1.0082 | -0.7517 |
| | (0.0118) | (0.0068) | (0.0534) | (0.0173) | (0.0172) | (0.0158) | (1.3867) | (1.3851) | (1.1534) |
| U.S. | -0.0356* | -0.0013 | 0.1472* | 0.0179 | 0.0213 | 0.0192 | -0.3071 | -0.4703 | -0.3241 |
| | (0.0188) | (0.0084) | (0.0864) | (0.0292) | (0.0291) | (0.0282) | (2.3132) | (2.3090) | (1.9951) |
| International Asian | -0.0828*** | -0.0214*** | 0.1644** | 0.0260 | 0.0298 | -0.0127 | -3.7699** | -3.9523** | 1.5152 |
| | (0.0148) | (0.0073) | (0.0691) | (0.0212) | (0.0213) | (0.0195) | (1.7114) | (1.7195) | (1.4490) |
| GRE Verbal Percentile | 0.0019*** | 0.0001 | -0.0015 | 0.0008 | 0.0005 | 0.0001 | -0.0944** | -0.0708* | -0.0151 |
| | (0.0004) | (0.0002) | (0.0017) | (0.0005) | (0.0005) | (0.0005) | (0.0379) | (0.0388) | (0.0337) |
| GRE Quantitative Percentile | 0.0095*** | 0.0011** | -0.0147** | 0.0038*** | 0.0028** | 0.0006 | -0.4096*** | -0.3468*** | -0.1114* |
| | (0.0014) | (0.0005) | (0.0063) | (0.0012) | (0.0011) | (0.0011) | (0.0757) | (0.0762) | (0.0670) |
| GRE Analytic Percentile | 0.0005** | 0.0002 | -0.0017 | -0.0003 | -0.0004 | -0.0003 | 0.0316 | 0.0395 | 0.0073 |
| | (0.0003) | (0.0002) | (0.0013) | (0.0004) | (0.0004) | (0.0004) | (0.0322) | (0.0323) | (0.0283) |
| Undergraduate GPA | 0.0910* | 0.0195 | -0.1026 | -0.0610 | -0.0721 | -0.1096** | 2.3156 | 3.3404 | 7.9232** |
| | (0.0499) | (0.0235) | (0.1820) | (0.0600) | (0.0587) | (0.0558) | (4.8911) | (4.8499) | (3.5564) |
| TOEFL/IELTS | 0.0303 | -0.0022 | -0.0631 | 0.0273 | 0.0219 | 0.0409* | -1.0847 | -0.6246 | -3.5273* |
| | (0.0209) | (0.0111) | (0.0776) | (0.0241) | (0.0238) | (0.0221) | (2.1145) | (2.0935) | (1.8033) |
| Elite U.S. University | 0.0524*** | -0.0034 | -0.0708 | -0.0024 | -0.0130 | -0.0192 | -1.1467 | -0.1712 | -0.1584 |
| | (0.0198) | (0.0111) | (0.0816) | (0.0327) | (0.0326) | (0.0296) | (2.5514) | (2.5479) | (2.0632) |
| Elite U.S. Liberal Arts College | 0.0012 | 0.0101 | 0.1409 | 0.0112 | 0.0123 | -0.0127 | -1.9025 | -1.8336 | -1.5611 |
| | (0.0327) | (0.0126) | (0.1197) | (0.0517) | (0.0513) | (0.0496) | (4.2692) | (4.2369) | (3.7802) |
| Elite Foreign University | 0.0089 | -0.0102 | -0.0096 | 0.0186 | 0.0115 | 0.0122 | -1.3354 | -0.7428 | -1.0164 |
| | (0.0381) | (0.0182) | (0.1597) | (0.0497) | (0.0497) | (0.0445) | (3.6022) | (3.5986) | (2.8173) |
| Other Foreign University | 0.0381* | 0.0118 | 0.1049 | 0.0228 | 0.0167 | 0.0200 | -3.2576 | -2.7560 | -4.1894** |
| | (0.0216) | (0.0119) | (0.0890) | (0.0301) | (0.0301) | (0.0279) | (2.3945) | (2.3912) | (1.9955) |
| Missing Undergraduate Institution | 0.0233 | 0.0046 | -0.0975 | 0.0352 | 0.0318 | 0.0223 | -3.8495 | -3.6329 | -3.7531* |
| | (0.0222) | (0.0111) | (0.1014) | (0.0309) | (0.0308) | (0.0296) | (2.4639) | (2.4635) | (2.0698) |
| Graduate Degree | 0.0123 | -0.0038 | -0.0420 | -0.0062 | -0.0141 | -0.0231 | -1.5811 | -0.9755 | 1.2034 |
| | (0.0128) | (0.0071) | (0.0712) | (0.0189) | (0.0194) | (0.0172) | (1.5257) | (1.5461) | (1.2606) |
| Work Experience | 0.0557*** | 0.0140** | 0.1591** | 0.0353* | 0.0271 | 0.0358** | -1.7325 | -1.0872 | -2.9554** |
| | (0.0135) | (0.0064) | (0.0700) | (0.0192) | (0.0194) | (0.0178) | (1.5432) | (1.5570) | (1.3344) |
| #Professor Recommenders | -0.0027 | -0.0033 | 0.0510 | -0.0072 | -0.0078 | -0.0069 | 0.4225 | 0.4806 | 0.3384 |
| | (0.0077) | (0.0045) | (0.0437) | (0.0115) | (0.0115) | (0.0109) | (0.9881) | (0.9881) | (0.7878) |
| #Prolific Recommenders | 0.0158*** | 0.0039 | -0.0426 | 0.0155* | 0.0137* | 0.0076 | -0.7048 | -0.5767 | 0.0792 |
| | (0.0058) | (0.0031) | (0.0285) | (0.0081) | (0.0081) | (0.0074) | (0.6595) | (0.6589) | (0.5373) |
| Missing GRE | -0.1102 | 0.0188 | | -0.0693 | -0.0497 | 0.0022 | 5.5529 | 4.1174 | -1.7034 |
| | (0.0706) | (0.0212) | | (0.0538) | (0.0541) | (0.0479) | (4.1627) | (4.1727) | (2.8358) |
| Missing Undergraduate GPA | -0.0246 | -0.0052 | 0.0114 | -0.0155 | -0.0098 | -0.0065 | 1.1711 | 0.7352 | -0.3999 |
| | (0.0158) | (0.0082) | (0.0703) | (0.0240) | (0.0242) | (0.0221) | (1.9187) | (1.9201) | (1.5412) |
| Missing TOEFL/IELTS | 0.0338** | 0.0070 | -0.0104 | -0.0194 | -0.0221 | -0.0008 | 1.9295 | 2.0863 | -1.0922 |
| | (0.0147) | (0.0078) | (0.0839) | (0.0194) | (0.0195) | (0.0182) | (1.6007) | (1.6020) | (1.3613) |
| Committee Score | | 0.0788*** | -0.0371 | | 0.0144** | 0.0007 | | -1.2091** | 0.3160 |
| | | (0.0045) | (0.0322) | | (0.0069) | (0.0062) | | (0.5716) | (0.4649) |
| #Math Courses | -0.0068 | -0.0050 | 0.0107 | 0.0096 | 0.0101 | 0.0001 | -1.1485 | -1.1832 | 0.4185 |
| | (0.0055) | (0.0053) | (0.0239) | (0.0116) | (0.0117) | (0.0112) | (0.8692) | (0.8747) | (0.6922) |
| Advanced Math | 0.0453** | 0.0246 | -0.1950** | -0.0296 | -0.0317 | -0.0287 | 1.4653 | 1.6660 | 0.9161 |
| | (0.0216) | (0.0168) | (0.0793) | (0.0399) | (0.0402) | (0.0372) | (3.1061) | (3.1022) | (2.5621) |
| Letters Length/100 | 0.0097*** | 0.0020 | -0.0129 | 0.0010 | -0.0003 | -0.0019 | -0.1250 | -0.0038 | 0.3308 |
| | (0.0037) | (0.0026) | (0.0100) | (0.0051) | (0.0052) | (0.0049) | (0.3978) | (0.4012) | (0.3253) |
| Letters Terms (%) PC1 | 0.0108 | 0.0055 | 0.0671** | 0.0056 | 0.0038 | 0.0159 | 0.7752 | 0.8855 | -0.2576 |
| | (0.0066) | (0.0054) | (0.0283) | (0.0112) | (0.0114) | (0.0115) | (0.9905) | (0.9892) | (0.8348) |
| Letters Terms (%) PC2 | -0.0065 | -0.0093 | 0.0387 | -0.0188 | -0.0183 | -0.0179 | 0.3431 | 0.2706 | 0.0530 |
| | (0.0105) | (0.0077) | (0.0371) | (0.0178) | (0.0179) | (0.0160) | (1.3495) | (1.3535) | (1.0765) |
| CV Length/100 | 0.0026 | 0.0010 | -0.0282 | 0.0046 | 0.0042 | 0.0044 | -0.5040 | -0.4877 | -0.3768 |
| | (0.0044) | (0.0022) | (0.0190) | (0.0086) | (0.0086) | (0.0075) | (0.6374) | (0.6328) | (0.4706) |
| CV Coding | -0.0031 | -0.0044* | -0.0414** | 0.0110 | 0.0108 | 0.0125 | -0.9448 | -0.9129 | -0.9246 |
| | (0.0055) | (0.0026) | (0.0193) | (0.0082) | (0.0081) | (0.0082) | (0.7454) | (0.7403) | (0.6139) |
| CV Publication/Working Paper | 0.0441* | 0.0154 | -0.0237 | 0.0451 | 0.0433 | 0.0165 | -5.5511 | -5.2459 | -2.1210 |
| | (0.0244) | (0.0173) | (0.0820) | (0.0460) | (0.0463) | (0.0427) | (3.5094) | (3.5215) | (3.0760) |
| Essay Length/100 | 0.0125* | 0.0041 | 0.0253 | -0.0001 | -0.0001 | -0.0090 | -0.2293 | -0.2579 | 0.8458 |
| | (0.0067) | (0.0073) | (0.0219) | (0.0115) | (0.0117) | (0.0103) | (0.8988) | (0.9022) | (0.7451) |
| Essay Specific Professor | 0.0275 | 0.0077 | 0.0974 | 0.0329 | 0.0271 | 0.0362 | -3.3108 | -2.8484 | -3.4539 |
| | (0.0208) | (0.0130) | (0.0724) | (0.0369) | (0.0368) | (0.0351) | (2.9044) | (2.8921) | (2.4207) |
| Essay Economics Terms (%) | -0.0012 | 0.0001 | 0.0366** | -0.0173** | -0.0175** | -0.0104 | 1.7613*** | 1.7965*** | 0.9962* |
| | (0.0039) | (0.0016) | (0.0162) | (0.0082) | (0.0081) | (0.0078) | (0.6048) | (0.6027) | (0.5134) |
| Essay Research Terms (%) | -0.0001 | -0.0009 | -0.0266 | 0.0107 | 0.0098 | 0.0019 | -1.7875** | -1.7733** | -0.4855 |
| | (0.0053) | (0.0031) | (0.0206) | (0.0097) | (0.0098) | (0.0085) | (0.7813) | (0.7808) | (0.6382) |
| Essay Positive Terms (%) | -0.0014 | -0.0013 | -0.0072 | 0.0111 | 0.0118* | 0.0123* | -0.4461 | -0.4785 | -0.3335 |
| | (0.0037) | (0.0019) | (0.0189) | (0.0069) | (0.0070) | (0.0068) | (0.5609) | (0.5541) | (0.4475) |
| Essay Negative Terms (%) | -0.0011 | -0.0016 | 0.0003 | -0.0059 | -0.0069 | -0.0060 | 0.3935 | 0.4548 | 0.4364 |
| | (0.0055) | (0.0029) | (0.0152) | (0.0102) | (0.0101) | (0.0094) | (0.7865) | (0.7837) | (0.6420) |
| Missing Committee Score | | -0.0096 | | | -0.0528** | -0.0277 | | 4.4451*** | 1.3124 |
| | | (0.0120) | | | (0.0212) | (0.0191) | | (1.6677) | (1.3892) |
| Missing Application Pdf | -0.0723*** | 0.0036 | -0.1054 | 0.0224 | 0.0410 | 0.0305 | -2.4877 | -4.0722 | -3.1533 |
| | (0.0227) | (0.0149) | (0.0858) | (0.0375) | (0.0379) | (0.0348) | (2.9090) | (2.9614) | (2.4962) |
| Open House | | | 0.2615*** | | | | | | |
| | | | (0.0512) | | | | | | |
| Program Rank | | | | | | -0.0011*** | | | 0.0832*** |
| | | | | | | (0.0003) | | | (0.0185) |
| Missing Program Rank | | | | | | -0.3965*** | | | 41.0729*** |
| | | | | | | (0.0512) | | | (2.6010) |
| Sample | All | All | Accept = 1 | All | All | All | All | All | All |
| Observations | 2329 | 2329 | 257 | 2329 | 2329 | 2329 | 2329 | 2329 | 2329 |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Accept ($p < .1$):** 2017 (-), 2018 (-), 2019 (-), Female (+), International Asian (-), GRE Quantitative Percentile (+), Work Experience (+), Committee Score (+), CV Coding (-)
- **Success ($p < .1$):** Winsorized Age (+), Undergraduate GPA (-), TOEFL/IELTS (+), Work Experience (+), Essay Positive Terms (+), Program Rank (-), Missing Program Rank (-)

▸▸ Acceptance, Matriculation, and Success AMEs

# A4: Economics Department Logistic Regression Results (Synthetic)

Table 6a: Economics Department Logistic Regression Results (Synthetic)

| Dependent Variable | Accept | Accept | Matriculate | T100 Placement | T100 Placement | T100 Placement | Placement Rank | Placement Rank | Placement Rank |
|---|---|---|---|---|---|---|---|---|---|
| 2016 | -0.2091 | -0.1311 | 0.1291 | 0.0997 | 0.0892 | 0.1020 | -0.8042 | -0.5831 | -1.3927 |
| | (0.2288) | (0.2577) | (0.6478) | (0.1812) | (0.1814) | (0.1970) | (4.5726) | (4.5702) | (4.1436) |
| 2017 | -0.3355 | -0.5461** | 0.6767 | 0.2827 | 0.2903 | 0.1565 | -12.4495*** | -12.2002*** | -6.3639 |
| | (0.2390) | (0.2627) | (0.6543) | (0.1761) | (0.1767) | (0.1925) | (4.4305) | (4.4347) | (3.9262) |
| 2018 | 0.2889 | -0.2993 | 0.2947 | 0.1627 | 0.2192 | 0.1272 | -9.3709** | -9.2465** | -6.0098 |
| | (0.2056) | (0.2212) | (0.5517) | (0.1806) | (0.1827) | (0.1972) | (4.4278) | (4.4749) | (3.9975) |
| 2019 | -0.1933 | -0.3208 | -0.0662 | 0.0116 | 0.0023 | -0.1488 | -5.3768 | -4.7378 | 1.0400 |
| | (0.2227) | (0.2490) | (0.6069) | (0.1927) | (0.1933) | (0.2061) | (4.4927) | (4.5026) | (3.9171) |
| Winsorized Age | -0.0508** | -0.0414* | 0.0065 | -0.0141 | -0.0150 | -0.0094 | 0.4466 | 0.4383 | 0.2496 |
| | (0.0220) | (0.0243) | (0.0600) | (0.0170) | (0.0170) | (0.0185) | (0.4147) | (0.4141) | (0.3766) |
| Female | -0.1826 | -0.1535 | 0.0691 | 0.1351 | 0.1379 | 0.2109* | -1.5097 | -1.6305 | -4.7144* |
| | (0.1446) | (0.1590) | (0.4006) | (0.1167) | (0.1168) | (0.1259) | (2.9357) | (2.9318) | (2.6789) |
| U.S. | 0.0002 | -0.0549 | 0.0998 | 0.0198 | 0.0241 | -0.0123 | 2.8625 | 2.5697 | 4.2450 |
| | (0.2240) | (0.2468) | (0.6663) | (0.1794) | (0.1801) | (0.1997) | (4.4691) | (4.4669) | (4.1868) |
| International Asian | -0.1230 | -0.0646 | -0.1205 | 0.2014 | 0.1972 | 0.1700 | -0.7475 | -0.9433 | 1.1783 |
| | (0.1688) | (0.1838) | (0.5028) | (0.1394) | (0.1392) | (0.1541) | (3.3754) | (3.3732) | (3.1231) |
| GRE Verbal Percentile | -0.0025 | -0.0001 | 0.0008 | 0.0004 | 0.0001 | -0.0021 | -0.0668 | -0.0623 | 0.0181 |
| | (0.0034) | (0.0036) | (0.0089) | (0.0026) | (0.0026) | (0.0029) | (0.0661) | (0.0663) | (0.0595) |
| GRE Quantitative Percentile | -0.0028 | -0.0005 | -0.0125 | 0.0145** | 0.0143** | -0.0003 | -0.5982*** | -0.5916*** | -0.0184 |
| | (0.0075) | (0.0080) | (0.0195) | (0.0065) | (0.0065) | (0.0069) | (0.1570) | (0.1569) | (0.1376) |
| GRE Analytic Percentile | 0.0041 | 0.0019 | -0.0010 | 0.0009 | 0.0011 | 0.0019 | 0.0181 | 0.0156 | -0.0008 |
| | (0.0031) | (0.0033) | (0.0074) | (0.0024) | (0.0024) | (0.0026) | (0.0587) | (0.0587) | (0.0529) |
| Undergraduate GPA | -0.4230 | -0.3157 | 0.9077 | 0.4075 | 0.3986 | 0.4627 | -2.7870 | -2.7400 | -5.8113 |
| | (0.3862) | (0.4536) | (1.8572) | (0.3984) | (0.3990) | (0.4326) | (9.4800) | (9.4624) | (8.8952) |
| TOEFL/IELTS | 0.2329 | 0.2776 | -0.2073 | -0.0682 | -0.0625 | -0.0142 | 2.4237 | 2.2404 | -0.4376 |
| | (0.2154) | (0.2202) | (0.4100) | (0.1509) | (0.1507) | (0.1670) | (3.7070) | (3.7021) | (3.1595) |
| Elite U.S. University | -0.0473 | -0.1157 | -0.2546 | 0.3145 | 0.3119 | 0.3760* | -6.7562 | -6.6352 | -7.1987 |
| | (0.2618) | (0.2943) | (0.7160) | (0.2054) | (0.2062) | (0.2230) | (5.3790) | (5.3875) | (4.8870) |
| Elite U.S. Liberal Arts College | -0.6642 | -0.6053 | | 0.0069 | -0.0044 | -0.0958 | -2.2643 | -2.2283 | 3.3486 |
| | (0.5448) | (0.5744) | | (0.3656) | (0.3659) | (0.3964) | (8.9098) | (8.8713) | (7.5578) |
| Elite Foreign University | -0.1797 | -0.1798 | 0.5156 | -0.0538 | -0.0692 | -0.0298 | -1.0302 | -0.9315 | -0.2765 |
| | (0.3900) | (0.4341) | (1.2147) | (0.3273) | (0.3294) | (0.3495) | (8.2453) | (8.2793) | (7.1254) |
| Other Foreign University | -0.0406 | -0.1089 | 0.6613 | 0.1055 | 0.1125 | 0.0046 | -6.0523 | -6.2650 | -0.6409 |
| | (0.2078) | (0.2262) | (0.5235) | (0.1688) | (0.1689) | (0.1840) | (4.2123) | (4.2076) | (3.8417) |
| Missing Undergraduate Institution | -0.0012 | 0.0377 | -0.6367 | 0.1037 | 0.0966 | 0.0297 | -5.9887 | -5.9936 | -2.3404 |
| | (0.2217) | (0.2413) | (0.5829) | (0.1844) | (0.1842) | (0.1977) | (4.5947) | (4.5865) | (4.1622) |
| Graduate Degree | -0.0846 | -0.1354 | -0.0100 | 0.1515 | 0.1611 | 0.1906 | -1.4756 | -1.6341 | -2.0725 |
| | (0.1458) | (0.1626) | (0.3834) | (0.1166) | (0.1165) | (0.1252) | (2.8643) | (2.8619) | (2.5933) |
| Work Experience | 0.2547* | 0.2149 | -0.2181 | 0.0750 | 0.0805 | 0.1443 | 0.4084 | 0.3542 | -2.7442 |
| | (0.1463) | (0.1588) | (0.3573) | (0.1161) | (0.1160) | (0.1250) | (2.8682) | (2.8651) | (2.6268) |
| #Professor Recommenders | 0.0194 | -0.0565 | 0.1806 | 0.0038 | 0.0074 | -0.0302 | -0.2230 | -0.1761 | 1.2584 |
| | (0.0950) | (0.1021) | (0.2725) | (0.0759) | (0.0761) | (0.0794) | (1.9087) | (1.9104) | (1.7006) |
| #Prolific Recommenders | -0.0388 | -0.0071 | 0.0168 | 0.0027 | -0.0000 | 0.0325 | 0.3327 | 0.3170 | -1.1079 |
| | (0.0671) | (0.0713) | (0.1685) | (0.0521) | (0.0522) | (0.0575) | (1.3245) | (1.3242) | (1.2261) |
| Missing GRE | 0.3002 | 0.0736 | | 0.3136 | 0.3314 | 0.2674 | -12.0035 | -11.9570 | -11.5860 |
| | (0.3757) | (0.3872) | | (0.2858) | (0.2857) | (0.3297) | (8.1036) | (8.0998) | (8.0641) |
| Missing Undergraduate GPA | 0.1997 | 0.1720 | 0.2910 | -0.1321 | -0.1290 | -0.0816 | 5.2305 | 5.1674 | 4.3351 |
| | (0.1863) | (0.2016) | (0.5443) | (0.1391) | (0.1389) | (0.1534) | (3.6167) | (3.6124) | (3.3094) |
| Missing TOEFL/IELTS | -0.1740 | -0.0792 | -0.3009 | 0.1948 | 0.1815 | 0.2286* | -1.5099 | -1.4121 | -3.1183 |
| | (0.1609) | (0.1741) | (0.4256) | (0.1280) | (0.1281) | (0.1380) | (3.1162) | (3.1192) | (2.7616) |
| Committee Score | | 0.0372 | -0.1004 | | 0.0465 | 0.0330 | | -1.6627 | -0.4906 |
| | | (0.0423) | (0.1060) | | (0.0505) | (0.0558) | | (1.1655) | (1.1645) |
| #Math Courses | 0.0632 | 0.0756 | -0.1683 | -0.1269 | -0.1305* | -0.1764** | 1.1826 | 1.2073 | 3.0509** |
| | (0.0687) | (0.0733) | (0.1601) | (0.0786) | (0.0780) | (0.0863) | (1.7287) | (1.7209) | (1.5010) |
| Advanced Math | 0.0222 | -0.0336 | 0.9873 | -0.1639 | -0.1630 | -0.2842 | 1.1644 | 1.2290 | 6.1671 |
| | (0.2590) | (0.2747) | (0.7209) | (0.3004) | (0.3013) | (0.3256) | (6.4828) | (6.4875) | (5.6351) |
| Letters Length/100 | -0.0254 | -0.0128 | -0.0575 | -0.1049*** | -0.1057*** | -0.1218*** | 1.6052** | 1.6308** | 2.1010*** |
| | (0.0333) | (0.0361) | (0.0885) | (0.0381) | (0.0376) | (0.0428) | (0.7873) | (0.7869) | (0.6934) |
| Letters Terms (%) PC1 | 0.0280 | -0.0073 | 0.0381 | 0.0574 | 0.0643 | 0.0566 | 1.1364 | 1.1816 | 1.0356 |
| | (0.0779) | (0.0821) | (0.2097) | (0.0828) | (0.0834) | (0.0930) | (1.8933) | (1.9000) | (1.6894) |
| Letters Terms (%) PC2 | -0.2328* | -0.2369* | -0.4921 | 0.0802 | 0.0718 | 0.1318 | -2.8589 | -2.7586 | -5.0669** |
| | (0.1226) | (0.1255) | (0.3028) | (0.1205) | (0.1211) | (0.1250) | (2.6915) | (2.7005) | (2.1872) |
| CV Length/100 | 0.0971** | 0.0795 | 0.1022 | -0.0358 | -0.0336 | -0.0095 | 2.4879** | 2.5172** | 0.5354 |
| | (0.0492) | (0.0508) | (0.1340) | (0.0548) | (0.0551) | (0.0641) | (1.2406) | (1.2458) | (1.0685) |
| CV Coding | -0.0198 | -0.0473 | -0.0834 | 0.0688 | 0.0720* | 0.0795* | -1.6529 | -1.6778 | -1.0250 |
| | (0.0411) | (0.0448) | (0.1069) | (0.0434) | (0.0434) | (0.0474) | (1.0435) | (1.0451) | (0.8904) |
| CV Publication/Working Paper | 0.3079 | 0.3005 | 1.3491* | 0.1111 | 0.1183 | 0.1111 | 2.7524 | 2.4246 | 5.5955 |
| | (0.2852) | (0.2962) | (0.7475) | (0.3264) | (0.3272) | (0.3482) | (7.3505) | (7.3749) | (6.2211) |
| Essay Length/100 | 0.0679 | 0.0790 | -0.0558 | -0.0667 | -0.0636 | -0.0694 | 0.5523 | 0.4398 | 0.3231 |
| | (0.0770) | (0.0831) | (0.1489) | (0.0780) | (0.0794) | (0.0871) | (1.7254) | (1.7321) | (1.3973) |
| Essay Specific Professor | -0.1756 | -0.2494 | -0.1115 | -0.0547 | -0.0524 | 0.0009 | 1.2639 | 1.2219 | -3.1507 |
| | (0.2564) | (0.2714) | (0.6302) | (0.2774) | (0.2798) | (0.3022) | (6.3588) | (6.3580) | (5.5747) |
| Essay Economics Terms (%) | -0.0123 | -0.0094 | -0.0406 | -0.0091 | -0.0082 | -0.0222 | 0.2276 | 0.2960 | 0.6769 |
| | (0.0542) | (0.0572) | (0.1334) | (0.0539) | (0.0547) | (0.0624) | (1.3179) | (1.3189) | (1.1799) |
| Essay Research Terms (%) | 0.0303 | 0.0066 | -0.0464 | 0.1042 | 0.1084 | 0.1176 | -1.8679 | -1.9069 | -1.8170 |
| | (0.0656) | (0.0702) | (0.1578) | (0.0703) | (0.0705) | (0.0723) | (1.5218) | (1.5245) | (1.2596) |
| Essay Positive Terms (%) | 0.0180 | 0.0074 | 0.3699*** | 0.0871* | 0.0868* | 0.0947 | -1.5542 | -1.4775 | -1.1629 |
| | (0.0474) | (0.0484) | (0.1213) | (0.0511) | (0.0517) | (0.0609) | (1.1518) | (1.1568) | (1.1006) |
| Essay Negative Terms (%) | 0.0333 | 0.0312 | -0.0731 | -0.0304 | -0.0280 | -0.0168 | 0.5011 | 0.4716 | 0.2911 |
| | (0.0314) | (0.0351) | (0.0769) | (0.0359) | (0.0359) | (0.0382) | (0.8608) | (0.8614) | (0.7100) |
| Missing Committee Score | | -4.3141*** | | | 0.2635** | 0.4129*** | | -0.9440 | -6.1957** |
| | | (0.4312) | | | (0.1267) | (0.1388) | | (3.0258) | (2.8364) |
| Missing Application Pdf | -0.9321*** | -0.0588 | 0.9473 | 0.0574 | -0.0560 | -0.1100 | 4.5136 | 4.8399 | 5.6614 |
| | (0.2534) | (0.2753) | (0.6793) | (0.2628) | (0.2678) | (0.2943) | (5.9238) | (6.0503) | (5.4160) |
| Open House | | | 1.0643 | | | | | | |
| | | | (0.8268) | | | | | | |
| Program Rank | | | | | | -0.0049** | | | 0.1176*** |
| | | | | | | (0.0019) | | | (0.0386) |
| Missing Program Rank | | | | | | -2.5925*** | | | 94.5255*** |
| | | | | | | (0.3273) | | | (5.6183) |
| Sample | All | All | Accept = 1 | All | All | All | All | All | All |
| Observations | 2329 | 2329 | 237 | 2329 | 2329 | 2329 | 2329 | 2329 | 2329 |
| Pseudo-$R^2$ | 0.0552 | 0.2804 | 0.1402 | 0.0221 | 0.0245 | 0.1935 | 0.0040 | 0.0042 | 0.0953 |
| CV Pseudo-$R^2$ | 0.0028 | 0.2137 | -0.6295 | -0.0225 | -0.0212 | 0.1432 | -0.0033 | -0.0033 | 0.0890 |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Accept ($p < .1$):** 2017 (-), Winsorized Age (-), Letters Terms (%) PC2, Missing Committee Score (-)
- **Success ($p < .1$):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

Acceptance, Matriculation, and Success AMEs

# A4: Economics Department Logistic Regression AMEs (Synthetic)

Table 6b: Economics Department Logistic Regression AMEs (Synthetic)

| Dependent Variable | Accept | Accept | Matriculate | T100 Placement | T100 Placement | T100 Placement | Placement Rank | Placement Rank | Placement Rank |
|---|---|---|---|---|---|---|---|---|---|
| 2016 | -0.0185 | -0.0100 | 0.0201 | 0.0136 | 0.0122 | 0.0120 | -0.3764 | -0.2729 | -0.6112 |
| | (0.0203) | (0.0197) | (0.1005) | (0.0248) | (0.0248) | (0.0233) | (2.1401) | (2.1390) | (1.8189) |
| 2017 | -0.0297 | -0.0417** | 0.1052 | 0.0387 | 0.0396 | 0.0185 | -5.8265*** | -5.7099*** | -2.7929 |
| | (0.0212) | (0.0199) | (0.1005) | (0.0241) | (0.0241) | (0.0227) | (2.0748) | (2.0770) | (1.7228) |
| 2018 | 0.0256 | -0.0229 | 0.0458 | 0.0223 | 0.0299 | 0.0150 | -4.3857** | -4.3275** | -2.6375 |
| | (0.0182) | (0.0168) | (0.0856) | (0.0247) | (0.0250) | (0.0233) | (2.0732) | (2.0958) | (1.7555) |
| 2019 | -0.0171 | -0.0245 | -0.0103 | 0.0016 | 0.0003 | -0.0176 | -2.5164 | -2.2174 | 0.4564 |
| | (0.0197) | (0.0189) | (0.0944) | (0.0264) | (0.0264) | (0.0243) | (2.1013) | (2.1061) | (1.7192) |
| Winsorized Age | -0.0045** | -0.0032* | 0.0010 | -0.0019 | -0.0020 | -0.0011 | 0.2090 | 0.2051 | 0.1095 |
| | (0.0020) | (0.0019) | (0.0093) | (0.0023) | (0.0023) | (0.0022) | (0.1941) | (0.1938) | (0.1654) |
| Female | -0.0162 | -0.0117 | 0.0107 | 0.0185 | 0.0188 | 0.0249* | -0.7066 | -0.7631 | -2.0690* |
| | (0.0128) | (0.0121) | (0.0621) | (0.0160) | (0.0159) | (0.0149) | (1.3745) | (1.3728) | (1.1767) |
| U.S. | 0.0000 | -0.0042 | 0.0155 | 0.0027 | 0.0033 | -0.0015 | 1.3397 | 1.2026 | 1.8630 |
| | (0.0198) | (0.0189) | (0.1035) | (0.0246) | (0.0246) | (0.0236) | (2.0914) | (2.0905) | (1.8379) |
| International Asian | -0.0109 | -0.0049 | -0.0187 | 0.0276 | 0.0269 | 0.0201 | -0.3498 | -0.4415 | 0.5171 |
| | (0.0150) | (0.0140) | (0.0782) | (0.0191) | (0.0190) | (0.0182) | (1.5798) | (1.5789) | (1.3706) |
| GRE Verbal Percentile | -0.0002 | -0.0000 | 0.0001 | 0.0001 | 0.0000 | -0.0002 | -0.0313 | -0.0291 | 0.0080 |
| | (0.0003) | (0.0003) | (0.0014) | (0.0004) | (0.0004) | (0.0003) | (0.0309) | (0.0310) | (0.0261) |
| GRE Quantitative Percentile | -0.0002 | -0.0000 | -0.0019 | 0.0020** | 0.0020** | -0.0000 | -0.2800*** | -0.2769*** | -0.0081 |
| | (0.0007) | (0.0006) | (0.0030) | (0.0009) | (0.0009) | (0.0008) | (0.0734) | (0.0734) | (0.0604) |
| GRE Analytic Percentile | 0.0004 | 0.0001 | -0.0002 | 0.0001 | 0.0002 | 0.0002 | 0.0085 | 0.0073 | -0.0004 |
| | (0.0003) | (0.0003) | (0.0011) | (0.0003) | (0.0003) | (0.0003) | (0.0275) | (0.0275) | (0.0232) |
| Undergraduate GPA | -0.0375 | -0.0241 | 0.1411 | 0.0558 | 0.0544 | 0.0546 | -1.3043 | -1.2824 | -2.5504 |
| | (0.0342) | (0.0346) | (0.2875) | (0.0545) | (0.0545) | (0.0510) | (4.4373) | (4.4292) | (3.9043) |
| TOEFL/IELTS | 0.0206 | 0.0212 | -0.0322 | -0.0093 | -0.0085 | -0.0017 | 1.1343 | 1.0486 | -0.1921 |
| | (0.0191) | (0.0168) | (0.0637) | (0.0207) | (0.0206) | (0.0197) | (1.7354) | (1.7331) | (1.3866) |
| Elite U.S. University | -0.0042 | -0.0088 | -0.0396 | 0.0431 | 0.0426 | 0.0444* | -3.1620 | -3.1054 | -3.1593 |
| | (0.0232) | (0.0225) | (0.1109) | (0.0281) | (0.0282) | (0.0263) | (2.5179) | (2.5220) | (2.1436) |
| Elite U.S. Liberal Arts College | -0.0588 | -0.0462 | | 0.0009 | -0.0006 | -0.0113 | -1.0597 | -1.0429 | 1.4696 |
| | (0.0483) | (0.0439) | | (0.0500) | (0.0500) | (0.0468) | (4.1696) | (4.1517) | (3.3177) |
| Elite Foreign University | -0.0159 | -0.0137 | 0.0801 | -0.0074 | -0.0095 | -0.0035 | -0.4822 | -0.4360 | -0.1214 |
| | (0.0346) | (0.0332) | (0.1891) | (0.0448) | (0.0450) | (0.0412) | (3.8590) | (3.8749) | (3.1271) |
| Other Foreign University | -0.0036 | -0.0083 | 0.1028 | 0.0144 | 0.0154 | 0.0005 | -2.8325 | -2.9321 | -0.2813 |
| | (0.0184) | (0.0173) | (0.0813) | (0.0231) | (0.0231) | (0.0217) | (1.9702) | (1.9680) | (1.6858) |
| Missing Undergraduate Institution | -0.0001 | 0.0029 | -0.0989 | 0.0142 | 0.0132 | 0.0035 | -2.8028 | -2.8051 | -1.0271 |
| | (0.0196) | (0.0184) | (0.0902) | (0.0252) | (0.0251) | (0.0233) | (2.1499) | (2.1460) | (1.8264) |
| Graduate Degree | -0.0075 | -0.0103 | -0.0015 | 0.0207 | 0.0220 | 0.0225 | -0.6906 | -0.7648 | -0.9095 |
| | (0.0129) | (0.0124) | (0.0596) | (0.0160) | (0.0159) | (0.0147) | (1.3406) | (1.3395) | (1.1381) |
| Work Experience | 0.0226* | 0.0164 | -0.0339 | 0.0103 | 0.0110 | 0.0170 | 0.1911 | 0.1658 | -1.2044 |
| | (0.0129) | (0.0121) | (0.0551) | (0.0159) | (0.0158) | (0.0147) | (1.3422) | (1.3408) | (1.1533) |
| #Professor Recommenders | 0.0017 | -0.0043 | 0.0281 | 0.0005 | 0.0010 | -0.0036 | -0.1044 | -0.0824 | 0.5523 |
| | (0.0084) | (0.0078) | (0.0423) | (0.0104) | (0.0104) | (0.0094) | (0.8933) | (0.8941) | (0.7465) |
| #Prolific Recommenders | -0.0034 | -0.0005 | 0.0026 | 0.0004 | -0.0000 | 0.0038 | 0.1557 | 0.1484 | -0.4862 |
| | (0.0059) | (0.0054) | (0.0262) | (0.0071) | (0.0071) | (0.0068) | (0.6197) | (0.6196) | (0.5384) |
| Missing GRE | 0.0266 | 0.0056 | | 0.0429 | 0.0453 | 0.0316 | -5.6177 | -5.5961 | -5.0848 |
| | (0.0333) | (0.0296) | | (0.0391) | (0.0390) | (0.0389) | (3.7950) | (3.7934) | (3.5412) |
| Missing Undergraduate GPA | 0.0177 | 0.0131 | 0.0452 | -0.0181 | -0.0176 | -0.0096 | 2.4479 | 2.4184 | 1.9026 |
| | (0.0165) | (0.0154) | (0.0847) | (0.0190) | (0.0190) | (0.0181) | (1.6942) | (1.6921) | (1.4526) |
| Missing TOEFL/IELTS | -0.0154 | -0.0060 | -0.0468 | 0.0267 | 0.0248 | 0.0270* | -0.7067 | -0.6609 | -1.3685 |
| | (0.0143) | (0.0133) | (0.0659) | (0.0175) | (0.0175) | (0.0162) | (1.4590) | (1.4604) | (1.2133) |
| Committee Score | | 0.0028 | -0.0156 | | 0.0064 | 0.0039 | | -0.7782 | -0.2153 |
| | | (0.0032) | (0.0165) | | (0.0069) | (0.0066) | | (0.5455) | (0.5111) |
| #Math Courses | 0.0056 | 0.0058 | -0.0262 | -0.0174 | -0.0178* | -0.0208** | 0.5535 | 0.5650 | 1.3390** |
| | (0.0061) | (0.0056) | (0.0248) | (0.0107) | (0.0106) | (0.0101) | (0.8095) | (0.8059) | (0.6591) |
| Advanced Math | 0.0020 | -0.0026 | 0.1534 | -0.0224 | -0.0223 | -0.0335 | 0.5450 | 0.5752 | 2.7066 |
| | (0.0229) | (0.0210) | (0.1122) | (0.0411) | (0.0411) | (0.0384) | (3.0339) | (3.0362) | (2.4720) |
| Letters Length/100 | -0.0022 | -0.0010 | -0.0089 | -0.0144*** | -0.0144*** | -0.0144*** | 0.7512** | 0.7632** | 0.9221*** |
| | (0.0029) | (0.0028) | (0.0137) | (0.0052) | (0.0051) | (0.0050) | (0.3694) | (0.3693) | (0.3050) |
| Letters Terms (%) PC1 | 0.0025 | -0.0006 | 0.0059 | 0.0079 | 0.0088 | 0.0067 | 0.5318 | 0.5530 | 0.4545 |
| | (0.0069) | (0.0063) | (0.0325) | (0.0113) | (0.0114) | (0.0110) | (0.8858) | (0.8889) | (0.7413) |
| Letters Terms (%) PC2 | -0.0206* | -0.0181* | -0.0765* | 0.0110 | 0.0098 | 0.0156 | -1.3380 | -1.2911 | -2.2237** |
| | (0.0108) | (0.0095) | (0.0462) | (0.0165) | (0.0165) | (0.0147) | (1.2609) | (1.2651) | (0.9607) |
| CV Length/100 | 0.0086** | 0.0061 | 0.0159 | -0.0049 | -0.0046 | -0.0011 | 1.1643** | 1.1781** | 0.2350 |
| | (0.0043) | (0.0039) | (0.0210) | (0.0075) | (0.0075) | (0.0076) | (0.5802) | (0.5825) | (0.4688) |
| CV Coding | -0.0018 | -0.0036 | -0.0130 | 0.0094 | 0.0098* | 0.0094* | -0.7736 | -0.7852 | -0.4499 |
| | (0.0036) | (0.0034) | (0.0167) | (0.0059) | (0.0059) | (0.0056) | (0.4887) | (0.4894) | (0.3909) |
| CV Publication/Working Paper | 0.0273 | 0.0230 | 0.2097* | 0.0152 | 0.0162 | 0.0131 | 1.2881 | 1.1348 | 2.4557 |
| | (0.0252) | (0.0226) | (0.1128) | (0.0447) | (0.0447) | (0.0411) | (3.4400) | (3.4515) | (2.7307) |
| Essay Length/100 | 0.0060 | 0.0060 | -0.0087 | -0.0091 | -0.0087 | -0.0082 | 0.2585 | 0.2058 | 0.1418 |
| | (0.0068) | (0.0063) | (0.0231) | (0.0107) | (0.0108) | (0.0103) | (0.8075) | (0.8107) | (0.6133) |
| Essay Specific Professor | -0.0156 | -0.0190 | -0.0173 | -0.0075 | -0.0072 | 0.0001 | 0.5915 | 0.5719 | -1.3828 |
| | (0.0227) | (0.0207) | (0.0978) | (0.0380) | (0.0382) | (0.0357) | (2.9760) | (2.9757) | (2.4463) |
| Essay Economics Terms (%) | -0.0011 | -0.0007 | -0.0063 | -0.0012 | -0.0011 | -0.0026 | 0.1065 | 0.1385 | 0.2971 |
| | (0.0048) | (0.0044) | (0.0207) | (0.0074) | (0.0075) | (0.0074) | (0.6168) | (0.6173) | (0.5179) |
| Essay Research Terms (%) | 0.0027 | 0.0005 | -0.0072 | 0.0143 | 0.0148 | 0.0139 | -0.8742 | -0.8925 | -0.7975 |
| | (0.0058) | (0.0054) | (0.0245) | (0.0096) | (0.0096) | (0.0085) | (0.7127) | (0.7141) | (0.5530) |
| Essay Positive Terms (%) | 0.0016 | 0.0006 | 0.0575*** | 0.0119* | 0.0118* | 0.0112 | -0.7274 | -0.6915 | -0.5104 |
| | (0.0042) | (0.0037) | (0.0182) | (0.0070) | (0.0071) | (0.0072) | (0.5394) | (0.5418) | (0.4832) |
| Essay Negative Terms (%) | 0.0030 | 0.0024 | -0.0114 | -0.0042 | -0.0038 | -0.0020 | 0.2345 | 0.2207 | 0.1278 |
| | (0.0028) | (0.0027) | (0.0120) | (0.0049) | (0.0049) | (0.0045) | (0.4029) | (0.4032) | (0.3117) |
| Missing Committee Score | | -0.3296*** | | | 0.0360** | 0.0487*** | | -0.4418 | -2.7191** |
| | | (0.0330) | | | (0.0173) | (0.0162) | | (1.4166) | (1.2447) |
| Missing Application Pdf | -0.0826*** | -0.0045 | 0.1472 | 0.0079 | -0.0076 | -0.0130 | 2.1124 | 2.2652 | 2.4846 |
| | (0.0225) | (0.0210) | (0.1064) | (0.0360) | (0.0366) | (0.0347) | (2.7720) | (2.8317) | (2.3765) |
| Open House | | | 0.1654 | | | | | | |
| | | | (0.1278) | | | | | | |
| Program Rank | | | | | | -0.0006** | | | 0.0516*** |
| | | | | | | (0.0002) | | | (0.0169) |
| Missing Program Rank | | | | | | -0.3060*** | | | 41.4849*** |
| | | | | | | (0.0381) | | | (2.4608) |
| Sample | All | All | Accept = 1 | All | All | All | All | All | All |
| Observations | 2329 | 2329 | 237 | 2329 | 2329 | 2329 | 2329 | 2329 | 2329 |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Accept ($p < .1$):** 2017 (-), Winsorized Age (-), Letters Terms (%) PC2, Missing Committee Score (-)
- **Success ($p < .1$):** Female (+), Elite U.S. University (+), Missing TOEFL/IELTS (+), #Math Courses (-), Letters Length/100 (-), CV Coding (+), Missing Committee Score (+), Program Rank (-), Missing Program Rank (-)

Acceptance, Matriculation, and Success AMEs

# A4: Economics Department Dynamic Estimates

## Table 7: Economics Department Dynamic Estimates

### (a) Unrestricted

| | Committee Score | Transition | | Accept | | Success | |
|---|---|---|---|---|---|---|---|
| | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -2.7966 | -23.0602*** | -3.4656*** | -28.5276*** | -1.2765*** | -3.7142 | -0.5347 |
| | (2.1751) | (2.0492) | (0.2371) | (6.9081) | (0.2933) | (3.3294) | (0.4783) |
| 2016 | 0.0915 | -0.2896 | -0.0435 | -0.5294 | -0.0237 | 0.5121 | 0.0737 |
| | (0.1856) | (0.1794) | (0.0268) | (0.5264) | (0.0230) | (0.3233) | (0.0460) |
| 2017 | 0.2354 | -0.1533 | -0.0230 | -1.3366** | -0.0598** | 0.6368* | 0.0917** |
| | (0.2130) | (0.1726) | (0.0260) | (0.5312) | (0.0235) | (0.3294) | (0.0466) |
| 2018 | 0.5999*** | 1.5429*** | 0.2319*** | -3.6854*** | -0.1649*** | 0.0600 | 0.0086 |
| | (0.1777) | (0.1858) | (0.0251) | (0.6281) | (0.0209) | (0.3133) | (0.0451) |
| 2019 | 0.6118*** | 0.2905 | 0.0437 | -3.0765*** | -0.1377*** | 0.3498 | 0.0504 |
| | (0.1907) | (0.1833) | (0.0273) | (0.6899) | (0.0257) | (0.3214) | (0.0459) |
| Winsorized Age | -0.0621*** | 0.0628*** | 0.0094*** | -0.0133 | -0.0006 | 0.0799** | 0.0115** |
| | (0.0235) | (0.0218) | (0.0032) | (0.0612) | (0.0027) | (0.0351) | (0.0050) |
| Female | -0.0543 | 0.3919*** | 0.0589*** | 0.8909** | 0.0399*** | 0.1872 | 0.0270 |
| | (0.1125) | (0.1109) | (0.0164) | (0.3672) | (0.0154) | (0.1856) | (0.0267) |
| U.S. | -0.5071** | -0.1124 | -0.0169 | 0.0293 | 0.0013 | 0.6453* | 0.0929* |
| | (0.2043) | (0.1894) | (0.0285) | (0.4978) | (0.0223) | (0.3470) | (0.0494) |
| International Asian | -0.7530*** | 0.0087 | 0.0013 | -1.1484*** | -0.0514*** | -0.0233 | -0.0033 |
| | (0.1452) | (0.1396) | (0.0210) | (0.4133) | (0.0172) | (0.2426) | (0.0349) |
| GRE Verbal Percentile | 0.0066* | 0.0326*** | 0.0049*** | 0.0039 | 0.0002 | -0.0005 | -0.0001 |
| | (0.0036) | (0.0034) | (0.0005) | (0.0131) | (0.0006) | (0.0062) | (0.0009) |
| GRE Quantitative Percentile | 0.0447*** | 0.1829*** | 0.0275*** | 0.0402 | 0.0018 | -0.0064 | -0.0009 |
| | (0.0111) | (0.0163) | (0.0018) | (0.0326) | (0.0015) | (0.0238) | (0.0034) |
| GRE Analytic Percentile | 0.0029 | 0.0152*** | 0.0023*** | 0.0078 | 0.0004 | -0.0064 | -0.0009 |
| | (0.0027) | (0.0026) | (0.0004) | (0.0090) | (0.0004) | (0.0043) | (0.0006) |
| TOEFL/IELTS | 0.4407*** | -0.0097 | -0.0015 | -0.1286 | -0.0058 | 0.3738 | 0.0538 |
| | (0.1696) | (0.1632) | (0.0245) | (0.5621) | (0.0250) | (0.2358) | (0.0338) |
| Elite U.S. University | 0.5507*** | 0.9066*** | 0.1363*** | -0.5317 | -0.0238 | 0.1629 | 0.0234 |
| | (0.1994) | (0.2103) | (0.0312) | (0.6438) | (0.0293) | (0.3360) | (0.0484) |
| Elite U.S. Liberal Arts College | -0.0711 | 0.3268 | 0.0491 | 0.3253 | 0.0146 | 0.1414 | 0.0204 |
| | (0.3683) | (0.3402) | (0.0510) | (0.8010) | (0.0356) | (0.5521) | (0.0795) |
| Elite Foreign University | 0.2902 | 0.0518 | 0.0078 | -0.9107 | -0.0408 | 0.2454 | 0.0353 |
| | (0.2991) | (0.2959) | (0.0445) | (0.9288) | (0.0424) | (0.4927) | (0.0709) |
| Other Foreign University | 0.4797** | 0.1161 | 0.0174 | 0.2293 | 0.0103 | 0.2550 | 0.0367 |
| | (0.2022) | (0.1969) | (0.0296) | (0.6888) | (0.0306) | (0.3455) | (0.0497) |
| Missing Undergraduate Institution | 0.5097** | -0.4337** | -0.0652** | 0.0505 | 0.0023 | 0.1146 | 0.0165 |
| | (0.2157) | (0.1996) | (0.0300) | (0.6688) | (0.0299) | (0.3847) | (0.0554) |
| Graduate Degree | -0.0757 | 1.0229*** | 0.1537*** | -0.3419 | -0.0153 | -0.3618* | -0.0521* |
| | (0.1279) | (0.1262) | (0.0177) | (0.3883) | (0.0171) | (0.2120) | (0.0302) |
| Work Experience | 0.3458** | 0.7870*** | 0.1183*** | 0.5914* | 0.0265* | 0.0458 | 0.0066 |
| | (0.1345) | (0.1256) | (0.0179) | (0.3412) | (0.0154) | (0.2252) | (0.0324) |
| #Professor Recommenders | 0.0618 | -0.0211 | -0.0032 | -0.1896 | -0.0085 | -0.0215 | -0.0031 |
| | (0.0769) | (0.0839) | (0.0126) | (0.2574) | (0.0113) | (0.1331) | (0.0192) |
| #Prolific Recommenders | 0.1667*** | 0.1325*** | 0.0199*** | 0.1903 | 0.0085 | 0.0961 | 0.0138 |
| | (0.0548) | (0.0513) | (0.0077) | (0.1757) | (0.0077) | (0.0843) | (0.0121) |
| Missing GRE | -1.6087* | -1.8142*** | -0.2727*** | 1.6533 | 0.0740 | -0.3074 | -0.0443 |
| | (0.9422) | (0.5268) | (0.0772) | (1.8091) | (0.0811) | (0.9426) | (0.1357) |
| Missing TOEFL/IELTS | 0.3199** | 0.0947 | 0.0142 | 0.4511 | 0.0202 | -0.2725 | -0.0392 |
| | (0.1319) | (0.1295) | (0.0194) | (0.4238) | (0.0189) | (0.2221) | (0.0317) |
| Committee Score | | | | 3.7089*** | 0.1660*** | 0.0176 | 0.0025 |
| | | | | (0.3984) | (0.0065) | (0.0510) | (0.0073) |
| #Math Courses | -0.0790 | | | -0.2494 | -0.0112 | -0.0676 | -0.0097 |
| | (0.0519) | | | (0.2386) | (0.0108) | (0.1076) | (0.0155) |
| Advanced Math | 0.2979 | | | 1.1003 | 0.0492 | -0.1678 | -0.0242 |
| | (0.1889) | | | (0.7916) | (0.0347) | (0.3465) | (0.0498) |
| Letters Length/100 | 0.0858*** | | | 0.0770 | 0.0034 | -0.0252 | -0.0036 |
| | (0.0245) | | | (0.1080) | (0.0048) | (0.0431) | (0.0062) |
| Letters Terms (%) PC1 | 0.1027* | | | 0.2451 | 0.0110 | 0.1496 | 0.0215 |
| | (0.0551) | | | (0.2489) | (0.0112) | (0.1082) | (0.0155) |
| Letters Terms (%) PC2 | -0.0208 | | | -0.3767 | -0.0169 | -0.0395 | -0.0057 |
| | (0.0823) | | | (0.3450) | (0.0153) | (0.1470) | (0.0212) |
| CV Length/100 | 0.0513 | | | 0.0391 | 0.0017 | 0.0778 | 0.0112 |
| | (0.0329) | | | (0.1060) | (0.0047) | (0.0700) | (0.0100) |
| CV Coding | 0.0381 | | | -0.2228* | -0.0100* | 0.0863 | 0.0124 |
| | (0.0424) | | | (0.1301) | (0.0060) | (0.0788) | (0.0113) |
| CV Publication/Working Paper | 0.5482*** | | | 0.7611 | 0.0341 | 0.1054 | 0.0152 |
| | (0.1912) | | | (0.8128) | (0.0364) | (0.3992) | (0.0575) |
| Essay Length/100 | 0.0927 | | | 0.2047 | 0.0092 | -0.0987 | -0.0142 |
| | (0.0597) | | | (0.3483) | (0.0157) | (0.0948) | (0.0136) |
| Essay Specific Professor | 0.1411 | | | 0.4125 | 0.0185 | 0.0869 | 0.0125 |
| | (0.1744) | | | (0.6138) | (0.0275) | (0.3148) | (0.0453) |
| Essay Economics Terms (%) | 0.0287 | | | 0.0154 | 0.0007 | -0.1091 | -0.0157 |
| | (0.0312) | | | (0.0814) | (0.0036) | (0.0736) | (0.0105) |
| Essay Research Terms (%) | 0.0396 | | | -0.0751 | -0.0034 | 0.0033 | 0.0005 |
| | (0.0426) | | | (0.1492) | (0.0067) | (0.0771) | (0.0111) |
| Essay Positive Terms (%) | 0.0171 | | | -0.0689 | -0.0031 | 0.0948 | 0.0137 |
| | (0.0345) | | | (0.0968) | (0.0043) | (0.0589) | (0.0084) |
| Essay Negative Terms (%) | 0.0441 | | | -0.0863 | -0.0039 | -0.1146 | -0.0165 |
| | (0.0443) | | | (0.1368) | (0.0061) | (0.0779) | (0.0111) |
| Missing Application Pdf | -0.5376*** | | | 0.2614 | 0.0117 | 0.3292 | 0.0474 |
| | (0.1841) | | | (0.7005) | (0.0311) | (0.3321) | (0.0477) |
| Program Rank | | | | | | -0.0093*** | -0.0013*** |
| | | | | | | (0.0033) | (0.0005) |
| Missing Program Rank | | | | | | -3.3225*** | -0.4783*** |
| | | | | | | (0.6860) | (0.0991) |
| $\sqrt{\text{MSE}}$ | 1.6467*** | | | | | | |
| | (0.0370) | | | | | | |
| Obs \| Param \| LL \| AIC | 2329 | 581 | -13672.6 | 28507.3 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

### (b) Restricted

| | Committee Score | Cutoff 1 | $L(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success | Sigma |
|---|---|---|---|---|---|---|---|
| Intercept | -5.6290** | -1.1004*** | 0.3327 | -0.7250*** | 32.63% | -0.8104** | |
| | (2.3609) | (0.1469) | (0.2495, 0.4437) | (0.2165) | (24.06%, 42.54%) | (0.4033) | |
| 2016 | -0.0236 | 0.1593 | 0.3902 | 0.2502 | 38.35% | 0.2397 | |
| | (0.1869) | (0.2367) | (0.2622, 0.5807) | (0.3792) | (24.57%, 54.29%) | (0.3786) | |
| 2017 | 0.4614** | 0.3272** | 0.4615 | 0.5346** | 45.25% | 0.5202* | |
| | (0.2164) | (0.1661) | (0.3928, 0.5423) | (0.2664) | (38.04%, 52.68%) | (0.2671) | |
| 2018 | 0.9169*** | 0.0427 | 0.3473 | 0.0677 | 34.14% | 0.0247 | |
| | (0.1990) | (0.1826) | (0.2828, 0.4263) | (0.2765) | (27.30%, 41.70%) | (0.2854) | |
| 2019 | 0.5987*** | 0.2264 | 0.4173 | 0.3636 | 41.06% | 0.3243 | |
| | (0.1972) | (0.1689) | (0.3429, 0.5078) | (0.2663) | (33.39%, 49.19%) | (0.2681) | |
| Winsorized Age | -0.0459* | | | | | -0.0004 | |
| | (0.0247) | | | | | (0.0009) | |
| Female | -0.1023 | | | | | 0.0103 | |
| | (0.1052) | | | | | (0.0091) | |
| U.S. | -0.3174 | | | | | 0.0033 | |
| | (0.2109) | | | | | (0.0076) | |
| International Asian | -0.3625** | | | | | -0.0044 | |
| | (0.1586) | | | | | (0.0064) | |
| GRE Verbal Percentile | 0.0110*** | | | | | 0.0002 | |
| | (0.0036) | | | | | (0.0002) | |
| GRE Quantitative Percentile | 0.0578*** | | | | | 0.0006 | |
| | (0.0108) | | | | | (0.0007) | |
| GRE Analytic Percentile | 0.0026 | | | | | 0.0001 | |
| | (0.0028) | | | | | (0.0001) | |
| TOEFL/IELTS | 0.3186* | | | | | 0.0003 | |
| | (0.1714) | | | | | (0.0067) | |
| Elite U.S. University | 0.5964*** | | | | | -0.0097 | |
| | (0.1777) | | | | | (0.0105) | |
| Elite U.S. Liberal Arts College | -0.4482 | | | | | 0.0038 | |
| | (0.3240) | | | | | (0.0137) | |
| Elite Foreign University | 0.2282 | | | | | -0.0076 | |
| | (0.2741) | | | | | (0.0126) | |
| Other Foreign University | 0.4355** | | | | | 0.0043 | |
| | (0.1947) | | | | | (0.0101) | |
| Missing Undergraduate Institution | 0.3910* | | | | | -0.0001 | |
| | (0.2050) | | | | | (0.0089) | |
| Graduate Degree | -0.0558 | | | | | -0.0018 | |
| | (0.1512) | | | | | (0.0058) | |
| Work Experience | 0.0776 | | | | | 0.0077 | |
| | (0.1419) | | | | | (0.0073) | |
| #Professor Recommenders | 0.1021 | | | | | -0.0002 | |
| | (0.0784) | | | | | (0.0035) | |
| #Prolific Recommenders | 0.0557 | | | | | -0.0005 | |
| | (0.0601) | | | | | (0.0024) | |
| Missing GRE | -1.2131 | | | | | 0.0197 | |
| | (0.9076) | | | | | (0.0219) | |
| Missing TOEFL/IELTS | 0.2158* | | | | | 0.0029 | |
| | (0.1255) | | | | | (0.0054) | |
| Committee Score | | | | | | 0.0462 | |
| | | | | | | (0.0359) | |
| #Math Courses | -0.1437 | | | | | -0.0026 | |
| | (0.0926) | | | | | (0.0044) | |
| Advanced Math | 0.4116 | | | | | -0.0019 | |
| | (0.3021) | | | | | (0.0100) | |
| Letters Length/100 | 0.0902** | | | | | 0.0004 | |
| | (0.0439) | | | | | (0.0018) | |
| Letters Terms (%) PC1 | 0.2113** | | | | | 0.0049 | |
| | (0.0952) | | | | | (0.0054) | |
| Letters Terms (%) PC2 | -0.1512 | | | | | -0.0038 | |
| | (0.1994) | | | | | (0.0060) | |
| CV Length/100 | 0.0276 | | | | | 0.0012 | |
| | (0.0720) | | | | | (0.0019) | |
| CV Coding | 0.0310 | | | | | -0.0041 | |
| | (0.0978) | | | | | (0.0041) | |
| CV Publication/Working Paper | 1.1626*** | | | | | 0.0011 | |
| | (0.4332) | | | | | (0.0123) | |
| Essay Length/100 | 0.2680** | | | | | 0.0072 | |
| | (0.1053) | | | | | (0.0086) | |
| Essay Specific Professor | 0.1400 | | | | | -0.0055 | |
| | (0.2233) | | | | | (0.0128) | |
| Essay Economics Terms (%) | 0.0664 | | | | | 0.0003 | |
| | (0.0551) | | | | | (0.0014) | |
| Essay Research Terms (%) | 0.1357* | | | | | -0.0002 | |
| | (0.0821) | | | | | (0.0028) | |
| Essay Positive Terms (%) | 0.0413 | | | | | -0.0004 | |
| | (0.0725) | | | | | (0.0027) | |
| Essay Negative Terms (%) | 0.1022 | | | | | 0.0009 | |
| | (0.0774) | | | | | (0.0032) | |
| Missing Application Pdf | -0.6972** | | | | | -0.0242 | |
| | (0.2751) | | | | | (0.0184) | |
| Program Rank | | | | | | -0.0087*** | |
| | | | | | | (0.0030) | |
| Missing Program Rank | | | | | | -3.1650*** | |
| | | | | | | (0.6793) | |
| $\sqrt{\text{MSE}}$ | 1.7301*** | | | | | | |
| | (0.0480) | | | | | | |
| Sigma | | | | | | | 0.0018 |
| | | | | | | | (0.0014) |
| Obs \| Param \| LL \| AIC | 2329 | 528 | -13667.0 | 28390.0 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Dynamic Estimates

# A4: Economics Department Static Estimates

## Table 8: Economics Department Static Estimates

### (a) Unrestricted

| | Transition | | Accept | | Success 1 | | Success 2 | |
|---|---|---|---|---|---|---|---|---|
| | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -23.0601*** | -3.4656*** | -28.5276*** | -1.2765*** | -4.6847*** | -0.5998*** | -3.7142 | -0.5347 |
| | (2.0491) | (0.2371) | (6.9081) | (0.2933) | (1.6180) | (0.2054) | (3.3294) | (0.4783) |
| 2016 | -0.2896 | -0.0435 | -0.5294 | -0.0237 | 0.0683 | 0.0087 | 0.5121 | 0.0737 |
| | (0.1794) | (0.0268) | (0.5264) | (0.0230) | (0.1852) | (0.0237) | (0.3233) | (0.0460) |
| 2017 | -0.1533 | -0.0230 | -1.3366** | -0.0598** | 0.1166 | 0.0149 | 0.6368* | 0.0917** |
| | (0.1726) | (0.0260) | (0.5312) | (0.0235) | (0.1877) | (0.0240) | (0.3294) | (0.0466) |
| 2018 | 1.5429*** | 0.2319*** | -3.6854*** | -0.1649*** | 0.0112 | 0.0014 | 0.0600 | 0.0086 |
| | (0.1858) | (0.0251) | (0.6281) | (0.0209) | (0.1920) | (0.0246) | (0.3133) | (0.0451) |
| 2019 | 0.2905 | 0.0437 | -3.0765*** | -0.1377*** | 0.0177 | 0.0023 | 0.3498 | 0.0504 |
| | (0.1833) | (0.0273) | (0.6899) | (0.0257) | (0.1913) | (0.0245) | (0.3214) | (0.0459) |
| Winsorized Age | 0.0628*** | 0.0094*** | -0.0133 | -0.0006 | 0.0439* | 0.0056* | 0.0799** | 0.0115** |
| | (0.0218) | (0.0032) | (0.0612) | (0.0027) | (0.0229) | (0.0029) | (0.0351) | (0.0050) |
| Female | 0.3919*** | 0.0589*** | 0.8909** | 0.0399*** | -0.0342 | -0.0044 | 0.1872 | 0.0270 |
| | (0.1109) | (0.0164) | (0.3672) | (0.0154) | (0.1232) | (0.0158) | (0.1856) | (0.0267) |
| U.S. | -0.1124 | -0.0169 | 0.0293 | 0.0013 | 0.1634 | 0.0209 | 0.6453* | 0.0929* |
| | (0.1894) | (0.0285) | (0.4978) | (0.0223) | (0.2198) | (0.0281) | (0.3470) | (0.0494) |
| International Asian | 0.0087 | 0.0013 | -1.1484*** | -0.0514*** | -0.0894 | -0.0114 | -0.0233 | -0.0033 |
| | (0.1396) | (0.0210) | (0.4133) | (0.0172) | (0.1489) | (0.0191) | (0.2426) | (0.0349) |
| GRE Verbal Percentile | 0.0326*** | 0.0049*** | 0.0039 | 0.0002 | 0.0015 | 0.0002 | -0.0005 | -0.0001 |
| | (0.0034) | (0.0005) | (0.0131) | (0.0006) | (0.0034) | (0.0004) | (0.0062) | (0.0009) |
| GRE Quantitative Percentile | 0.1829*** | 0.0275*** | 0.0402 | 0.0018 | 0.0065 | 0.0008 | -0.0064 | -0.0009 |
| | (0.0163) | (0.0018) | (0.0326) | (0.0015) | (0.0076) | (0.0010) | (0.0238) | (0.0034) |
| GRE Analytic Percentile | 0.0152*** | 0.0023*** | 0.0078 | 0.0004 | -0.0011 | -0.0001 | -0.0064 | -0.0009 |
| | (0.0026) | (0.0004) | (0.0090) | (0.0004) | (0.0028) | (0.0004) | (0.0043) | (0.0006) |
| TOEFL/IELTS | -0.0097 | -0.0015 | -0.1286 | -0.0058 | 0.3267* | 0.0418* | 0.3738 | 0.0538 |
| | (0.1632) | (0.0245) | (0.5621) | (0.0250) | (0.1721) | (0.0220) | (0.2358) | (0.0338) |
| Elite U.S. University | 0.9066*** | 0.1363*** | -0.5317 | -0.0238 | -0.0506 | -0.0065 | 0.1629 | 0.0234 |
| | (0.2103) | (0.0312) | (0.6438) | (0.0293) | (0.2283) | (0.0292) | (0.3360) | (0.0484) |
| Elite U.S. Liberal Arts College | 0.3268 | 0.0491 | 0.3253 | 0.0146 | -0.0812 | -0.0104 | 0.1414 | 0.0204 |
| | (0.3402) | (0.0510) | (0.8010) | (0.0356) | (0.3863) | (0.0495) | (0.5521) | (0.0795) |
| Elite Foreign University | 0.0518 | 0.0078 | -0.9107 | -0.0408 | 0.0910 | 0.0117 | 0.2454 | 0.0353 |
| | (0.2959) | (0.0445) | (0.9288) | (0.0424) | (0.3413) | (0.0437) | (0.4927) | (0.0709) |
| Other Foreign University | 0.1161 | 0.0174 | 0.2293 | 0.0103 | 0.2023 | 0.0259 | 0.2550 | 0.0367 |
| | (0.1969) | (0.0296) | (0.6888) | (0.0306) | (0.2140) | (0.0274) | (0.3455) | (0.0497) |
| Missing Undergraduate Institution | -0.4337** | -0.0652** | 0.0505 | 0.0023 | 0.1941 | 0.0249 | 0.1146 | 0.0165 |
| | (0.1996) | (0.0300) | (0.6688) | (0.0299) | (0.2292) | (0.0293) | (0.3847) | (0.0554) |
| Graduate Degree | 1.0229*** | 0.1537*** | -0.3419 | -0.0153 | -0.1377 | -0.0176 | -0.3618* | -0.0521* |
| | (0.1262) | (0.0177) | (0.3883) | (0.0171) | (0.1312) | (0.0168) | (0.2120) | (0.0302) |
| Work Experience | 0.7870*** | 0.1183*** | 0.5914* | 0.0265* | 0.3144** | 0.0403** | 0.0458 | 0.0066 |
| | (0.1256) | (0.0179) | (0.3412) | (0.0154) | (0.1361) | (0.0173) | (0.2252) | (0.0324) |
| #Professor Recommenders | -0.0211 | -0.0032 | -0.1896 | -0.0085 | -0.0570 | -0.0073 | -0.0215 | -0.0031 |
| | (0.0839) | (0.0126) | (0.2574) | (0.0113) | (0.0846) | (0.0108) | (0.1331) | (0.0192) |
| #Prolific Recommenders | 0.1325*** | 0.0199*** | 0.1903 | 0.0085 | 0.0559 | 0.0072 | 0.0961 | 0.0138 |
| | (0.0513) | (0.0077) | (0.1757) | (0.0077) | (0.0574) | (0.0073) | (0.0843) | (0.0121) |
| Missing GRE | -1.8142*** | -0.2727*** | 1.6533 | 0.0740 | -0.0087 | -0.0011 | -0.3074 | -0.0443 |
| | (0.5268) | (0.0772) | (1.8091) | (0.0811) | (0.3716) | (0.0476) | (0.9426) | (0.1357) |
| Missing TOEFL/IELTS | 0.0947 | 0.0142 | 0.4511 | 0.0202 | -0.0080 | -0.0010 | -0.2725 | -0.0392 |
| | (0.1295) | (0.0194) | (0.4238) | (0.0189) | (0.1419) | (0.0182) | (0.2221) | (0.0317) |
| Committee Score | | | 3.7089*** | 0.1660*** | | | 0.0176 | 0.0025 |
| | | | (0.3984) | (0.0065) | | | (0.0510) | (0.0073) |
| #Math Courses | | | -0.2494 | -0.0112 | | | -0.0676 | -0.0097 |
| | | | (0.2386) | (0.0108) | | | (0.1076) | (0.0155) |
| Advanced Math | | | 1.1003 | 0.0492 | | | -0.1678 | -0.0242 |
| | | | (0.7916) | (0.0347) | | | (0.3465) | (0.0498) |
| Letters Length/100 | | | 0.0770 | 0.0034 | | | -0.0252 | -0.0036 |
| | | | (0.1080) | (0.0048) | | | (0.0431) | (0.0062) |
| Letters Terms (%) PC1 | | | 0.2451 | 0.0110 | | | 0.1496 | 0.0215 |
| | | | (0.2489) | (0.0112) | | | (0.1082) | (0.0155) |
| Letters Terms (%) PC2 | | | -0.3767 | -0.0169 | | | -0.0395 | -0.0057 |
| | | | (0.3450) | (0.0153) | | | (0.1470) | (0.0212) |
| CV Length/100 | | | 0.0391 | 0.0017 | | | 0.0778 | 0.0112 |
| | | | (0.1060) | (0.0047) | | | (0.0700) | (0.0100) |
| CV Coding | | | -0.2228* | -0.0100* | | | 0.0863 | 0.0124 |
| | | | (0.1301) | (0.0060) | | | (0.0788) | (0.0113) |
| CV Publication/Working Paper | | | 0.7611 | 0.0341 | | | 0.1054 | 0.0152 |
| | | | (0.8128) | (0.0364) | | | (0.3992) | (0.0575) |
| Essay Length/100 | | | 0.2047 | 0.0092 | | | -0.0987 | -0.0142 |
| | | | (0.3483) | (0.0157) | | | (0.0948) | (0.0136) |
| Essay Specific Professor | | | 0.4125 | 0.0185 | | | 0.0869 | 0.0125 |
| | | | (0.6138) | (0.0275) | | | (0.3148) | (0.0453) |
| Essay Economics Terms (%) | | | 0.0154 | 0.0007 | | | -0.1091 | -0.0157 |
| | | | (0.0814) | (0.0036) | | | (0.0736) | (0.0105) |
| Essay Research Terms (%) | | | -0.0751 | -0.0034 | | | 0.0033 | 0.0005 |
| | | | (0.1492) | (0.0067) | | | (0.0771) | (0.0111) |
| Essay Positive Terms (%) | | | -0.0689 | -0.0031 | | | 0.0948 | 0.0137 |
| | | | (0.0968) | (0.0043) | | | (0.0589) | (0.0084) |
| Essay Negative Terms (%) | | | -0.0863 | -0.0039 | | | -0.1146 | -0.0165 |
| | | | (0.1368) | (0.0061) | | | (0.0779) | (0.0111) |
| Missing Application Pdf | | | 0.2614 | 0.0117 | | | 0.3292 | 0.0474 |
| | | | (0.7005) | (0.0311) | | | (0.3321) | (0.0477) |
| Program Rank | | | | | -0.0084*** | -0.0011*** | -0.0093*** | -0.0013*** |
| | | | | | (0.0020) | (0.0003) | (0.0033) | (0.0005) |
| Missing Program Rank | | | | | -3.1385*** | -0.4018*** | -3.3225*** | -0.4784*** |
| | | | | | (0.4051) | (0.0517) | (0.6860) | (0.0991) |
| Obs \| Param \| LL \| AIC | 2329 | 132 | -2501.6 | 5267.1 | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

### (b) Restricted

| | Cutoff 1 | $P(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success 1 | Success 2 | Sigma |
|---|---|---|---|---|---|---|---|
| Intercept | -0.8795*** | 29.33% | -0.5593*** | 36.37% | -1.1693*** | -0.9173** | |
| | (0.1627) | (23.18%, 36.34%) | (0.1850) | (28.46%, 45.10%) | (0.4372) | (0.4528) | |
| 2016 | -0.0075 | 29.17% | 0.3004 | 43.56% | -0.0151 | 0.2884 | |
| | (0.2468) | (22.20%, 37.28%) | (0.2694) | (34.01%, 53.62%) | (0.2479) | (0.2705) | |
| 2017 | 0.3491* | 37.04% | 0.4187* | 46.49% | 0.3460* | 0.3888 | |
| | (0.1962) | (32.09%, 42.28%) | (0.2507) | (37.65%, 55.56%) | (0.1963) | (0.2507) | |
| 2018 | 0.3979** | 38.19% | -0.0958 | 34.19% | 0.4378** | -0.1814 | |
| | (0.1853) | (34.05%, 42.50%) | (0.2524) | (26.12%, 43.29%) | (0.1859) | (0.2502) | |
| 2019 | 0.2181 | 34.04% | 0.2241 | 41.70% | 0.2264 | 0.1532 | |
| | (0.2016) | (28.67%, 39.85%) | (0.2499) | (33.20%, 50.72%) | (0.2017) | (0.2511) | |
| Winsorized Age | | | | | 0.0016 | 0.0001 | |
| | | | | | (0.0011) | (0.0014) | |
| Female | | | | | 0.0104 | 0.0216 | |
| | | | | | (0.0070) | (0.0149) | |
| U.S. | | | | | -0.0030 | 0.0026 | |
| | | | | | (0.0052) | (0.0119) | |
| International Asian | | | | | -0.0005 | -0.0264 | |
| | | | | | (0.0036) | (0.0175) | |
| GRE Verbal Percentile | | | | | 0.0008 | 0.0001 | |
| | | | | | (0.0005) | (0.0003) | |
| GRE Quantitative Percentile | | | | | 0.0049 | 0.0009 | |
| | | | | | (0.0031) | (0.0009) | |
| GRE Analytic Percentile | | | | | 0.0004 | 0.0001 | |
| | | | | | (0.0003) | (0.0002) | |
| TOEFL/IELTS | | | | | -0.0008 | -0.0025 | |
| | | | | | (0.0041) | (0.0129) | |
| Elite U.S. University | | | | | 0.0222 | -0.0116 | |
| | | | | | (0.0150) | (0.0165) | |
| Elite U.S. Liberal Arts College | | | | | 0.0077 | 0.0082 | |
| | | | | | (0.0102) | (0.0197) | |
| Elite Foreign University | | | | | 0.0011 | -0.0195 | |
| | | | | | (0.0076) | (0.0248) | |
| Other Foreign University | | | | | 0.0033 | 0.0064 | |
| | | | | | (0.0055) | (0.0165) | |
| Missing Undergraduate Institution | | | | | -0.0105 | 0.0020 | |
| | | | | | (0.0078) | (0.0158) | |
| Graduate Degree | | | | | 0.0270 | -0.0091 | |
| | | | | | (0.0170) | (0.0106) | |
| Work Experience | | | | | 0.0207 | 0.0138 | |
| | | | | | (0.0136) | (0.0111) | |
| #Professor Recommenders | | | | | -0.0004 | -0.0047 | |
| | | | | | (0.0022) | (0.0065) | |
| #Prolific Recommenders | | | | | 0.0034 | 0.0041 | |
| | | | | | (0.0025) | (0.0048) | |
| Missing GRE | | | | | -0.0497 | 0.0385 | |
| | | | | | (0.0346) | (0.0471) | |
| Missing TOEFL/IELTS | | | | | 0.0026 | 0.0100 | |
| | | | | | (0.0037) | (0.0111) | |
| Committee Score | | | | | | 0.0859* | |
| | | | | | | (0.0492) | |
| #Math Courses | | | | | | -0.0061 | |
| | | | | | | (0.0064) | |
| Advanced Math | | | | | | 0.0257 | |
| | | | | | | (0.0225) | |
| Letters Length/100 | | | | | | 0.0017 | |
| | | | | | | (0.0026) | |
| Letters Terms (%) PC1 | | | | | | 0.0059 | |
| | | | | | | (0.0067) | |
| Letters Terms (%) PC2 | | | | | | -0.0088 | |
| | | | | | | (0.0095) | |
| CV Length/100 | | | | | | 0.0010 | |
| | | | | | | (0.0024) | |
| CV Coding | | | | | | -0.0050 | |
| | | | | | | (0.0042) | |
| CV Publication/Working Paper | | | | | | 0.0182 | |
| | | | | | | (0.0223) | |
| Essay Length/100 | | | | | | 0.0046 | |
| | | | | | | (0.0081) | |
| Essay Specific Professor | | | | | | 0.0088 | |
| | | | | | | (0.0154) | |
| Essay Economics Terms (%) | | | | | | 0.0001 | |
| | | | | | | (0.0019) | |
| Essay Research Terms (%) | | | | | | -0.0017 | |
| | | | | | | (0.0036) | |
| Essay Positive Terms (%) | | | | | | -0.0014 | |
| | | | | | | (0.0023) | |
| Essay Negative Terms (%) | | | | | | -0.0020 | |
| | | | | | | (0.0035) | |
| Missing Application Pdf | | | | | | 0.0063 | |
| | | | | | | (0.0167) | |
| Program Rank | | | | | -0.0085*** | -0.0083*** | |
| | | | | | (0.0021) | (0.0030) | |
| Missing Program Rank | | | | | -3.0033*** | -3.2457*** | |
| | | | | | (0.4079) | (0.6738) | |
| Sigma | | | | | | | 0.0056 |
| | | | | | | | (0.0035) |
| Obs \| Param \| LL \| AIC | 2329 | 79 | -2528.4 | 5214.8 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

▸▸ Restricted Static Estimates

## Table 9: Economics Department Myopic Estimates

### (a) Unrestricted

| | Committee Score | Transition | | Accept | | Success | |
|---|---|---|---|---|---|---|---|
| | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | 4.0405*** | -23.0601*** | -3.4656*** | -28.5204*** | -1.2762*** | -3.7142 | -0.5347 |
| | (0.9990) | (2.0491) | (0.2371) | (6.9074) | (0.2933) | (3.3294) | (0.4783) |
| 2016 | 0.1701 | -0.2896 | -0.0435 | -0.5295 | -0.0237 | 0.5121 | 0.0737 |
| | (0.1966) | (0.1794) | (0.0268) | (0.5264) | (0.0230) | (0.3233) | (0.0460) |
| 2017 | 0.4355** | -0.1533 | -0.0230 | -1.3366** | -0.0598** | 0.6368* | 0.0917** |
| | (0.2214) | (0.1726) | (0.0260) | (0.5312) | (0.0235) | (0.3294) | (0.0466) |
| 2018 | 0.8410*** | 1.5429*** | 0.2319*** | -3.6854*** | -0.1649*** | 0.0600 | 0.0086 |
| | (0.1873) | (0.1858) | (0.0251) | (0.6281) | (0.0209) | (0.3133) | (0.0451) |
| 2019 | 0.7560*** | 0.2905 | 0.0437 | -3.0764*** | -0.1377*** | 0.3498 | 0.0504 |
| | (0.1987) | (0.1833) | (0.0273) | (0.6899) | (0.0257) | (0.3214) | (0.0459) |
| Winsorized Age | | 0.0628*** | 0.0094*** | -0.0134 | -0.0006 | 0.0799** | 0.0115** |
| | | (0.0218) | (0.0032) | (0.0612) | (0.0027) | (0.0351) | (0.0050) |
| Female | | 0.3920*** | 0.0589*** | 0.8909** | 0.0399*** | 0.1872 | 0.0270 |
| | | (0.1109) | (0.0164) | (0.3672) | (0.0154) | (0.1856) | (0.0267) |
| U.S. | | -0.1124 | -0.0169 | 0.0292 | 0.0013 | 0.6453* | 0.0929* |
| | | (0.1894) | (0.0285) | (0.4978) | (0.0223) | (0.3470) | (0.0494) |
| International Asian | | 0.0087 | 0.0013 | -1.1484*** | -0.0514*** | -0.0233 | -0.0033 |
| | | (0.1396) | (0.0210) | (0.4133) | (0.0172) | (0.2426) | (0.0349) |
| GRE Verbal Percentile | | 0.0326*** | 0.0049*** | 0.0039 | 0.0002 | -0.0005 | -0.0001 |
| | | (0.0034) | (0.0005) | (0.0131) | (0.0006) | (0.0062) | (0.0009) |
| GRE Quantitative Percentile | | 0.1829*** | 0.0275*** | 0.0402 | 0.0018 | -0.0064 | -0.0009 |
| | | (0.0163) | (0.0018) | (0.0326) | (0.0015) | (0.0238) | (0.0034) |
| GRE Analytic Percentile | | 0.0152*** | 0.0023*** | 0.0078 | 0.0004 | -0.0064 | -0.0009 |
| | | (0.0026) | (0.0004) | (0.0090) | (0.0004) | (0.0043) | (0.0006) |
| TOEFL/IELTS | | -0.0097 | -0.0015 | -0.1290 | -0.0058 | 0.3738 | 0.0538 |
| | | (0.1632) | (0.0245) | (0.5621) | (0.0250) | (0.2358) | (0.0338) |
| Elite U.S. University | | 0.9066*** | 0.1363*** | -0.5318 | -0.0238 | 0.1629 | 0.0234 |
| | | (0.2103) | (0.0312) | (0.6438) | (0.0293) | (0.3360) | (0.0484) |
| Elite U.S. Liberal Arts College | | 0.3268 | 0.0491 | 0.3252 | 0.0146 | 0.1414 | 0.0204 |
| | | (0.3402) | (0.0510) | (0.8010) | (0.0356) | (0.5521) | (0.0795) |
| Elite Foreign University | | 0.0518 | 0.0078 | -0.9106 | -0.0407 | 0.2454 | 0.0353 |
| | | (0.2959) | (0.0445) | (0.9288) | (0.0424) | (0.4927) | (0.0709) |
| Other Foreign University | | 0.1161 | 0.0174 | 0.2293 | 0.0103 | 0.2550 | 0.0367 |
| | | (0.1969) | (0.0296) | (0.6888) | (0.0306) | (0.3455) | (0.0497) |
| Missing Undergraduate Institution | | -0.4337** | -0.0652** | 0.0504 | 0.0023 | 0.1146 | 0.0165 |
| | | (0.1996) | (0.0300) | (0.6688) | (0.0299) | (0.3847) | (0.0554) |
| Graduate Degree | | 1.0229*** | 0.1537*** | -0.3418 | -0.0153 | -0.3618* | -0.0521* |
| | | (0.1262) | (0.0177) | (0.3883) | (0.0171) | (0.2120) | (0.0302) |
| Work Experience | | 0.7870*** | 0.1183*** | 0.5914* | 0.0265* | 0.0458 | 0.0066 |
| | | (0.1256) | (0.0179) | (0.3412) | (0.0154) | (0.2252) | (0.0324) |
| #Professor Recommenders | | -0.0211 | -0.0032 | -0.1896 | -0.0085 | -0.0215 | -0.0031 |
| | | (0.0839) | (0.0126) | (0.2574) | (0.0113) | (0.1331) | (0.0192) |
| #Prolific Recommenders | | 0.1325*** | 0.0199*** | 0.1903 | 0.0085 | 0.0961 | 0.0138 |
| | | (0.0513) | (0.0077) | (0.1757) | (0.0077) | (0.0843) | (0.0121) |
| Missing GRE | | -1.8142*** | -0.2727*** | 1.6530 | 0.0740 | -0.3074 | -0.0443 |
| | | (0.5268) | (0.0772) | (1.8090) | (0.0811) | (0.9426) | (0.1357) |
| Missing TOEFL/IELTS | | 0.0947 | 0.0142 | 0.4510 | 0.0202 | -0.2725 | -0.0392 |
| | | (0.1295) | (0.0194) | (0.4238) | (0.0189) | (0.2221) | (0.0317) |
| Committee Score | | | | 3.7088*** | 0.1660*** | 0.0176 | 0.0025 |
| | | | | (0.3984) | (0.0065) | (0.0510) | (0.0073) |
| #Math Courses | -0.0804 | | | -0.2494 | -0.0112 | -0.0676 | -0.0097 |
| | (0.0520) | | | (0.2386) | (0.0108) | (0.1076) | (0.0155) |
| Advanced Math | 0.3560* | | | 1.1003 | 0.0492 | -0.1678 | -0.0242 |
| | (0.1909) | | | (0.7915) | (0.0347) | (0.3465) | (0.0498) |
| Letters Length/100 | 0.1225*** | | | 0.0770 | 0.0034 | -0.0252 | -0.0036 |
| | (0.0242) | | | (0.1080) | (0.0048) | (0.0431) | (0.0062) |
| Letters Terms (%) PC1 | 0.0974* | | | 0.2451 | 0.0110 | 0.1496 | 0.0215 |
| | (0.0536) | | | (0.2488) | (0.0112) | (0.1082) | (0.0155) |
| Letters Terms (%) PC2 | -0.0925 | | | -0.3767 | -0.0169 | -0.0395 | -0.0057 |
| | (0.0856) | | | (0.3450) | (0.0153) | (0.1470) | (0.0212) |
| CV Length/100 | 0.0588* | | | 0.0391 | 0.0017 | 0.0778 | 0.0112 |
| | (0.0323) | | | (0.1060) | (0.0047) | (0.0700) | (0.0100) |
| CV Coding | 0.0551 | | | -0.2228* | -0.0100* | 0.0863 | 0.0124 |
| | (0.0417) | | | (0.1301) | (0.0060) | (0.0788) | (0.0113) |
| CV Publication/Working Paper | 0.5324*** | | | 0.7610 | 0.0341 | 0.1054 | 0.0152 |
| | (0.1975) | | | (0.8127) | (0.0364) | (0.3992) | (0.0575) |
| Essay Length/100 | 0.0763 | | | 0.2047 | 0.0092 | -0.0987 | -0.0142 |
| | (0.0567) | | | (0.3483) | (0.0157) | (0.0948) | (0.0136) |
| Essay Specific Professor | 0.1442 | | | 0.4124 | 0.0185 | 0.0869 | 0.0125 |
| | (0.1767) | | | (0.6138) | (0.0275) | (0.3148) | (0.0453) |
| Essay Economics Terms (%) | 0.0042 | | | 0.0154 | 0.0007 | -0.1091 | -0.0157 |
| | (0.0314) | | | (0.0814) | (0.0036) | (0.0736) | (0.0105) |
| Essay Research Terms (%) | 0.0304 | | | -0.0752 | -0.0034 | 0.0033 | 0.0005 |
| | (0.0446) | | | (0.1492) | (0.0067) | (0.0771) | (0.0111) |
| Essay Positive Terms (%) | 0.0175 | | | -0.0689 | -0.0031 | 0.0948 | 0.0137 |
| | (0.0373) | | | (0.0968) | (0.0043) | (0.0589) | (0.0084) |
| Essay Negative Terms (%) | 0.0429 | | | -0.0863 | -0.0039 | -0.1146 | -0.0165 |
| | (0.0460) | | | (0.1368) | (0.0061) | (0.0779) | (0.0111) |
| Missing Application Pdf | -0.4006** | | | 0.2614 | 0.0117 | 0.3292 | 0.0474 |
| | (0.1850) | | | (0.7005) | (0.0311) | (0.3321) | (0.0477) |
| Program Rank | | | | | | -0.0093*** | -0.0013*** |
| | | | | | | (0.0033) | (0.0005) |
| Missing Program Rank | | | | | | -3.3225*** | -0.4784*** |
| | | | | | | (0.6860) | (0.0991) |
| $\sqrt{\text{MSE}}$ | 1.7525*** | | | | | | |
| | (0.0403) | | | | | | |
| Obs \| Param \| LL \| AIC | 2329 | 296 | -14007.8 | 28607.5 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

### (b) Restricted

| | Committee Score | Cutoff 1 | $L(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success | Sigma |
|---|---|---|---|---|---|---|---|
| Intercept | 3.5032*** | -1.0719*** | 0.3424 | -0.6968*** | 33.25% | -1.2920** | |
| | (1.2668) | (0.1515) | (0.2544, 0.4607) | (0.2303) | (24.08%, 43.90%) | (0.5375) | |
| 2016 | 0.0794 | 0.1267 | 0.3886 | 0.1977 | 37.78% | 0.1867 | |
| | (0.2040) | (0.2580) | (0.2616, 0.5772) | (0.4150) | (23.97%, 53.90%) | (0.4143) | |
| 2017 | 0.6881*** | 0.3114* | 0.4674 | 0.5092* | 45.32% | 0.4878* | |
| | (0.2283) | (0.1671) | (0.4104, 0.5324) | (0.2687) | (39.09%, 51.71%) | (0.2707) | |
| 2018 | 1.0665*** | 0.0059 | 0.3444 | 0.0142 | 33.57% | -0.0389 | |
| | (0.1782) | (0.1860) | (0.2772, 0.4279) | (0.2844) | (26.49%, 41.47%) | (0.2888) | |
| 2019 | 0.7839*** | 0.1487 | 0.3972 | 0.2375 | 38.72% | 0.1780 | |
| | (0.2074) | (0.2024) | (0.3060, 0.5157) | (0.3235) | (29.02%, 49.40%) | (0.3262) | |
| Winsorized Age | | | | | | 0.0009 | |
| | | | | | | (0.0006) | |
| Female | | | | | | 0.0119* | |
| | | | | | | (0.0065) | |
| U.S. | | | | | | 0.0010 | |
| | | | | | | (0.0044) | |
| International Asian | | | | | | -0.0067 | |
| | | | | | | (0.0048) | |
| GRE Verbal Percentile | | | | | | 0.0007* | |
| | | | | | | (0.0004) | |
| GRE Quantitative Percentile | | | | | | 0.0035* | |
| | | | | | | (0.0018) | |
| GRE Analytic Percentile | | | | | | 0.0003* | |
| | | | | | | (0.0002) | |
| TOEFL/IELTS | | | | | | 0.0005 | |
| | | | | | | (0.0037) | |
| Elite U.S. University | | | | | | 0.0097 | |
| | | | | | | (0.0069) | |
| Elite U.S. Liberal Arts College | | | | | | 0.0055 | |
| | | | | | | (0.0077) | |
| Elite Foreign University | | | | | | -0.0011 | |
| | | | | | | (0.0069) | |
| Other Foreign University | | | | | | 0.0059 | |
| | | | | | | (0.0055) | |
| Missing Undergraduate Institution | | | | | | -0.0032 | |
| | | | | | | (0.0049) | |
| Graduate Degree | | | | | | 0.0177* | |
| | | | | | | (0.0092) | |
| Work Experience | | | | | | 0.0167* | |
| | | | | | | (0.0089) | |
| #Professor Recommenders | | | | | | -0.0012 | |
| | | | | | | (0.0020) | |
| #Prolific Recommenders | | | | | | 0.0013 | |
| | | | | | | (0.0015) | |
| Missing GRE | | | | | | -0.0395 | |
| | | | | | | (0.0277) | |
| Missing TOEFL/IELTS | | | | | | 0.0009 | |
| | | | | | | (0.0031) | |
| Committee Score | | | | | | 0.0614** | |
| | | | | | | (0.0311) | |
| #Math Courses | -0.0602 | | | | | -0.0010 | |
| | (0.0688) | | | | | (0.0066) | |
| Advanced Math | 0.3634 | | | | | 0.0026 | |
| | (0.2725) | | | | | (0.0149) | |
| Letters Length/100 | 0.1461*** | | | | | 0.0020 | |
| | (0.0317) | | | | | (0.0027) | |
| Letters Terms (%) PC1 | 0.1129 | | | | | 0.0016 | |
| | (0.0786) | | | | | (0.0084) | |
| Letters Terms (%) PC2 | -0.1002 | | | | | -0.0061 | |
| | (0.1155) | | | | | (0.0085) | |
| CV Length/100 | 0.0620 | | | | | 0.0018 | |
| | (0.0408) | | | | | (0.0025) | |
| CV Coding | 0.0448 | | | | | -0.0047 | |
| | (0.0577) | | | | | (0.0035) | |
| CV Publication/Working Paper | 0.7302** | | | | | -0.0012 | |
| | (0.2876) | | | | | (0.0144) | |
| Essay Length/100 | 0.1332 | | | | | 0.0074 | |
| | (0.0855) | | | | | (0.0119) | |
| Essay Specific Professor | 0.1790 | | | | | 0.0039 | |
| | (0.3042) | | | | | (0.0199) | |
| Essay Economics Terms (%) | -0.0022 | | | | | -0.0017 | |
| | (0.0395) | | | | | (0.0020) | |
| Essay Research Terms (%) | 0.0632 | | | | | 0.0020 | |
| | (0.0598) | | | | | (0.0041) | |
| Essay Positive Terms (%) | 0.0394 | | | | | 0.0008 | |
| | (0.0475) | | | | | (0.0031) | |
| Essay Negative Terms (%) | 0.0458 | | | | | -0.0007 | |
| | (0.0575) | | | | | (0.0046) | |
| Missing Application Pdf | -0.9256*** | | | | | -0.0324** | |
| | (0.2410) | | | | | (0.0163) | |
| Program Rank | | | | | | -0.0084*** | |
| | | | | | | (0.0031) | |
| Missing Program Rank | | | | | | -3.1555*** | |
| | | | | | | (0.6820) | |
| $\sqrt{\text{MSE}}$ | 1.8063*** | | | | | | |
| | (0.0435) | | | | | | |
| Sigma | | | | | | | 0.0024** |
| | | | | | | | (0.0012) |
| Obs \| Param \| LL \| AIC | 2329 | 243 | -14072.6 | 28631.2 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.


Restricted Dynamic Estimates

# A4: Economics Department Matriculation-Dynamic Estimates

Table 10: Economics Department Matriculation-Dynamic Estimates

## (a) Unrestricted

| | Committee Score | Transition | | Accept | | Success | | Matriculate | |
|---|---|---|---|---|---|---|---|---|---|
| | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -2.7966 | -23.0601*** | -3.4656*** | -28.5277*** | -1.2765*** | -3.7142 | -0.5347 | 21.0916*** | 3.3374*** |
| | (2.1751) | (2.0492) | (0.2371) | (6.9081) | (0.2933) | (3.3294) | (0.4783) | (6.9868) | (1.0163) |
| 2016 | 0.0915 | -0.2896 | -0.0435 | -0.5294 | -0.0237 | 0.5121 | 0.0737 | 0.3206 | 0.0507 |
| | (0.1856) | (0.1794) | (0.0268) | (0.5264) | (0.0230) | (0.3233) | (0.0460) | (0.6530) | (0.1030) |
| 2017 | 0.2354 | -0.1533 | -0.0230 | -1.3366** | -0.0598** | 0.6368* | 0.0917** | 0.9971 | 0.1578 |
| | (0.2130) | (0.1726) | (0.0260) | (0.5312) | (0.0235) | (0.3294) | (0.0466) | (0.6410) | (0.1008) |
| 2018 | 0.5999*** | 1.5429*** | 0.2319*** | -3.6854*** | -0.1649*** | 0.0600 | 0.0086 | 1.6496*** | 0.2610*** |
| | (0.1777) | (0.1858) | (0.0251) | (0.6281) | (0.0209) | (0.3133) | (0.0451) | (0.5920) | (0.0886) |
| 2019 | 0.6118*** | 0.2905 | 0.0437 | -3.0765*** | -0.1377*** | 0.3498 | 0.0504 | 1.0916* | 0.1727* |
| | (0.1907) | (0.1833) | (0.0273) | (0.6899) | (0.0257) | (0.3214) | (0.0459) | (0.6190) | (0.0971) |
| Winsorized Age | -0.0621*** | 0.0628*** | 0.0094*** | -0.0133 | -0.0006 | 0.0799** | 0.0115** | -0.1396* | -0.0221* |
| | (0.0235) | (0.0218) | (0.0032) | (0.0612) | (0.0027) | (0.0351) | (0.0050) | (0.0738) | (0.0113) |
| Female | -0.0543 | 0.3919*** | 0.0589*** | 0.8909** | 0.0399*** | 0.1872 | 0.0270 | -0.4562 | -0.0722 |
| | (0.1125) | (0.1109) | (0.0164) | (0.3672) | (0.0154) | (0.1856) | (0.0267) | (0.3550) | (0.0556) |
| U.S. | -0.5071** | -0.1124 | -0.0169 | 0.0293 | 0.0013 | 0.6453* | 0.0929* | 0.9160 | 0.1449 |
| | (0.2043) | (0.1894) | (0.0285) | (0.4978) | (0.0223) | (0.3470) | (0.0494) | (0.5700) | (0.0896) |
| International Asian | -0.7530*** | 0.0087 | 0.0013 | -1.1484*** | -0.0514*** | -0.0233 | -0.0033 | 1.1667** | 0.1846*** |
| | (0.1452) | (0.1396) | (0.0210) | (0.4133) | (0.0172) | (0.2426) | (0.0349) | (0.4531) | (0.0689) |
| GRE Verbal Percentile | 0.0066* | 0.0326*** | 0.0049*** | 0.0039 | 0.0002 | -0.0005 | -0.0001 | -0.0043 | -0.0007 |
| | (0.0036) | (0.0034) | (0.0005) | (0.0131) | (0.0006) | (0.0062) | (0.0009) | (0.0115) | (0.0018) |
| GRE Quantitative Percentile | 0.0447*** | 0.1829*** | 0.0275*** | 0.0402 | 0.0018 | -0.0064 | -0.0009 | -0.1358*** | -0.0215*** |
| | (0.0111) | (0.0163) | (0.0018) | (0.0326) | (0.0015) | (0.0238) | (0.0034) | (0.0444) | (0.0064) |
| GRE Analytic Percentile | 0.0029 | 0.0152*** | 0.0023*** | 0.0078 | 0.0004 | -0.0064 | -0.0009 | -0.0185** | -0.0029** |
| | (0.0027) | (0.0026) | (0.0004) | (0.0090) | (0.0004) | (0.0043) | (0.0006) | (0.0087) | (0.0013) |
| TOEFL/IELTS | 0.4407*** | -0.0097 | -0.0015 | -0.1286 | -0.0058 | 0.3738 | 0.0538 | -0.5140 | -0.0813 |
| | (0.1696) | (0.1632) | (0.0245) | (0.5621) | (0.0250) | (0.2358) | (0.0338) | (0.5175) | (0.0807) |
| Elite U.S. University | 0.5507*** | 0.9066*** | 0.1363*** | -0.5317 | -0.0238 | 0.1629 | 0.0234 | -0.4342 | -0.0687 |
| | (0.1994) | (0.2103) | (0.0312) | (0.6438) | (0.0293) | (0.3360) | (0.0484) | (0.5475) | (0.0858) |
| Elite U.S. Liberal Arts College | -0.0711 | 0.3268 | 0.0491 | 0.3253 | 0.0146 | 0.1414 | 0.0204 | 1.0020 | 0.1586 |
| | (0.3683) | (0.3402) | (0.0510) | (0.8010) | (0.0356) | (0.5521) | (0.0795) | (0.7518) | (0.1185) |
| Elite Foreign University | 0.2902 | 0.0518 | 0.0078 | -0.9107 | -0.0408 | 0.2454 | 0.0353 | -0.2759 | -0.0437 |
| | (0.2991) | (0.2959) | (0.0445) | (0.9288) | (0.0424) | (0.4927) | (0.0709) | (0.9777) | (0.1548) |
| Other Foreign University | 0.4797** | 0.1161 | 0.0174 | 0.2293 | 0.0103 | 0.2550 | 0.0367 | 0.4536 | 0.0718 |
| | (0.2022) | (0.1969) | (0.0296) | (0.6888) | (0.0306) | (0.3455) | (0.0497) | (0.6285) | (0.0992) |
| Missing Undergraduate Institution | 0.5097** | -0.4337** | -0.0652** | 0.0505 | 0.0023 | 0.1146 | 0.0165 | -0.8208 | -0.1299 |
| | (0.2157) | (0.1996) | (0.0300) | (0.6688) | (0.0299) | (0.3847) | (0.0554) | (0.6805) | (0.1058) |
| Graduate Degree | -0.0757 | 1.0229*** | 0.1537*** | -0.3419 | -0.0153 | -0.3618* | -0.0521* | -0.0653 | -0.0103 |
| | (0.1279) | (0.1262) | (0.0177) | (0.3883) | (0.0171) | (0.2120) | (0.0302) | (0.4719) | (0.0746) |
| Work Experience | 0.3458** | 0.7870*** | 0.1183*** | 0.5914* | 0.0265* | 0.0458 | 0.0066 | 0.9166** | 0.1450** |
| | (0.1345) | (0.1256) | (0.0179) | (0.3412) | (0.0154) | (0.2252) | (0.0324) | (0.4411) | (0.0667) |
| #Professor Recommenders | 0.0618 | -0.0211 | -0.0032 | -0.1896 | -0.0085 | -0.0215 | -0.0031 | 0.3588 | 0.0568 |
| | (0.0769) | (0.0839) | (0.0126) | (0.2574) | (0.0113) | (0.1331) | (0.0192) | (0.2789) | (0.0433) |
| #Prolific Recommenders | 0.1667*** | 0.1325*** | 0.0199*** | 0.1903 | 0.0085 | 0.0961 | 0.0138 | -0.2845 | -0.0450 |
| | (0.0548) | (0.0513) | (0.0077) | (0.1757) | (0.0077) | (0.0843) | (0.0121) | (0.1875) | (0.0288) |
| Missing GRE | -1.6087* | -1.8142*** | -0.2727*** | 1.6533 | 0.0740 | -0.3074 | -0.0443 | | |
| | (0.9422) | (0.5268) | (0.0772) | (1.8092) | (0.0811) | (0.9426) | (0.1357) | | |
| Missing TOEFL/IELTS | 0.3199** | 0.0947 | 0.0142 | 0.4511 | 0.0202 | -0.2725 | -0.0392 | -0.0091 | -0.0014 |
| | (0.1319) | (0.1295) | (0.0194) | (0.4238) | (0.0189) | (0.2221) | (0.0317) | (0.5591) | (0.0885) |
| Committee Score | | | | 3.7089*** | 0.1660*** | 0.0176 | 0.0025 | -0.1637 | -0.0259 |
| | | | | (0.3984) | (0.0065) | (0.0510) | (0.0073) | (0.2099) | (0.0336) |
| #Math Courses | -0.0790 | | | -0.2494 | -0.0112 | -0.0676 | -0.0097 | -0.0182 | -0.0029 |
| | (0.0519) | | | (0.2386) | (0.0108) | (0.1076) | (0.0155) | (0.1671) | (0.0265) |
| Advanced Math | 0.2979 | | | 1.1003 | 0.0492 | -0.1678 | -0.0242 | -1.1143** | -0.1763** |
| | (0.1889) | | | (0.7916) | (0.0347) | (0.3465) | (0.0498) | (0.5578) | (0.0877) |
| Letters Length/100 | 0.0858*** | | | 0.0770 | 0.0034 | -0.0252 | -0.0036 | -0.0744 | -0.0118 |
| | (0.0245) | | | (0.1080) | (0.0048) | (0.0431) | (0.0062) | (0.0711) | (0.0113) |
| Letters Terms (%) PC1 | 0.1027* | | | 0.2451 | 0.0110 | 0.1496 | 0.0215 | 0.3997** | 0.0632** |
| | (0.0551) | | | (0.2489) | (0.0112) | (0.1082) | (0.0155) | (0.1903) | (0.0287) |
| Letters Terms (%) PC2 | -0.0208 | | | -0.3767 | -0.0169 | -0.0395 | -0.0057 | 0.2897 | 0.0458 |
| | (0.0823) | | | (0.3450) | (0.0153) | (0.1470) | (0.0212) | (0.2460) | (0.0388) |
| CV Length/100 | 0.0513 | | | 0.0391 | 0.0017 | 0.0778 | 0.0112 | -0.1109 | -0.0175 |
| | (0.0329) | | | (0.1060) | (0.0047) | (0.0700) | (0.0100) | (0.1105) | (0.0175) |
| CV Coding | 0.0381 | | | -0.2228* | -0.0100* | 0.0863 | 0.0124 | -0.2901** | -0.0459** |
| | (0.0424) | | | (0.1301) | (0.0060) | (0.0788) | (0.0113) | (0.1176) | (0.0181) |
| CV Publication/Working Paper | 0.5482*** | | | 0.7611 | 0.0341 | 0.1054 | 0.0152 | -0.3451 | -0.0546 |
| | (0.1912) | | | (0.8128) | (0.0364) | (0.3992) | (0.0575) | (0.5102) | (0.0812) |
| Essay Length/100 | 0.0927 | | | 0.2047 | 0.0092 | -0.0987 | -0.0142 | 0.0882 | 0.0140 |
| | (0.0597) | | | (0.3483) | (0.0157) | (0.0948) | (0.0136) | (0.1386) | (0.0220) |
| Essay Specific Professor | 0.1411 | | | 0.4125 | 0.0185 | 0.0869 | 0.0125 | 0.6812 | 0.1078 |
| | (0.1744) | | | (0.6138) | (0.0275) | (0.3148) | (0.0453) | (0.4976) | (0.0773) |
| Essay Economics Terms (%) | 0.0287 | | | 0.0154 | 0.0007 | -0.1091 | -0.0157 | 0.2244** | 0.0355** |
| | (0.0312) | | | (0.0814) | (0.0036) | (0.0736) | (0.0105) | (0.1075) | (0.0169) |
| Essay Research Terms (%) | 0.0396 | | | -0.0751 | -0.0034 | 0.0033 | 0.0005 | -0.1904 | -0.0301 |
| | (0.0426) | | | (0.1492) | (0.0067) | (0.0771) | (0.0111) | (0.1440) | (0.0226) |
| Essay Positive Terms (%) | 0.0171 | | | -0.0689 | -0.0031 | 0.0948 | 0.0137 | -0.0170 | -0.0027 |
| | (0.0345) | | | (0.0968) | (0.0043) | (0.0589) | (0.0084) | (0.1151) | (0.0182) |
| Essay Negative Terms (%) | 0.0441 | | | -0.0863 | -0.0039 | -0.1146 | -0.0165 | -0.0176 | -0.0028 |
| | (0.0443) | | | (0.1368) | (0.0061) | (0.0779) | (0.0111) | (0.1043) | (0.0165) |
| Missing Application Pdf | -0.5376*** | | | 0.2614 | 0.0117 | 0.3292 | 0.0474 | -0.7440 | -0.1177 |
| | (0.1841) | | | (0.7005) | (0.0311) | (0.3321) | (0.0477) | (0.5803) | (0.0919) |
| Program Rank | | | | | | -0.0093*** | -0.0013*** | | |
| | | | | | | (0.0033) | (0.0005) | | |
| Missing Program Rank | | | | | | -3.3225*** | -0.4784*** | | |
| | | | | | | (0.6860) | (0.0991) | | |
| $\sqrt{\text{MSE}}$ | 1.6467*** | | | | | | | | |
| | (0.0370) | | | | | | | | |
| Obs \| Param \| LL \| AIC | 2329 | 620 | -13796.1 | 28832.3 | | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

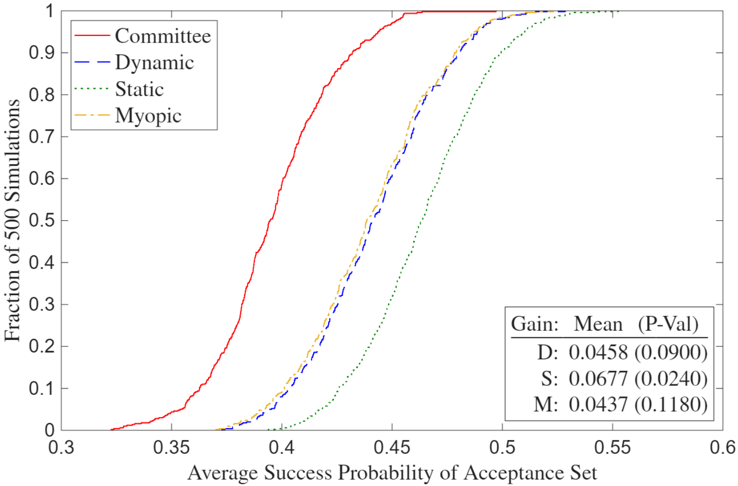
## (b) Restricted

| | Committee Score | Cutoff 1 | $L(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success | Matriculate | Sigma |
|---|---|---|---|---|---|---|---|---|
| Intercept | -6.5744*** | -5.0258*** | 0.0066 | -0.7845*** | 31.33% | -0.9724** | 2.0177 | |
| | (2.5443) | (0.5433) | (0.0023, 0.0190) | (0.2501) | (21.85%, 42.69%) | (0.4154) | (5.1883) | |
| 2016 | 0.2797 | -0.0027 | 0.0065 | 0.4378 | 41.42% | 0.4344 | -1.3526* | |
| | (0.2028) | (0.1805) | (0.0024, 0.0180) | (0.3469) | (30.36%, 53.41%) | (0.3480) | (0.7087) | |
| 2017 | 0.6354*** | 0.3991** | 0.0098 | 0.6101** | 45.65% | 0.5944** | -0.3904 | |
| | (0.2429) | (0.1775) | (0.0036, 0.0267) | (0.2914) | (39.08%, 52.37%) | (0.2914) | (0.8385) | |
| 2018 | 0.8480*** | 0.2676 | 0.0086 | 0.0302 | 31.99% | -0.0271 | 1.2580* | |
| | (0.1893) | (0.1910) | (0.0032, 0.0231) | (0.3068) | (24.21%, 40.91%) | (0.3129) | (0.7179) | |
| 2019 | 0.5712*** | 0.3053 | 0.0089 | 0.3696 | 39.77% | 0.3215 | 0.7374 | |
| | (0.2115) | (0.2124) | (0.0031, 0.0253) | (0.3145) | (31.52%, 48.65%) | (0.3164) | (0.8671) | |
| Winsorized Age | -0.0572** | | | | | 0.0000 | -0.0328 | |
| | (0.0251) | | | | | (0.0010) | (0.0536) | |
| Female | -0.0789 | | | | | 0.0109 | 0.0997 | |
| | (0.1146) | | | | | (0.0082) | (0.2090) | |
| U.S. | -0.3691* | | | | | -0.0026 | 0.4429 | |
| | (0.2091) | | | | | (0.0078) | (0.4392) | |
| International Asian | -0.5014*** | | | | | -0.0092 | 0.5847* | |
| | (0.1609) | | | | | (0.0078) | (0.3315) | |
| GRE Verbal Percentile | 0.0082** | | | | | 0.0000 | 0.0093 | |
| | (0.0037) | | | | | (0.0002) | (0.0072) | |
| GRE Quantitative Percentile | 0.0707*** | | | | | 0.0008 | -0.0640** | |
| | (0.0106) | | | | | (0.0008) | (0.0321) | |
| GRE Analytic Percentile | 0.0015 | | | | | 0.0001 | 0.0025 | |
| | (0.0029) | | | | | (0.0002) | (0.0063) | |
| TOEFL/IELTS | 0.2866 | | | | | -0.0022 | -0.0613 | |
| | (0.1752) | | | | | (0.0098) | (0.2699) | |
| Elite U.S. University | 0.6413*** | | | | | -0.0090 | -0.0833 | |
| | (0.1891) | | | | | (0.0138) | (0.3856) | |
| Elite U.S. Liberal Arts College | -0.3529 | | | | | 0.0121 | -0.2907 | |
| | (0.3495) | | | | | (0.0166) | (0.6651) | |
| Elite Foreign University | 0.1882 | | | | | -0.0171 | 0.4510 | |
| | (0.2890) | | | | | (0.0186) | (0.5277) | |
| Other Foreign University | 0.5558*** | | | | | 0.0038 | 0.2319 | |
| | (0.2053) | | | | | (0.0141) | (0.3737) | |
| Missing Undergraduate Institution | 0.5487*** | | | | | 0.0031 | -0.3893 | |
| | (0.2110) | | | | | (0.0131) | (0.4032) | |
| Graduate Degree | -0.1898 | | | | | -0.0061 | 0.5020 | |
| | (0.1552) | | | | | (0.0078) | (0.3403) | |
| Work Experience | 0.0771 | | | | | 0.0074 | 0.5609 | |
| | (0.1492) | | | | | (0.0068) | (0.3613) | |
| #Professor Recommenders | 0.0815 | | | | | -0.0019 | 0.2138 | |
| | (0.0822) | | | | | (0.0052) | (0.1566) | |
| #Prolific Recommenders | 0.0932 | | | | | -0.0007 | -0.1801* | |
| | (0.0590) | | | | | (0.0032) | (0.1040) | |
| Missing GRE | -1.3773 | | | | | 0.0129 | | |
| | (0.8760) | | | | | (0.0183) | | |
| Missing TOEFL/IELTS | 0.2646** | | | | | 0.0019 | 0.0865 | |
| | (0.1307) | | | | | (0.0060) | (0.2435) | |
| Committee Score | | | | | | 0.0604** | 0.5114*** | |
| | | | | | | (0.0302) | (0.1474) | |
| #Math Courses | -0.1654 | | | | | -0.0045 | -0.1737 | |
| | (0.1024) | | | | | (0.0054) | (0.1598) | |
| Advanced Math | 0.7448* | | | | | 0.0244 | 0.1226 | |
| | (0.4190) | | | | | (0.0194) | (0.5210) | |
| Letters Length/100 | 0.1083** | | | | | 0.0010 | -0.0124 | |
| | (0.0473) | | | | | (0.0021) | (0.0644) | |
| Letters Terms (%) PC1 | 0.2634*** | | | | | 0.0084 | 0.4294*** | |
| | (0.0994) | | | | | (0.0070) | (0.1635) | |
| Letters Terms (%) PC2 | -0.3064* | | | | | -0.0100 | -0.2706 | |
| | (0.1751) | | | | | (0.0086) | (0.2153) | |
| CV Length/100 | 0.0248 | | | | | 0.0003 | -0.0592 | |
| | (0.0758) | | | | | (0.0022) | (0.0814) | |
| CV Coding | 0.0414 | | | | | -0.0046 | -0.0645 | |
| | (0.1090) | | | | | (0.0043) | (0.1016) | |
| CV Publication/Working Paper | 1.3563*** | | | | | 0.0185 | -0.2243 | |
| | (0.3391) | | | | | (0.0187) | (0.5277) | |
| Essay Length/100 | 0.2764* | | | | | 0.0061 | 0.1399 | |
| | (0.1467) | | | | | (0.0092) | (0.1527) | |
| Essay Specific Professor | 0.3968 | | | | | 0.0116 | 0.6573 | |
| | (0.3698) | | | | | (0.0163) | (0.4045) | |
| Essay Economics Terms (%) | 0.0876 | | | | | 0.0008 | 0.0836 | |
| | (0.0634) | | | | | (0.0021) | (0.0804) | |
| Essay Research Terms (%) | 0.1309 | | | | | 0.0004 | -0.1743 | |
| | (0.0843) | | | | | (0.0027) | (0.1268) | |
| Essay Positive Terms (%) | 0.0530 | | | | | -0.0007 | 0.0065 | |
| | (0.0763) | | | | | (0.0023) | (0.0960) | |
| Essay Negative Terms (%) | 0.0804 | | | | | -0.0017 | -0.0464 | |
| | (0.0842) | | | | | (0.0035) | (0.1030) | |
| Missing Application Pdf | -0.3982 | | | | | 0.0113 | -1.5898*** | |
| | (0.2791) | | | | | (0.0117) | (0.2207) | |
| Program Rank | | | | | | -0.0084*** | | |
| | | | | | | (0.0031) | | |
| Missing Program Rank | | | | | | -3.1674*** | | |
| | | | | | | (0.6814) | | |
| $\sqrt{\text{MSE}}$ | 1.7639*** | | | | | | | |
| | (0.0466) | | | | | | | |
| Sigma | | | | | | | | 0.0019** |
| | | | | | | | | (0.0009) |
| Obs \| Param \| LL \| AIC | 2329 | 567 | -13796.0 | 28726.0 | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

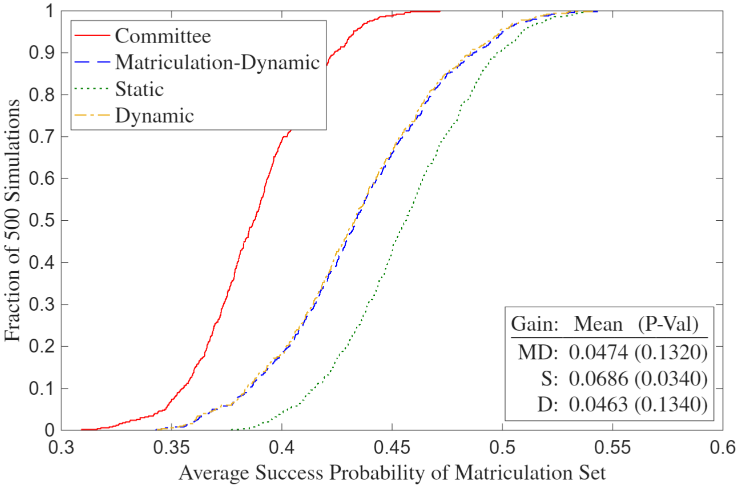
Restricted Matriculation-Dynamic Estimates

# A4: Economics Department Deviation Gains



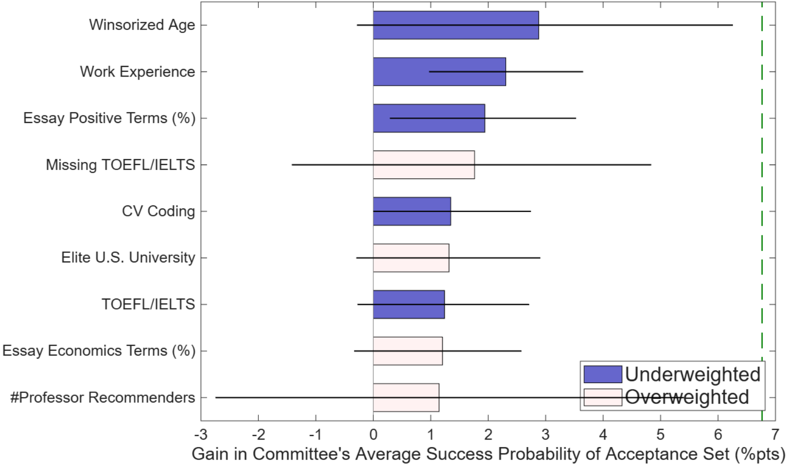

Deviation Gains

# A4: Economics Department Deviation Gains (Matriculation)



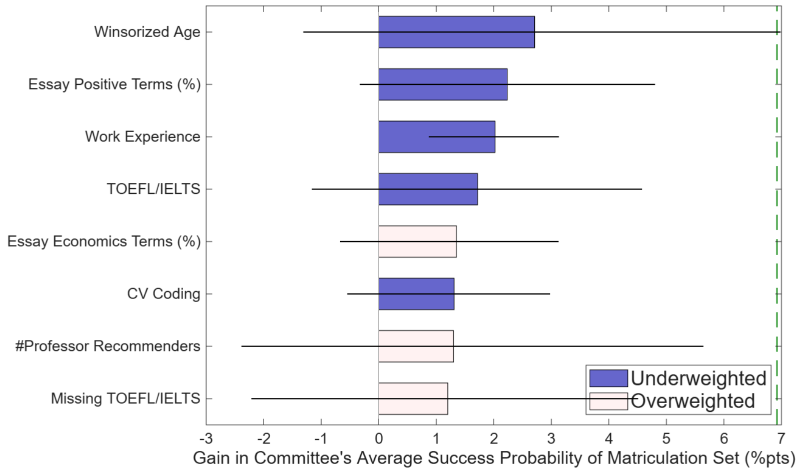

Deviation Gains (Matriculation)

# A4: Economics Department Gains Attribution




Deviation Gains Attribution

# A4: Economics Department Gains Attribution (Matriculation)




Deviation Gains Attribution (Matriculation)

# A4: Economics Department Static Attribution

Table 11: Economics Department Static Attribution

| | Gain (%pt) | Sample Mean | Committee Mean | 1-Variable Optimal Mean | Joint Optimal Mean | 1-Variable Difference (SD) | Joint Difference (SD) |
|---|---|---|---|---|---|---|---|
| Winsorized Age | 2.8788 | 25.9910 | 26.2840 | 29.9572 | 29.2918 | 1.1674 | 0.9559 |
| Work Experience | 2.3065 | 0.5247 | 0.7276 | 0.9844 | 0.8833 | 0.5141 | 0.3116 |
| Essay Positive Terms (%) | 1.9410 | 11.7499 | 11.6521 | 12.5376 | 12.0896 | 0.7858 | 0.3882 |
| Missing TOEFL/IELTS | 1.7635 | 0.6076 | 0.6654 | 0.0739 | 0.4086 | -1.2110 | -0.5258 |
| CV Coding | 1.3478 | 2.8001 | 2.7820 | 3.3399 | 2.8488 | 0.6244 | 0.0748 |
| Elite U.S. University | 1.3179 | 0.1267 | 0.2335 | 0.0078 | 0.1323 | -0.6784 | -0.3041 |
| TOEFL/IELTS | 1.2400 | 7.4376 | 7.5675 | 7.7759 | 7.5990 | 0.5957 | 0.0903 |
| Essay Economics Terms (%) | 1.2043 | 9.7228 | 9.7423 | 9.0284 | 9.5511 | -0.6411 | -0.1717 |
| #Professor Recommenders | 1.1455 | 2.6230 | 2.6148 | 1.3852 | 2.3191 | -1.6680 | -0.4012 |

Deviation Gains Attribution

# A4: Economics Department Matriculation-Static Attribution

Table 12: Economics Department Matriculation-Static Attribution

| | Gain (% Points) | Sample Mean | Committee Mean | 1-Variable Optimal Mean | Joint Optimal Mean | 1-Variable Difference (SD) | Joint Difference (SD) |
|---|---|---|---|---|---|---|---|
| Winsorized Age | 2.7099 | 25.9910 | 26.3284 | 30.9159 | 29.6161 | 1.4580 | 1.0449 |
| Essay Positive Terms (%) | 2.2339 | 11.7499 | 11.6107 | 12.9136 | 12.0979 | 1.1534 | 0.4323 |
| Work Experience | 2.0220 | 0.5247 | 0.7938 | 0.9949 | 0.9391 | 0.4077 | 0.2910 |
| TOEFL/IELTS | 1.7179 | 7.4376 | 7.4939 | 7.8501 | 7.5082 | 1.0238 | 0.0407 |
| Essay Economics Terms (%) | 1.3503 | 9.7228 | 9.9994 | 9.0856 | 9.5668 | -0.8271 | -0.3885 |
| CV Coding | 1.3099 | 2.8001 | 2.6661 | 3.3395 | 2.7049 | 0.7618 | 0.0434 |
| #Professor Recommenders | 1.3005 | 2.6230 | 2.6171 | 1.4173 | 2.3702 | -1.6321 | -0.3350 |
| Missing TOEFL/IELTS | 1.1999 | 0.6076 | 0.6858 | 0.0442 | 0.4801 | -1.3075 | -0.4212 |

Deviation Gains Attribution (Matriculation)

# A4: Graduate School Means by Program

Table 13a: Graduate School Means by Program

| | Economics | Government | History | Linguistics | Psychology |
|---|---|---|---|---|---|
| Accept | 0.1121 | 0.0430 | 0.0986 | 0.0678 | 0.0535 |
| Covid Year | 0.2478 | 0.2792 | 0.3029 | 0.3460 | 0.3004 |
| Winsorized Age | 26.2782 | 27.9633 | 26.8478 | 28.0692 | 25.8230 |
| Female | 0.4201 | 0.4182 | 0.4232 | 0.6133 | 0.7490 |
| U.S. African/Hispanic/Native | 0.0289 | 0.0766 | 0.1159 | 0.0665 | 0.1728 |
| U.S. Asian | 0.0313 | 0.0421 | 0.0304 | 0.0353 | 0.0576 |
| International Asian | 0.5322 | 0.2510 | 0.1058 | 0.2714 | 0.1523 |
| International Other | 0.2517 | 0.2967 | 0.2203 | 0.3256 | 0.1584 |
| GRE Verbal Percentile | 71.8677 | 81.3486 | 82.6054 | 73.5451 | 75.6192 |
| GRE Quantitative Percentile | 86.2163 | 63.3428 | 49.3295 | 57.8455 | 58.0000 |
| GRE Analytic Percentile | 52.6351 | 72.1905 | 75.0460 | 63.6824 | 69.2077 |
| Undergraduate GPA | 3.6534 | 3.5675 | 3.6531 | 3.6567 | 3.5695 |
| TOEFL/IELTS | 7.5232 | 7.5422 | 7.4252 | 7.5444 | 7.3158 |
| Elite U.S. University | 0.1516 | 0.1847 | 0.2565 | 0.1954 | 0.2840 |
| Elite U.S. Liberal Arts College | 0.0423 | 0.0672 | 0.0696 | 0.0231 | 0.0535 |
| Elite Foreign University | 0.0717 | 0.0578 | 0.0478 | 0.0488 | 0.0206 |
| Other Foreign University | 0.5736 | 0.3953 | 0.2333 | 0.5102 | 0.1852 |
| Graduate Degree | 0.7478 | 0.7553 | 0.6652 | 0.7544 | 0.4259 |
| Work Experience | 0.5938 | 0.7463 | 0.6638 | 0.7544 | 0.7593 |
| #Professor Recommenders | 2.5173 | 2.3469 | 2.6261 | 2.4355 | 2.1831 |
| #Prolific Recommenders | 1.0760 | 0.6912 | 0.5043 | 0.7069 | 0.3128 |
| #Math Courses | 6.6396 | | | | |
| Advanced Math | 0.4827 | | | | |
| Missing GRE | 0.0255 | 0.4612 | 0.6217 | 0.6839 | 0.4650 |
| Missing Undergraduate GPA | 0.6607 | 0.4827 | 0.3072 | 0.5726 | 0.2490 |
| Missing TOEFL/IELTS | 0.5958 | 0.7714 | 0.8449 | 0.7707 | 0.8827 |
| Observations | 2078 | 2231 | 690 | 737 | 486 |

- **Economics Highest:** Accept, International Asian, GRE Quantitative Percentile, Elite Foreign University, Other Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, U.S. African/Hispanic/Native, GRE Verbal Percentile, GRE Analytic Percentile, Elite U.S. University, Work Experience, Missing GRE, Missing TOEFL/IELTS

Acceptance, Matriculation, and Success AMEs

# A4: Graduate School Means by Program, Accept = 1

Table 13b: Graduate School Means by Program, Accept = 1

| | Economics | Government | History | Linguistics | Psychology |
|---|---|---|---|---|---|
| Covid Year | 0.1888 | 0.2188 | 0.2059 | 0.2200 | 0.2308 |
| Winsorized Age | 25.0815 | 27.7292 | 27.2794 | 27.1000 | 25.7692 |
| Female | 0.4850 | 0.5000 | 0.5588 | 0.5800 | 0.6154 |
| U.S. African/Hispanic/Native | 0.0386 | 0.1458 | 0.1765 | 0.0400 | 0.0769 |
| U.S. Asian | 0.0429 | 0.0312 | 0.0147 | 0.0600 | 0.1154 |
| International Asian | 0.4163 | 0.1667 | 0.0882 | 0.3600 | 0.0769 |
| International Other | 0.2575 | 0.2292 | 0.3235 | 0.2000 | 0.1538 |
| GRE Verbal Percentile | 84.3980 | 88.8725 | 86.6328 | 77.9161 | 85.7729 |
| GRE Quantitative Percentile | 91.9289 | 68.3708 | 54.8113 | 64.5504 | 62.3462 |
| GRE Analytic Percentile | 69.0355 | 82.7222 | 78.0879 | 68.8958 | 74.6169 |
| Undergraduate GPA | 3.7336 | 3.6404 | 3.7204 | 3.7004 | 3.6341 |
| TOEFL/IELTS | 7.6419 | 7.6050 | 7.4730 | 7.6237 | 7.3178 |
| Elite U.S. University | 0.2833 | 0.3125 | 0.3382 | 0.2200 | 0.5385 |
| Elite U.S. Liberal Arts College | 0.1030 | 0.1458 | 0.1176 | 0.0200 | 0.1154 |
| Elite Foreign University | 0.1030 | 0.0521 | 0.0588 | 0.1000 | 0.0385 |
| Other Foreign University | 0.3863 | 0.2396 | 0.2647 | 0.4000 | 0.1154 |
| Graduate Degree | 0.5837 | 0.6771 | 0.7206 | 0.7400 | 0.4615 |
| Work Experience | 0.6309 | 0.8125 | 0.7647 | 0.7000 | 0.9231 |
| #Professor Recommenders | 2.5494 | 2.3750 | 2.8235 | 2.7200 | 2.2308 |
| #Prolific Recommenders | 1.2833 | 0.9167 | 0.6765 | 1.0800 | 0.4615 |
| #Math Courses | 7.2575 | | | | |
| Advanced Math | 0.6781 | | | | |
| Missing GRE | 0.0086 | 0.2917 | 0.6324 | 0.5800 | 0.1923 |
| Missing Undergraduate GPA | 0.4893 | 0.3125 | 0.3088 | 0.4800 | 0.1923 |
| Missing TOEFL/IELTS | 0.6567 | 0.8854 | 0.8529 | 0.7600 | 0.8846 |
| Observations | 233 | 96 | 68 | 50 | 26 |

- **Economics Highest:** International Asian, GRE Quantitative Percentile, Undergraduate GPA, TOEFL/IELTS, Elite Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, Winsorized Age, Female, U.S. African/Hispanic/Native, Work Experience, Missing GRE, Missing TOEFL/IELTS

Acceptance, Matriculation, and Success AMEs

# A4: Graduate School Correlation Matrix

Table 14: Graduate School Correlation Matrix

| | Covid | Age | Fem | USAHN | USAsn | IntAsn | IntOth | GREV | GREQ | GREA | GPA | T/I | EUSUni | EUSLA | EFor | OthFor | Grad | Work | Prof | Pro | MGRE | MGPA | MT/I |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Covid Year | 1.00 | | | | | | | | | | | | | | | | | | | | | | |
| Winsorized Age | 0.01 | 1.00 | | | | | | | | | | | | | | | | | | | | | |
| Female | 0.02 | -0.13 | 1.00 | | | | | | | | | | | | | | | | | | | | |
| U.S. African/Hispanic/Native | 0.00 | -0.04 | 0.04 | 1.00 | | | | | | | | | | | | | | | | | | | |
| U.S. Asian | -0.00 | -0.06 | 0.05 | -0.05 | 1.00 | | | | | | | | | | | | | | | | | | |
| International Asian | -0.05 | -0.07 | 0.02 | -0.19 | -0.14 | 1.00 | | | | | | | | | | | | | | | | | |
| International Other | 0.01 | 0.18 | -0.06 | -0.17 | -0.12 | -0.42 | 1.00 | | | | | | | | | | | | | | | | |
| GRE Verbal Percentile | 0.03 | -0.06 | -0.04 | 0.00 | 0.04 | -0.12 | -0.15 | 1.00 | | | | | | | | | | | | | | | |
| GRE Quantitative Percentile | -0.02 | -0.16 | -0.10 | -0.18 | 0.00 | 0.44 | -0.10 | 0.16 | 1.00 | | | | | | | | | | | | | | |
| GRE Analytic Percentile | 0.05 | -0.06 | 0.03 | 0.08 | 0.08 | -0.37 | -0.08 | 0.60 | -0.20 | 1.00 | | | | | | | | | | | | | |
| Undergraduate GPA | -0.02 | -0.29 | 0.07 | -0.13 | -0.02 | 0.02 | 0.00 | 0.06 | 0.12 | 0.05 | 1.00 | | | | | | | | | | | | |
| TOEFL/IELTS | -0.01 | -0.07 | 0.01 | -0.02 | 0.01 | 0.04 | 0.00 | 0.24 | 0.12 | 0.19 | 0.00 | 1.00 | | | | | | | | | | | |
| Elite U.S. University | 0.01 | -0.12 | 0.02 | 0.12 | 0.14 | -0.12 | -0.22 | 0.16 | -0.00 | 0.19 | -0.01 | -0.02 | 1.00 | | | | | | | | | | |
| Elite U.S. Liberal Arts College | -0.02 | -0.02 | 0.04 | 0.04 | 0.04 | -0.08 | -0.09 | 0.12 | -0.01 | 0.13 | -0.02 | -0.00 | -0.11 | 1.00 | | | | | | | | | |
| Elite Foreign University | -0.00 | -0.04 | -0.00 | -0.06 | -0.04 | 0.09 | 0.09 | 0.02 | 0.08 | -0.02 | 0.01 | 0.04 | -0.12 | -0.06 | 1.00 | | | | | | | | |
| Other Foreign University | -0.02 | 0.20 | -0.04 | -0.23 | -0.14 | 0.34 | 0.39 | -0.26 | 0.20 | -0.39 | 0.02 | 0.02 | -0.43 | -0.21 | -0.22 | 1.00 | | | | | | | |
| Graduate Degree | 0.00 | 0.41 | -0.08 | -0.13 | -0.09 | 0.21 | 0.15 | -0.14 | 0.05 | -0.20 | -0.22 | 0.03 | -0.21 | -0.10 | 0.05 | 0.37 | 1.00 | | | | | | |
| Work Experience | 0.01 | 0.33 | 0.05 | 0.05 | 0.01 | -0.18 | 0.11 | 0.01 | -0.15 | 0.11 | -0.08 | -0.03 | -0.02 | 0.03 | -0.03 | -0.01 | 0.10 | 1.00 | | | | | |
| #Professor Recommenders | -0.02 | -0.19 | -0.04 | -0.04 | 0.01 | 0.11 | -0.09 | 0.01 | 0.09 | -0.04 | 0.14 | 0.03 | -0.02 | 0.02 | -0.00 | -0.02 | -0.03 | -0.18 | 1.00 | | | | |
| #Prolific Recommenders | 0.01 | -0.06 | -0.02 | -0.07 | -0.01 | 0.24 | -0.10 | -0.01 | 0.25 | -0.11 | 0.04 | 0.10 | -0.01 | -0.01 | 0.06 | 0.08 | 0.17 | -0.09 | 0.19 | 1.00 | | | |
| Missing GRE | 0.09 | 0.15 | 0.05 | 0.11 | -0.01 | -0.22 | 0.17 | 0.09 | -0.33 | 0.17 | -0.09 | -0.10 | -0.03 | -0.04 | -0.03 | -0.01 | 0.02 | 0.10 | -0.08 | -0.20 | 1.00 | | |
| Missing Undergraduate GPA | -0.01 | 0.18 | -0.04 | -0.24 | -0.15 | 0.36 | 0.42 | -0.24 | 0.21 | -0.38 | 0.02 | 0.04 | -0.46 | -0.23 | 0.23 | 0.83 | 0.37 | -0.02 | -0.03 | 0.10 | -0.01 | 1.00 | |
| Missing TOEFL/IELTS | 0.05 | -0.07 | 0.03 | 0.16 | 0.11 | -0.32 | -0.22 | 0.16 | -0.24 | 0.32 | -0.02 | -0.02 | 0.29 | 0.14 | -0.04 | -0.58 | -0.20 | 0.05 | -0.07 | -0.05 | 0.12 | -0.58 | 1.00 |
| Observations | 6222 | | | | | | | | | | | | | | | | | | | | | | |

Acceptance, Matriculation, and Success AMEs

# A4: Graduate School Logistic Regression Results

Table 15a: Graduate School Logistic Regression Results

| Dependent Variable = Accept | Economics | Government | History | Linguistics | Psychology |
|---|---|---|---|---|---|
| Covid Year | -0.3966** | -0.4332* | -0.5752* | -0.5501 | -0.1766 |
| | (0.2010) | (0.2612) | (0.3479) | (0.4129) | (0.5545) |
| Winsorized Age | -0.0995*** | 0.0454* | 0.0380 | 0.0222 | -0.0520 |
| | (0.0358) | (0.0271) | (0.0327) | (0.0532) | (0.0899) |
| Female | 0.4079** | 0.6723*** | 0.6539** | 0.1311 | -0.5060 |
| | (0.1681) | (0.2299) | (0.2971) | (0.3452) | (0.5606) |
| U.S. African/Hispanic/Native | 0.4221 | 1.0081*** | 1.1601*** | 0.0120 | -0.8263 |
| | (0.5258) | (0.3545) | (0.4426) | (0.8411) | (1.1611) |
| U.S. Asian | -0.5394 | -0.5669 | -0.3260 | 0.8638 | 1.1711 |
| | (0.4199) | (0.6504) | (1.2512) | (0.7201) | (0.7276) |
| International Asian | -0.8445*** | -0.1853 | 0.9405 | 0.4890 | -0.7827 |
| | (0.2945) | (0.4146) | (0.7233) | (0.6729) | (0.9323) |
| International Other | 0.5034 | 0.3781 | 1.3523*** | -0.1061 | 0.6239 |
| | (0.3172) | (0.4131) | (0.5086) | (0.7483) | (1.1070) |
| GRE Verbal Percentile | 0.0155*** | 0.0281* | 0.0753** | -0.0092 | 0.0607*** |
| | (0.0060) | (0.0160) | (0.0344) | (0.0154) | (0.0218) |
| GRE Quantitative Percentile | 0.1547*** | 0.0161* | 0.0210* | 0.0211 | -0.0138 |
| | (0.0186) | (0.0086) | (0.0116) | (0.0150) | (0.0159) |
| GRE Analytic Percentile | 0.0179*** | 0.0199** | 0.0030 | 0.0175 | -0.0202 |
| | (0.0042) | (0.0094) | (0.0134) | (0.0127) | (0.0218) |
| Undergraduate GPA | 2.2212*** | 0.6968 | 1.3648** | 1.2645 | 1.7772* |
| | (0.7219) | (0.5476) | (0.5513) | (0.9703) | (0.9487) |
| TOEFL/IELTS | 0.3878 | 0.6456 | 1.0963** | 0.9655* | -0.4656 |
| | (0.2883) | (0.4708) | (0.5526) | (0.5456) | (0.6112) |
| Elite U.S. University | 0.8850*** | 0.3943 | 0.6541* | -0.6057 | 1.4303** |
| | (0.2960) | (0.3135) | (0.3802) | (0.4985) | (0.6192) |
| Elite U.S. Liberal Arts | 1.1173*** | 0.6276* | 1.0698** | -0.6088 | 2.3151** |
| | (0.3813) | (0.3754) | (0.5399) | (1.4186) | (1.0642) |
| Elite Foreign University | 1.0061* | 0.4759 | 1.7715** | 1.6195* | 0.3889 |
| | (0.6060) | (0.8224) | (0.7938) | (0.8587) | (1.9889) |
| Other Foreign University | 0.5442 | 0.5777 | 1.3733** | 0.4188 | -1.0737 |
| | (0.5765) | (0.7686) | (0.6992) | (0.7157) | (1.3016) |
| Graduate Degree | 0.0467 | 0.0733 | 0.3128 | 0.4049 | 0.4892 |
| | (0.2238) | (0.2583) | (0.3280) | (0.4896) | (0.6399) |
| Work Experience | 0.3932** | 0.4027 | 0.4972 | 0.1495 | 1.8201** |
| | (0.1889) | (0.2993) | (0.3295) | (0.3778) | (0.8564) |
| #Professor Recommenders | 0.0340 | -0.0179 | 0.5511** | 0.4781* | -0.0981 |
| | (0.1317) | (0.1339) | (0.2554) | (0.2662) | (0.2660) |
| #Prolific Recommenders | 0.2635*** | 0.3025** | 0.1880 | 0.3284** | 0.5901* |
| | (0.0854) | (0.1262) | (0.1576) | (0.1544) | (0.3145) |
| #Math Courses | 0.1689*** | | | | |
| | (0.0596) | | | | |
| Advanced Math | 0.2319 | | | | |
| | (0.1790) | | | | |
| Missing GRE | 0.1899 | -0.3259 | 0.6282 | -0.0831 | -1.1514** |
| | (0.7646) | (0.3100) | (0.4275) | (0.3761) | (0.5702) |
| Missing Undergraduate GPA | -0.4745 | -0.4917 | -1.5492** | -1.1097* | 0.4690 |
| | (0.4891) | (0.5440) | (0.6893) | (0.6481) | (1.1831) |
| MissingTOEFL/IELTS | -0.3786 | 0.6188 | 0.6745 | -0.0440 | -1.7224* |
| | (0.2432) | (0.5657) | (0.5190) | (0.5099) | (0.9271) |
| Concentration FE | No | Yes | Yes | Yes | Yes |
| Observations | 2078 | 2231 | 690 | 737 | 486 |
| Pseudo-$R^2$ | 0.2677 | 0.1525 | 0.1982 | 0.1871 | 0.2475 |
| CV Pseudo-$R^2$ | 0.2102 | 0.0720 | 0.0604 | -0.0265 | -0.2787 |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses

- **Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

▸▸ Acceptance, Matriculation, and Success AMEs

# A4: Graduate School Logistic Regression AMEs

Table 15b: Graduate School Logistic Regression AMEs

| Depdendent Variable = Accept | Economics | Government | History | Linguistics | Psychology | Non-Economics |
|---|---|---|---|---|---|---|
| Covid Year | -0.0303** | -0.0164 | -0.0431* | -0.0298 | -0.0075 | -0.0218*** |
| | (0.0153) | (0.0100) | (0.0261) | (0.0224) | (0.0238) | (0.0084) |
| Winsorized Age | -0.0076*** | 0.0017* | 0.0028 | 0.0012 | -0.0022 | 0.0016* |
| | (0.0027) | (0.0010) | (0.0025) | (0.0029) | (0.0038) | (0.0009) |
| Female | 0.0312** | 0.0254*** | 0.0490** | 0.0071 | -0.0216 | 0.0207*** |
| | (0.0129) | (0.0088) | (0.0220) | (0.0187) | (0.0239) | (0.0075) |
| U.S. African/Hispanic/Native | 0.0323 | 0.0381*** | 0.0869*** | 0.0007 | -0.0352 | 0.0384*** |
| | (0.0402) | (0.0137) | (0.0332) | (0.0455) | (0.0487) | (0.0119) |
| U.S. Asian | -0.0413 | -0.0215 | -0.0244 | 0.0468 | 0.0499 | 0.0062 |
| | (0.0322) | (0.0247) | (0.0936) | (0.0391) | (0.0313) | (0.0185) |
| International Asian | -0.0646*** | -0.0070 | 0.0704 | 0.0265 | -0.0334 | 0.0069 |
| | (0.0224) | (0.0157) | (0.0543) | (0.0363) | (0.0393) | (0.0132) |
| International Other | 0.0385 | 0.0143 | 0.1013*** | -0.0057 | 0.0266 | 0.0238* |
| | (0.0243) | (0.0157) | (0.0385) | (0.0406) | (0.0471) | (0.0132) |
| GRE Verbal Percentile | 0.0012*** | 0.0011* | 0.0056** | -0.0005 | 0.0026*** | 0.0015*** |
| | (0.0005) | (0.0006) | (0.0026) | (0.0008) | (0.0010) | (0.0005) |
| GRE Quantitative Percentile | 0.0118*** | 0.0006* | 0.0016* | 0.0011 | -0.0006 | 0.0006** |
| | (0.0014) | (0.0003) | (0.0009) | (0.0008) | (0.0007) | (0.0003) |
| GRE Analytic Percentile | 0.0014*** | 0.0008** | 0.0002 | 0.0009 | -0.0009 | 0.0007** |
| | (0.0003) | (0.0004) | (0.0010) | (0.0007) | (0.0009) | (0.0003) |
| Undergraduate GPA | 0.1699*** | 0.0264 | 0.1022** | 0.0684 | 0.0757* | 0.0488*** |
| | (0.0551) | (0.0208) | (0.0423) | (0.0522) | (0.0418) | (0.0176) |
| TOEFL/IELTS | 0.0297 | 0.0244 | 0.0821** | 0.0523* | -0.0198 | 0.0326*** |
| | (0.0219) | (0.0179) | (0.0417) | (0.0301) | (0.0259) | (0.0123) |
| Elite U.S. University | 0.0677*** | 0.0149 | 0.0490* | -0.0328 | 0.0610** | 0.0233** |
| | (0.0225) | (0.0119) | (0.0281) | (0.0264) | (0.0271) | (0.0098) |
| Elite U.S. Liberal Arts College | 0.0855*** | 0.0238* | 0.0801** | -0.0330 | 0.0987** | 0.0420*** |
| | (0.0288) | (0.0143) | (0.0402) | (0.0761) | (0.0480) | (0.0136) |
| Elite Foreign University | 0.0770* | 0.0180 | 0.1327** | 0.0877* | 0.0166 | 0.0479** |
| | (0.0463) | (0.0311) | (0.0591) | (0.0481) | (0.0850) | (0.0241) |
| Other Foreign University | 0.0416 | 0.0219 | 0.1028* | 0.0227 | -0.0458 | 0.0273 |
| | (0.0441) | (0.0291) | (0.0527) | (0.0391) | (0.0556) | (0.0221) |
| Graduate Degree | 0.0036 | 0.0028 | 0.0234 | 0.0219 | 0.0208 | 0.0106 |
| | (0.0171) | (0.0098) | (0.0247) | (0.0262) | (0.0273) | (0.0089) |
| Work Experience | 0.0301** | 0.0152 | 0.0372 | 0.0081 | 0.0776** | 0.0221** |
| | (0.0144) | (0.0113) | (0.0248) | (0.0205) | (0.0368) | (0.0089) |
| #Professor Recommenders | 0.0026 | -0.0007 | 0.0413** | 0.0259* | -0.0042 | 0.0063 |
| | (0.0101) | (0.0051) | (0.0190) | (0.0142) | (0.0113) | (0.0047) |
| #Prolific Recommenders | 0.0202*** | 0.0114** | 0.0141 | 0.0178** | 0.0251* | 0.0133*** |
| | (0.0065) | (0.0048) | (0.0118) | (0.0082) | (0.0130) | (0.0038) |
| #Math Courses | 0.0129*** | | | | | |
| | (0.0045) | | | | | |
| Advanced Math | 0.0177 | | | | | |
| | (0.0137) | | | | | |
| Missing GRE | 0.0145 | -0.0123 | 0.0470 | -0.0045 | -0.0491* | -0.0062 |
| | (0.0585) | (0.0118) | (0.0319) | (0.0204) | (0.0252) | (0.0088) |
| Missing Undergraduate GPA | -0.0363 | -0.0186 | -0.1160** | -0.0601* | 0.0200 | -0.0291 |
| | (0.0375) | (0.0206) | (0.0524) | (0.0355) | (0.0505) | (0.0179) |
| Missing TOEFL/IELTS | -0.0290 | 0.0234 | 0.0505 | -0.0024 | -0.0734* | 0.0108 |
| | (0.0188) | (0.0215) | (0.0388) | (0.0276) | (0.0417) | (0.0135) |
| Concentration FE | No | Yes | Yes | Yes | Yes | Yes |
| Observations | 2078 | 2231 | 690 | 737 | 486 | 4144 |
| Wald Statistic \| DF \| P-Value | 115.8837 | 23 | 0.0000 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Elite Foreign University, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .1$):** Covid Year, Winsorized Age, International Asian

Acceptance, Matriculation, and Success AMEs

# A4: Graduate School Logistic Regression Results (Synthetic)

Table 16a: Graduate School Logistic Regression Results (Synthetic)

| Dependent Variable = Accept | Economics | Government | History | Linguistics | Psychology |
|---|---|---|---|---|---|
| Covid Year | -0.1997 | -0.0323 | 0.2028 | -0.4791 | -0.5946 |
| | (0.1633) | (0.2215) | (0.3078) | (0.3477) | (0.4520) |
| Winsorized Age | -0.0509** | -0.0198 | 0.0576* | -0.0164 | 0.1203*** |
| | (0.0201) | (0.0174) | (0.0307) | (0.0357) | (0.0400) |
| Female | 0.1941 | 0.0923 | 0.1389 | 0.2595 | 0.2119 |
| | (0.1402) | (0.2004) | (0.2526) | (0.3255) | (0.4310) |
| U.S. African/Hispanic/Native | 0.3053 | -0.1457 | -0.2379 | -1.6401 | 0.5413 |
| | (0.3724) | (0.4303) | (0.4670) | (1.0720) | (0.5306) |
| U.S. Asian | 0.5663 | 0.6016 | -0.7934 | -0.4305 | 0.7122 |
| | (0.3642) | (0.4452) | (1.1675) | (0.7936) | (0.8863) |
| International Asian | -0.1428 | 0.1630 | 0.5780 | -0.4965 | 0.1868 |
| | (0.2169) | (0.2803) | (0.3740) | (0.4275) | (0.6866) |
| International Other | -0.1231 | 0.1395 | 0.1097 | -0.1602 | -1.0807 |
| | (0.2311) | (0.3046) | (0.3237) | (0.3832) | (0.7583) |
| GRE Verbal Percentile | 0.0167*** | 0.0247** | 0.0164 | -0.0029 | 0.0132 |
| | (0.0048) | (0.0117) | (0.0127) | (0.0100) | (0.0184) |
| GRE Quantitative Percentile | 0.0233*** | -0.0027 | 0.0026 | 0.0304** | 0.0097 |
| | (0.0090) | (0.0051) | (0.0088) | (0.0137) | (0.0099) |
| GRE Analytic Percentile | 0.0042 | 0.0004 | -0.0073 | -0.0042 | 0.0069 |
| | (0.0031) | (0.0059) | (0.0092) | (0.0101) | (0.0103) |
| Undergraduate GPA | 0.2292 | 0.6327 | 1.3552** | -1.3171** | -0.1002 |
| | (0.4745) | (0.4985) | (0.5796) | (0.5584) | (0.6152) |
| TOEFL/IELTS | -0.4673** | 0.5141 | -0.7648 | 0.2446 | -0.7354 |
| | (0.2157) | (0.4185) | (0.6814) | (0.7498) | (0.8025) |
| Elite U.S. University | -0.4850* | -0.2243 | 0.0572 | 0.4764 | 0.3941 |
| | (0.2560) | (0.3258) | (0.3421) | (0.4788) | (0.4782) |
| Elite U.S. Liberal Arts College | 0.2429 | 0.1759 | 0.2358 | 0.4860 | 0.8652 |
| | (0.3049) | (0.4042) | (0.4964) | (0.9631) | (0.7261) |
| Elite Foreign University | -0.1622 | -1.1929 | 0.7404 | 0.4801 | 3.0530*** |
| | (0.3015) | (0.7484) | (0.5668) | (0.7841) | (0.8673) |
| Other Foreign University | -0.1087 | 0.2043 | 0.2721 | 0.1731 | 0.3023 |
| | (0.1979) | (0.2732) | (0.3391) | (0.4292) | (0.5926) |
| Graduate Degree | -0.0198 | 0.1770 | -0.1601 | -0.8698*** | -0.2769 |
| | (0.1624) | (0.2439) | (0.2765) | (0.3320) | (0.3814) |
| Work Experience | 0.2525* | -0.0391 | -0.0736 | -0.2650 | -1.1936** |
| | (0.1441) | (0.2396) | (0.2629) | (0.3417) | (0.4805) |
| #Professor Recommenders | -0.1059 | 0.0911 | -0.1828 | 0.2181 | -0.4923** |
| | (0.0909) | (0.1138) | (0.1473) | (0.1817) | (0.1956) |
| #Prolific Recommenders | 0.0039 | 0.2189** | -0.1743 | 0.0673 | 0.5134** |
| | (0.0656) | (0.1108) | (0.1723) | (0.1642) | (0.2213) |
| #Math Courses | 0.0366 | | | | |
| | (0.0421) | | | | |
| Advanced Math | 0.1961 | | | | |
| | (0.1416) | | | | |
| Missing GRE | -0.1993 | -0.2029 | -0.0011 | -0.0433 | -0.6040 |
| | (0.4328) | (0.2195) | (0.2689) | (0.3503) | (0.4017) |
| Missing Undergraduate GPA | 0.1531 | 0.0895 | -0.2455 | 0.6298** | -0.3574 |
| | (0.1548) | (0.2291) | (0.3015) | (0.3125) | (0.4342) |
| Missing TOEFL/IELTS | -0.2950* | 0.4898 | 0.5752 | 0.1346 | -2.0781*** |
| | (0.1640) | (0.3480) | (0.5466) | (0.4345) | (0.7633) |
| Concentration FE | No | Yes | Yes | Yes | Yes |
| Observations | 2078 | 2231 | 690 | 737 | 486 |
| Pseudo-$R^2$ | 0.0524 | 0.0366 | 0.0753 | 0.1046 | 0.2230 |
| CV Pseudo-$R^2$ | 0.0139 | -0.0189 | -0.0887 | 0.0055 | -0.0014 |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative ($p < .1$):** Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

Acceptance, Matriculation, and Success AMEs

# A4: Graduate School Logistic Regression AMEs (Synthetic)

Table 16b: Graduate School Logistic Regression AMEs (Synthetic)

| Dependent Variable = Accept | Economics | Government | History | Linguistics | Psychology | Non-Economics |
|---|---|---|---|---|---|---|
| Covid Year | -0.0204 | -0.0014 | 0.0192 | -0.0298 | -0.0345 | -0.0089 |
| | (0.0167) | (0.0099) | (0.0291) | (0.0216) | (0.0262) | (0.0086) |
| Winsorized Age | -0.0052** | -0.0009 | 0.0054* | -0.0010 | 0.0070*** | 0.0004 |
| | (0.0021) | (0.0008) | (0.0029) | (0.0022) | (0.0023) | (0.0008) |
| Female | 0.0198 | 0.0041 | 0.0131 | 0.0162 | 0.0123 | 0.0099 |
| | (0.0143) | (0.0089) | (0.0239) | (0.0204) | (0.0251) | (0.0078) |
| U.S. African/Hispanic/Native | 0.0312 | -0.0065 | -0.0225 | -0.1021 | 0.0314 | -0.0137 |
| | (0.0381) | (0.0192) | (0.0441) | (0.0673) | (0.0306) | (0.0147) |
| U.S. Asian | 0.0579 | 0.0268 | -0.0750 | -0.0268 | 0.0413 | 0.0241 |
| | (0.0372) | (0.0199) | (0.1101) | (0.0495) | (0.0516) | (0.0179) |
| International Asian | -0.0146 | 0.0073 | 0.0546 | -0.0309 | 0.0108 | 0.0075 |
| | (0.0222) | (0.0125) | (0.0353) | (0.0263) | (0.0399) | (0.0112) |
| International Other | -0.0126 | 0.0062 | 0.0104 | -0.0100 | -0.0627 | -0.0014 |
| | (0.0236) | (0.0136) | (0.0306) | (0.0237) | (0.0443) | (0.0106) |
| GRE Verbal Percentile | 0.0017*** | 0.0011** | 0.0015 | -0.0002 | 0.0008 | 0.0008** |
| | (0.0005) | (0.0005) | (0.0012) | (0.0006) | (0.0010) | (0.0004) |
| GRE Quantitative Percentile | 0.0024*** | -0.0001 | 0.0003 | 0.0019** | 0.0006 | 0.0003 |
| | (0.0009) | (0.0002) | (0.0008) | (0.0009) | (0.0006) | (0.0002) |
| GRE Analytic Percentile | 0.0004 | 0.0000 | -0.0007 | -0.0003 | 0.0004 | 0.0000 |
| | (0.0003) | (0.0003) | (0.0009) | (0.0006) | (0.0006) | (0.0002) |
| Undergraduate GPA | 0.0234 | 0.0282 | 0.1281** | -0.0820** | -0.0058 | 0.0130 |
| | (0.0485) | (0.0223) | (0.0553) | (0.0358) | (0.0357) | (0.0158) |
| TOEFL/IELTS | -0.0478** | 0.0229 | -0.0723 | 0.0152 | -0.0427 | 0.0050 |
| | (0.0221) | (0.0187) | (0.0645) | (0.0466) | (0.0469) | (0.0187) |
| Elite U.S. University | -0.0496* | -0.0100 | 0.0054 | 0.0297 | 0.0229 | 0.0036 |
| | (0.0261) | (0.0145) | (0.0324) | (0.0297) | (0.0274) | (0.0108) |
| Elite U.S. Liberal Arts College | 0.0248 | 0.0078 | 0.0223 | 0.0303 | 0.0502 | 0.0217 |
| | (0.0312) | (0.0180) | (0.0469) | (0.0602) | (0.0418) | (0.0160) |
| Elite Foreign University | -0.0166 | -0.0532 | 0.0700 | 0.0299 | 0.1771*** | 0.0158 |
| | (0.0308) | (0.0337) | (0.0537) | (0.0490) | (0.0484) | (0.0179) |
| Other Foreign University | -0.0111 | 0.0091 | 0.0257 | 0.0108 | 0.0175 | 0.0116 |
| | (0.0202) | (0.0122) | (0.0321) | (0.0267) | (0.0344) | (0.0103) |
| Graduate Degree | -0.0020 | 0.0079 | -0.0151 | -0.0542*** | -0.0161 | -0.0093 |
| | (0.0166) | (0.0109) | (0.0261) | (0.0206) | (0.0220) | (0.0085) |
| Work Experience | 0.0258* | -0.0017 | -0.0070 | -0.0165 | -0.0692** | -0.0093 |
| | (0.0148) | (0.0107) | (0.0248) | (0.0213) | (0.0275) | (0.0086) |
| #Professor Recommenders | -0.0108 | 0.0041 | -0.0173 | 0.0136 | -0.0286*** | -0.0027 |
| | (0.0093) | (0.0051) | (0.0140) | (0.0114) | (0.0110) | (0.0042) |
| #Prolific Recommenders | 0.0004 | 0.0098* | -0.0165 | 0.0042 | 0.0298** | 0.0083* |
| | (0.0067) | (0.0050) | (0.0164) | (0.0102) | (0.0128) | (0.0045) |
| #Math Courses | 0.0037 | | | | | |
| | (0.0043) | | | | | |
| Advanced Math | 0.0201 | | | | | |
| | (0.0145) | | | | | |
| Missing GRE | -0.0204 | -0.0090 | -0.0001 | -0.0027 | -0.0350 | -0.0098 |
| | (0.0443) | (0.0098) | (0.0254) | (0.0218) | (0.0239) | (0.0079) |
| Missing Undergraduate GPA | 0.0157 | 0.0040 | -0.0232 | 0.0392* | -0.0207 | 0.0019 |
| | (0.0158) | (0.0102) | (0.0285) | (0.0201) | (0.0254) | (0.0083) |
| Missing TOEFL/IELTS | -0.0302* | 0.0218 | 0.0544 | 0.0084 | -0.1205*** | 0.0063 |
| | (0.0168) | (0.0156) | (0.0517) | (0.0272) | (0.0434) | (0.0130) |
| Concentration FE | No | Yes | Yes | Yes | Yes | Yes |
| Observations | 2078 | 2231 | 690 | 737 | 486 | 4144 |
| Wald Statistic \| DF \| P-Value | 37.4658 | 23 | 0.0290 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .1$):** GRE Verbal Percentile, GRE Quantitative Percentile, Work Experience
- **Economics Negative ($p < .1$):** Winsorized Age, TOEFL/IELTS, Elite U.S. University, Missing TOEFL/IELTS

Acceptance, Matriculation, and Success AMEs

# A5: Simulated Wald Test Results (Same)

Table 17a: Simulated Wald Test Results (Same)

| | Dependent Variable = Accept | | | | Dependent Variable = Success | | | |
|---|---|---|---|---|---|---|---|---|
| | Economics | | Non-Economics | | Economics | | Non-Economics | |
| | Coef | AME | Coef | AME | Coef | AME | Coef | AME |
| Intercept | -16.3848*** | -1.5822*** | -28.5524*** | -1.3677*** | -10.4586*** | -1.9105*** | -10.2610*** | -1.8514*** |
| | (1.1315) | (0.0956) | (1.4963) | (0.0500) | (0.7236) | (0.1070) | (0.4815) | (0.0628) |
| GRE Quantitative Percentile | 0.1509*** | 0.0146*** | 0.3245*** | 0.0155*** | 0.1058*** | 0.0193*** | 0.1154*** | 0.0208*** |
| | (0.0131) | (0.0012) | (0.0179) | (0.0007) | (0.0088) | (0.0014) | (0.0070) | (0.0011) |
| Gender | 0.4714*** | 0.0455*** | 0.7880*** | 0.0377*** | -0.1384 | -0.0253 | -0.0497 | -0.0090 |
| | (0.1475) | (0.0140) | (0.1696) | (0.0081) | (0.1095) | (0.0200) | (0.0895) | (0.0161) |
| Demographic 1 | -0.6287*** | -0.0607*** | -1.8488*** | -0.0886*** | -0.3486** | -0.0637** | -0.3079*** | -0.0556*** |
| | (0.2088) | (0.0200) | (0.2445) | (0.0115) | (0.1467) | (0.0267) | (0.1123) | (0.0202) |
| Demographic 2 | 0.1531 | 0.0148 | 0.3703* | 0.0177* | 0.1756 | 0.0321 | 0.0282 | 0.0051 |
| | (0.2072) | (0.0200) | (0.1929) | (0.0091) | (0.1558) | (0.0284) | (0.1069) | (0.0193) |
| Advanced Math | 0.2889* | 0.0279* | | | 0.3267*** | 0.0597*** | | |
| | (0.1612) | (0.0156) | | | (0.1112) | (0.0202) | | |
| Committee Score | 0.2241*** | 0.0216*** | 0.3994*** | 0.0191*** | 0.1026*** | 0.0187*** | 0.0757*** | 0.0137*** |
| | (0.0413) | (0.0039) | (0.0506) | (0.0023) | (0.0287) | (0.0052) | (0.0228) | (0.0041) |
| Non-Economics 1 | | | 0.8376*** | 0.0401*** | | | 1.9483*** | 0.3515*** |
| | | | (0.2297) | (0.0107) | | | (0.1236) | (0.0205) |
| Observations \| Pseudo-$R^2$ | 5000 | 0.3251 | | | 5000 | 0.1431 | | |
| Parameters \| LL \| AIC | 14 | -1130.4 | 2288.8 | | 14 | -2691.4 | 5410.8 | |
| Wald Statistic \| DF \| P-Value | 3.1018 | 5 | 0.6843 | | 3.0042 | 5 | 0.6993 | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

⏩ Acceptance, Matriculation, and Success AMEs

# A5: Simulated Wald Test Results (Different)

Table 17b: Simulated Wald Test Results (Different)

| | Dependent Variable = Accept | | | | Dependent Variable = Success | | | |
|---|---|---|---|---|---|---|---|---|
| | Economics | | Non-Economics | | Economics | | Non-Economics | |
| | Coef | AME | Coef | AME | Coef | AME | Coef | AME |
| Intercept | -16.3291*** | -1.5784*** | -8.4753*** | -0.3642*** | -10.4581*** | -1.9104*** | -3.2236*** | -0.5997*** |
| | (1.1274) | (0.0955) | (0.8673) | (0.0412) | (0.7237) | (0.1070) | (0.4018) | (0.0728) |
| GRE Quantitative Percentile | 0.1502*** | 0.0145*** | 0.0402*** | 0.0017*** | 0.1058*** | 0.0193*** | 0.0034 | 0.0006 |
| | (0.0131) | (0.0012) | (0.0128) | (0.0006) | (0.0088) | (0.0014) | (0.0062) | (0.0012) |
| Gender | 0.4702*** | 0.0454*** | 0.5942*** | 0.0255*** | -0.1372 | -0.0251 | 0.0375 | 0.0070 |
| | (0.1473) | (0.0140) | (0.1820) | (0.0079) | (0.1095) | (0.0200) | (0.0868) | (0.0161) |
| Demographic 1 | -0.6246*** | -0.0604*** | -0.2541 | -0.0109 | -0.3544** | -0.0647** | -0.1152 | -0.0214 |
| | (0.2085) | (0.0200) | (0.2393) | (0.0103) | (0.1467) | (0.0267) | (0.1103) | (0.0205) |
| Demographic 2 | 0.1555 | 0.0150 | 0.0081 | 0.0003 | 0.1713 | 0.0313 | -0.0562 | -0.0105 |
| | (0.2070) | (0.0200) | (0.2109) | (0.0091) | (0.1557) | (0.0284) | (0.1030) | (0.0192) |
| Advanced Math | 0.2886* | 0.0279* | | | 0.3284*** | 0.0600*** | | |
| | (0.1611) | (0.0156) | | | (0.1112) | (0.0202) | | |
| Committee Score | 0.2241*** | 0.0217*** | 0.3817*** | 0.0164*** | 0.1026*** | 0.0187*** | 0.1808*** | 0.0336*** |
| | (0.0413) | (0.0039) | (0.0528) | (0.0025) | (0.0287) | (0.0052) | (0.0228) | (0.0041) |
| Non-Economics 1 | | | -0.0452 | -0.0019 | | | 1.1393*** | 0.2120*** |
| | | | (0.1995) | (0.0086) | | | (0.1219) | (0.0219) |
| Observations \| Pseudo-$R^2$ | 5000 | 0.1488 | | | 5000 | 0.0758 | | |
| Parameters \| LL \| AIC | 14 | -1155.7 | 2339.5 | | 14 | -2747.2 | 5522.4 | |
| Wald Statistic \| DF \| P-Value | 112.5897 | 5 | 0.0000 | | 124.7379 | 5 | 0.0000 | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Acceptance, Matriculation, and Success AMEs

# A5: Simulated Unrestricted Dynamic Estimates (Optimal)

Table 18a: Simulated Unrestricted Dynamic Estimates (Optimal)

| | Advanced Math | | Committee Score | Transition | | Accept | | Success | |
|---|---|---|---|---|---|---|---|---|---|
| | Coefficient | AME | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -1.9001** | -0.4657** | 4.8636*** | -11.5885*** | -2.2558*** | -13.1542*** | -2.3735*** | -9.5556*** | -1.4872*** |
| | (0.7563) | (0.1830) | (1.3238) | (1.0232) | (0.1454) | (2.0077) | (0.2886) | (2.0721) | (0.2688) |
| GRE Quantitative Percentile | 0.0207** | 0.0051** | 0.0008 | 0.1236*** | 0.0241*** | 0.1372*** | 0.0248*** | 0.1161*** | 0.0181*** |
| | (0.0090) | (0.0022) | (0.0150) | (0.0118) | (0.0018) | (0.0223) | (0.0033) | (0.0227) | (0.0029) |
| Gender | -0.2335* | -0.0572* | -0.0783 | 0.2266 | 0.0441 | 0.1289 | 0.0233 | 0.1263 | 0.0197 |
| | (0.1309) | (0.0319) | (0.2113) | (0.1464) | (0.0283) | (0.2502) | (0.0451) | (0.2699) | (0.0419) |
| Demographic 1 | 0.3692** | 0.0905** | -0.3431 | -0.3013 | -0.0587 | -0.2638 | -0.0476 | -0.4605 | -0.0717 |
| | (0.1732) | (0.0421) | (0.2885) | (0.1988) | (0.0386) | (0.3743) | (0.0672) | (0.3933) | (0.0608) |
| Demographic 2 | 0.0925 | 0.0227 | -0.1806 | 0.1538 | 0.0299 | 0.1708 | 0.0308 | -0.2340 | -0.0364 |
| | (0.1911) | (0.0468) | (0.3001) | (0.2215) | (0.0431) | (0.3844) | (0.0693) | (0.4322) | (0.0672) |
| Advanced Math | | | 0.6870*** | | | 0.7736*** | 0.1396*** | 0.6276** | 0.0977** |
| | | | (0.2024) | | | (0.2535) | (0.0443) | (0.2749) | (0.0417) |
| Committee Score | | | | | | 0.0783 | 0.0141 | 0.0464 | 0.0072 |
| | | | | | | (0.0602) | (0.0108) | (0.0655) | (0.0102) |
| Year Transition More | 0.0663 | 0.0163 | 0.0932 | 0.9840*** | 0.1915*** | -0.9251*** | -0.1669*** | -0.0237 | -0.0037 |
| | (0.1281) | (0.0314) | (0.2069) | (0.1472) | (0.0263) | (0.2513) | (0.0427) | (0.2763) | (0.0430) |
| Program Rank | | | | | | | | -0.0282*** | -0.0044*** |
| | | | | | | | | (0.0027) | (0.0003) |
| $\sqrt{\text{MSE}}$ | | | 1.9465*** | | | | | | |
| | | | (0.0673) | | | | | | |
| Obs \| Param \| LL \| AIC | 1000 | 37 | -2408.7 | 4891.4 | | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

# A5: Simulated Unrestricted Dynamic Estimates (Suboptimal)

Table 18b: Simulated Unrestricted Dynamic Estimates (Suboptimal)

| | Advanced Math | | Committee Score | Transition | | Accept | | Success | |
|---|---|---|---|---|---|---|---|---|---|
| | Coefficient | AME | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -1.8086** | -0.4435** | 5.3561*** | -11.3342*** | -2.1952*** | -6.7719*** | -1.3374*** | -9.1414*** | -1.4363*** |
| | (0.7548) | (0.1830) | (1.2694) | (1.0218) | (0.1474) | (1.7405) | (0.3203) | (1.9229) | (0.2518) |
| GRE Quantitative Percentile | 0.0197** | 0.0048** | -0.0026 | 0.1238*** | 0.0240*** | 0.0555*** | 0.0110*** | 0.1113*** | 0.0175*** |
| | (0.0090) | (0.0022) | (0.0144) | (0.0118) | (0.0018) | (0.0191) | (0.0036) | (0.0212) | (0.0027) |
| Gender | -0.2332* | -0.0572* | -0.1025 | 0.2601* | 0.0504* | 0.0119 | 0.0024 | 0.0519 | 0.0082 |
| | (0.1309) | (0.0319) | (0.2051) | (0.1472) | (0.0283) | (0.2369) | (0.0468) | (0.2652) | (0.0416) |
| Demographic 1 | 0.3717** | 0.0911** | -0.4863* | -0.8309*** | -0.1609*** | -0.3354 | -0.0662 | -0.5014 | -0.0788 |
| | (0.1731) | (0.0421) | (0.2701) | (0.1972) | (0.0371) | (0.3396) | (0.0669) | (0.3621) | (0.0564) |
| Demographic 2 | 0.0905 | 0.0222 | -0.2750 | 0.0815 | 0.0158 | 0.1755 | 0.0347 | -0.4742 | -0.0745 |
| | (0.1909) | (0.0468) | (0.2740) | (0.2141) | (0.0415) | (0.3347) | (0.0659) | (0.3825) | (0.0597) |
| Advanced Math | | | 0.7219*** | | | 0.3555 | 0.0702 | 0.5330** | 0.0837** |
| | | | (0.1966) | | | (0.2376) | (0.0465) | (0.2687) | (0.0416) |
| Committee Score | | | | | | 0.2969*** | 0.0586*** | 0.0597 | 0.0094 |
| | | | | | | (0.0593) | (0.0106) | (0.0639) | (0.0100) |
| Year Transition More | 0.0642 | 0.0158 | 0.0002 | 1.1142*** | 0.2158*** | -0.7164*** | -0.1415*** | 0.1143 | 0.0180 |
| | (0.1280) | (0.0314) | (0.2010) | (0.1484) | (0.0257) | (0.2398) | (0.0455) | (0.2774) | (0.0434) |
| Program Rank | | | | | | | | -0.0280*** | -0.0044*** |
| | | | | | | | | (0.0028) | (0.0003) |
| $\sqrt{\text{MSE}}$ | | | 1.9186*** | | | | | | |
| | | | (0.0682) | | | | | | |
| Obs \| Param \| LL \| AIC | 1000 | 37 | -2452.7 | 4979.4 | | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

# A5: Simulated Restricted Dynamic Estimates (Optimal)

Table 19a: Simulated Restricted Dynamic Estimates (Optimal)

| | Advanced Math | Committee Score | Cutoff 1 | $L(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success | Sigma |
|---|---|---|---|---|---|---|---|---|
| Intercept | -2.1226*** | 4.7314*** | -0.3330*** | 0.7168 | -0.0067 | 49.83% | -9.5110*** | |
| | (0.7573) | (1.3235) | (0.0802) | (0.6125, 0.8387) | (0.2115) | (39.62%, 60.06%) | (1.9433) | |
| GRE Quantitative Percentile | 0.0235*** | 0.0024 | | | | | 0.1134*** | |
| | (0.0090) | (0.0150) | | | | | (0.0214) | |
| Gender | -0.2151* | -0.0652 | | | | | 0.2238* | |
| | (0.1305) | (0.2113) | | | | | (0.1229) | |
| Demographic 1 | 0.3587** | -0.3509 | | | | | -0.3323* | |
| | (0.1725) | (0.2903) | | | | | (0.1836) | |
| Demographic 2 | 0.0940 | -0.1813 | | | | | 0.0172 | |
| | (0.1905) | (0.3010) | | | | | (0.1773) | |
| Advanced Math | | 0.6919*** | | | | | 0.5658*** | |
| | | (0.2026) | | | | | (0.1714) | |
| Committee Score | | | | | | | 0.0496 | |
| | | | | | | | (0.0377) | |
| Year Transition More | 0.0730 | 0.0954 | -0.0244 | 0.6995 | 0.5755* | 63.85% | 0.0619 | |
| | (0.1277) | (0.2069) | (0.0931) | (0.5948, 0.8227) | (0.3212) | (52.94%, 73.50%) | (0.2670) | |
| Program Rank | | | | | | | -0.0282*** | |
| | | | | | | | (0.0027) | |
| $\sqrt{\text{MSE}}$ | | 1.9468*** | | | | | | |
| | | (0.0673) | | | | | | |
| Sigma | | | | | | | | 0.1367*** |
| | | | | | | | | (0.0247) |
| Obs \| Param \| LL \| AIC | 1000 | 28 | -2435.7 | 4927.5 | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

# A5: Simulated Restricted Dynamic Estimates (Suboptimal)

Table 19b: Simulated Restricted Dynamic Estimates (Suboptimal)

| | Advanced Math | Committee Score | Cutoff 1 | $L$(Cutoff)$_1$ | Cutoff 2 | $P$(Cutoff)$_2$ | Success | Sigma |
|---|---|---|---|---|---|---|---|---|
| Intercept | -2.1619*** | 4.4213*** | -0.3434*** | 0.7094 | -0.0905 | 47.74% | -7.5239*** | |
| | (0.7594) | (1.2675) | (0.1028) | (0.5800, 0.8676) | (0.2232) | (37.10%, 58.59%) | (2.2918) | |
| GRE Quantitative Percentile | 0.0240*** | 0.0086 | | | | | 0.0852*** | |
| | (0.0090) | (0.0144) | | | | | (0.0245) | |
| Gender | -0.2194* | -0.0603 | | | | | 0.1567 | |
| | (0.1307) | (0.2029) | | | | | (0.1225) | |
| Demographic 1 | 0.3468** | -0.5636** | | | | | -0.4967** | |
| | (0.1730) | (0.2718) | | | | | (0.2081) | |
| Demographic 2 | 0.1009 | -0.2620 | | | | | -0.0131 | |
| | (0.1908) | (0.2740) | | | | | (0.1789) | |
| Advanced Math | | 0.7335*** | | | | | 0.3527** | |
| | | (0.1973) | | | | | (0.1705) | |
| Committee Score | | | | | | | 0.1726*** | |
| | | | | | | | (0.0366) | |
| Year Transition More | 0.0693 | 0.0080 | -0.0615 | 0.6670 | 0.5395 | 61.04% | 0.1139 | |
| | (0.1278) | (0.2011) | (0.0978) | (0.5507, 0.8080) | (0.3422) | (49.37%, 71.57%) | (0.2774) | |
| Program Rank | | | | | | | -0.0279*** | |
| | | | | | | | (0.0027) | |
| $\sqrt{\text{MSE}}$ | | 1.9212*** | | | | | | |
| | | (0.0688) | | | | | | |
| Sigma | | | | | | | | 0.1458*** |
| | | | | | | | | (0.0395) |
| Obs \| Param \| LL \| AIC | 1000 | 28 | -2497.3 | 5050.5 | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

# A5: Simulated Unrestricted Static Estimates (Optimal)

Table 20a: Simulated Unrestricted Static Estimates (Optimal)

| | Transition | | Accept | | Success 1 | | Success 2 | |
|---|---|---|---|---|---|---|---|---|
| | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -11.5886*** | -2.2559*** | -13.1494*** | -2.3728*** | -8.1224*** | -1.1660*** | -9.5306*** | -1.4839*** |
| | (1.0232) | (0.1454) | (2.0072) | (0.2886) | (1.1371) | (0.1344) | (2.0699) | (0.2689) |
| GRE Quantitative Percentile | 0.1235*** | 0.0240*** | 0.1371*** | 0.0247*** | 0.1078*** | 0.0155*** | 0.1158*** | 0.0180*** |
| | (0.0118) | (0.0018) | (0.0223) | (0.0033) | (0.0133) | (0.0015) | (0.0227) | (0.0029) |
| Gender | 0.2262 | 0.0440 | 0.1290 | 0.0233 | 0.1396 | 0.0200 | 0.1262 | 0.0197 |
| | (0.1464) | (0.0283) | (0.2502) | (0.0451) | (0.1738) | (0.0249) | (0.2698) | (0.0419) |
| Demographic 1 | -0.2995 | -0.0583 | -0.2639 | -0.0476 | -0.2529 | -0.0363 | -0.4577 | -0.0713 |
| | (0.1988) | (0.0386) | (0.3743) | (0.0672) | (0.2358) | (0.0338) | (0.3931) | (0.0608) |
| Demographic 2 | 0.1552 | 0.0302 | 0.1711 | 0.0309 | -0.1180 | -0.0169 | -0.2318 | -0.0361 |
| | (0.2215) | (0.0431) | (0.3843) | (0.0693) | (0.2580) | (0.0371) | (0.4320) | (0.0672) |
| Advanced Math | | | 0.7733*** | 0.1396*** | | | 0.6278** | 0.0978** |
| | | | (0.2534) | (0.0443) | | | (0.2748) | (0.0417) |
| Committee Score | | | 0.0781 | 0.0141 | | | 0.0462 | 0.0072 |
| | | | (0.0602) | (0.0108) | | | (0.0655) | (0.0102) |
| Year Transition More | 0.9837*** | 0.1915*** | -0.9252*** | -0.1670*** | -0.0592 | -0.0085 | -0.0239 | -0.0037 |
| | (0.1472) | (0.0263) | (0.2512) | (0.0427) | (0.1689) | (0.0242) | (0.2762) | (0.0430) |
| Program Rank | | | | | -0.0310*** | -0.0045*** | -0.0282*** | -0.0044*** |
| | | | | | (0.0019) | (0.0002) | (0.0027) | (0.0003) |
| Obs \| Param \| LL \| AIC | 1000 | 30 | -1375.9 | 2811.7 | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

# A5: Simulated Unrestricted Static Estimates (Suboptimal)

Table 20b: Simulated Unrestricted Static Estimates (Suboptimal)

| | Transition | | Accept | | Success 1 | | Success 2 | |
|---|---|---|---|---|---|---|---|---|
| | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -11.2297*** | -2.1801*** | -6.7297*** | -1.3296*** | -8.2833*** | -1.1914*** | -9.6070*** | -1.4986*** |
| | (1.0144) | (0.1474) | (1.7376) | (0.3202) | (1.1393) | (0.1341) | (1.9611) | (0.2508) |
| GRE Quantitative Percentile | 0.1226*** | 0.0238*** | 0.0550*** | 0.0109*** | 0.1097*** | 0.0158*** | 0.1163*** | 0.0181*** |
| | (0.0117) | (0.0018) | (0.0191) | (0.0036) | (0.0133) | (0.0015) | (0.0216) | (0.0027) |
| Gender | 0.2576* | 0.0500* | 0.0097 | 0.0019 | 0.1355 | 0.0195 | 0.0584 | 0.0091 |
| | (0.1469) | (0.0283) | (0.2368) | (0.0468) | (0.1741) | (0.0250) | (0.2669) | (0.0416) |
| Demographic 1 | -0.8291*** | -0.1610*** | -0.3339 | -0.0660 | -0.2777 | -0.0399 | -0.5157 | -0.0804 |
| | (0.1966) | (0.0371) | (0.3394) | (0.0669) | (0.2344) | (0.0337) | (0.3648) | (0.0563) |
| Demographic 2 | 0.0798 | 0.0155 | 0.1762 | 0.0348 | -0.1808 | -0.0260 | -0.4742 | -0.0740 |
| | (0.2135) | (0.0414) | (0.3346) | (0.0659) | (0.2566) | (0.0369) | (0.3857) | (0.0598) |
| Advanced Math | | | 0.3581 | 0.0708 | | | 0.5369** | 0.0837** |
| | | | (0.2375) | (0.0465) | | | (0.2704) | (0.0416) |
| Committee Score | | | 0.2962*** | 0.0585*** | | | 0.0616 | 0.0096 |
| | | | (0.0592) | (0.0106) | | | (0.0643) | (0.0100) |
| Year Transition More | 1.1109*** | 0.2157*** | -0.7185*** | -0.1419*** | -0.0766 | -0.0110 | 0.1314 | 0.0205 |
| | (0.1479) | (0.0257) | (0.2397) | (0.0455) | (0.1688) | (0.0243) | (0.2795) | (0.0434) |
| Program Rank | | | | | -0.0304*** | -0.0044*** | -0.0279*** | -0.0044*** |
| | | | | | (0.0019) | (0.0002) | (0.0028) | (0.0003) |
| Obs \| Param \| LL \| AIC | 1000 | 30 | -1404.1 | 2868.1 | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

# A5: Simulated Restricted Static Estimates (Optimal)

Table 21a: Simulated Restricted Static Estimates (Optimal)

| | Cutoff 1 | $P(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success 1 | Success 2 | Sigma |
|---|---|---|---|---|---|---|---|
| Intercept | 0.6884*** | 66.56% | 0.1530 | 53.82% | -8.0150*** | -9.7263*** | |
| | (0.1830) | (58.17%, 74.02%) | (0.2411) | (42.08%, 65.15%) | (1.1629) | (1.7366) | |
| GRE Quantitative Percentile | | | | | 0.1057*** | 0.1158*** | |
| | | | | | (0.0134) | (0.0193) | |
| Gender | | | | | 0.1987* | 0.1142 | |
| | | | | | (0.1015) | (0.1673) | |
| Demographic 1 | | | | | -0.2471* | -0.3338 | |
| | | | | | (0.1346) | (0.2403) | |
| Demographic 2 | | | | | 0.0406 | -0.0047 | |
| | | | | | (0.1504) | (0.2423) | |
| Advanced Math | | | | | | 0.6315*** | |
| | | | | | | (0.1766) | |
| Committee Score | | | | | | 0.0546 | |
| | | | | | | (0.0421) | |
| Year Transition More | -0.8466*** | 46.05% | 0.7965** | 72.10% | -0.0628 | 0.0285 | |
| | (0.2184) | (39.60%, 52.64%) | (0.3769) | (58.90%, 82.34%) | (0.1667) | (0.2701) | |
| Program Rank | | | | | -0.0310*** | -0.0283*** | |
| | | | | | (0.0019) | (0.0027) | |
| Sigma | | | | | | | 0.1913*** |
| | | | | | | | (0.0252) |
| Obs \| Param \| LL \| AIC | 1000 | 21 | -1376.5 | 2795.0 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

# A5: Simulated Restricted Static Estimates (Suboptimal)

Table 21b: Simulated Restricted Static Estimates (Suboptimal)

| | Cutoff 1 | $P(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success 1 | Success 2 | Sigma |
|---|---|---|---|---|---|---|---|
| Intercept | 0.7340*** | 67.57% | 0.1682 | 54.20% | -8.0582*** | -6.3578*** | |
| | (0.2062) | (58.17%, 75.73%) | (0.2593) | (41.58%, 66.29%) | (1.2803) | (1.7689) | |
| GRE Quantitative Percentile | | | | | 0.1079*** | 0.0727*** | |
| | | | | | (0.0148) | (0.0189) | |
| Gender | | | | | 0.2185** | 0.0409 | |
| | | | | | (0.1042) | (0.1610) | |
| Demographic 1 | | | | | -0.5505*** | -0.3739 | |
| | | | | | (0.1395) | (0.2380) | |
| Demographic 2 | | | | | -0.0151 | -0.0921 | |
| | | | | | (0.1505) | (0.2356) | |
| Advanced Math | | | | | | 0.3897** | |
| | | | | | | (0.1686) | |
| Committee Score | | | | | | 0.1754*** | |
| | | | | | | (0.0378) | |
| Year Transition More | -0.9965*** | 43.47% | 0.6110 | 68.55% | -0.0772 | 0.0376 | |
| | (0.2396) | (37.12%, 50.05%) | (0.3768) | (55.51%, 79.20%) | (0.1671) | (0.2807) | |
| Program Rank | | | | | -0.0305*** | -0.0281*** | |
| | | | | | (0.0019) | (0.0027) | |
| Sigma | | | | | | | 0.2001*** |
| | | | | | | | (0.0325) |
| Obs \| Param \| LL \| AIC | 1000 | 21 | -1417.0 | 2875.9 | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

# A5: Simulated Unrestricted Myopic Estimates (Optimal)

Table 22a: Simulated Unrestricted Myopic Estimates (Optimal)

| | Advanced Math | | Committee Score | Transition | | Accept | | Success | |
|---|---|---|---|---|---|---|---|---|---|
| | Coefficient | AME | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | 0.0000 | 0.0000 | 4.6669*** | -11.5730*** | -2.2535*** | -12.9989*** | -2.3525*** | -9.3999*** | -1.4664*** |
| | (0.0894) | (0.0224) | (0.1940) | (1.0222) | (0.1454) | (1.9897) | (0.2885) | (2.0583) | (0.2690) |
| GRE Quantitative Percentile | | | | 0.1234*** | 0.0240*** | 0.1355*** | 0.0245*** | 0.1145*** | 0.0179*** |
| | | | | (0.0117) | (0.0018) | (0.0221) | (0.0033) | (0.0226) | (0.0029) |
| Gender | | | | 0.2264 | 0.0441 | 0.1267 | 0.0229 | 0.1249 | 0.0195 |
| | | | | (0.1463) | (0.0283) | (0.2495) | (0.0451) | (0.2693) | (0.0419) |
| Demographic 1 | | | | -0.3011 | -0.0586 | -0.2603 | -0.0471 | -0.4657 | -0.0727 |
| | | | | (0.1987) | (0.0386) | (0.3731) | (0.0672) | (0.3923) | (0.0608) |
| Demographic 2 | | | | 0.1542 | 0.0300 | 0.1699 | 0.0307 | -0.2420 | -0.0378 |
| | | | | (0.2214) | (0.0431) | (0.3832) | (0.0693) | (0.4309) | (0.0672) |
| Advanced Math | | | 0.6763*** | | | 0.7710*** | 0.1395*** | 0.6254** | 0.0976** |
| | | | (0.2006) | | | (0.2527) | (0.0443) | (0.2743) | (0.0417) |
| Committee Score | | | | | | 0.0772 | 0.0140 | 0.0454 | 0.0071 |
| | | | | | | (0.0599) | (0.0108) | (0.0654) | (0.0102) |
| Year Transition More | 0.0480 | 0.0120 | 0.1022 | 0.9845*** | 0.1917*** | -0.9255*** | -0.1675*** | -0.0272 | -0.0042 |
| | (0.1265) | (0.0316) | (0.2069) | (0.1472) | (0.0263) | (0.2505) | (0.0427) | (0.2757) | (0.0430) |
| Program Rank | | | | | | | | -0.0282*** | -0.0044*** |
| | | | | | | | | (0.0027) | (0.0003) |
| $\sqrt{\text{MSE}}$ | | | 1.9512*** | | | | | | |
| | | | (0.0674) | | | | | | |
| Obs \| Param \| LL \| AIC | 1000 | 29 | -2419.4 | 4896.8 | | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

# A5: Simulated Unrestricted Myopic Estimates (Suboptimal)

Table 22b: Simulated Unrestricted Myopic Estimates (Suboptimal)

| | Advanced Math | | Committee Score | Transition | | Accept | | Success | |
|---|---|---|---|---|---|---|---|---|---|
| | Coefficient | AME | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | 0.0000 | 0.0000 | 4.7588*** | -11.2089*** | -2.1776*** | -6.7003*** | -1.3243*** | -9.4427*** | -1.4775*** |
| | (0.0894) | (0.0224) | (0.1881) | (1.0127) | (0.1474) | (1.7354) | (0.3201) | (1.9470) | (0.2511) |
| GRE Quantitative Percentile | | | | 0.1224*** | 0.0238*** | 0.0547*** | 0.0108*** | 0.1144*** | 0.0179*** |
| | | | | (0.0117) | (0.0018) | (0.0190) | (0.0036) | (0.0214) | (0.0027) |
| Gender | | | | 0.2568* | 0.0499* | 0.0094 | 0.0019 | 0.0567 | 0.0089 |
| | | | | (0.1467) | (0.0283) | (0.2367) | (0.0468) | (0.2662) | (0.0416) |
| Demographic 1 | | | | -0.8281*** | -0.1609*** | -0.3330 | -0.0658 | -0.5034 | -0.0788 |
| | | | | (0.1965) | (0.0371) | (0.3392) | (0.0669) | (0.3636) | (0.0564) |
| Demographic 2 | | | | 0.0791 | 0.0154 | 0.1763 | 0.0348 | -0.4606 | -0.0721 |
| | | | | (0.2134) | (0.0414) | (0.3344) | (0.0659) | (0.3842) | (0.0598) |
| Advanced Math | | | 0.7040*** | | | 0.3576 | 0.0707 | 0.5331** | 0.0834** |
| | | | (0.1957) | | | (0.2374) | (0.0465) | (0.2696) | (0.0416) |
| Committee Score | | | | | | 0.2958*** | 0.0585*** | 0.0611 | 0.0096 |
| | | | | | | (0.0592) | (0.0106) | (0.0641) | (0.0100) |
| Year Transition More | 0.0480 | 0.0120 | 0.0218 | 1.1069*** | 0.2151*** | -0.7191*** | -0.1421*** | 0.1276 | 0.0200 |
| | (0.1265) | (0.0316) | (0.2022) | (0.1477) | (0.0257) | (0.2396) | (0.0455) | (0.2786) | (0.0434) |
| Program Rank | | | | | | | | -0.0279*** | -0.0044*** |
| | | | | | | | | (0.0028) | (0.0003) |
| $\sqrt{\text{MSE}}$ | | | 1.9289*** | | | | | | |
| | | | (0.0684) | | | | | | |
| Obs \| Param \| LL \| AIC | 1000 | 29 | -2464.5 | 4987.0 | | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

# A5: Simulated Restricted Myopic Estimates (Optimal)

Table 23a: Simulated Restricted Myopic Estimates (Optimal)

| | Advanced Math | Committee Score | Cutoff 1 | $L(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success | Sigma |
|---|---|---|---|---|---|---|---|---|
| Intercept | 0.0128 | 4.6811*** | -0.3383*** | 0.7130 | -0.0029 | 49.93% | -9.4033*** | |
| | (0.0889) | (0.1930) | (0.0804) | (0.6090, 0.8348) | (0.2105) | (39.76%, 60.10%) | (1.9406) | |
| GRE Quantitative Percentile | | | | | | | 0.1124*** | |
| | | | | | | | (0.0214) | |
| Gender | | | | | | | 0.2104* | |
| | | | | | | | (0.1205) | |
| Demographic 1 | | | | | | | -0.3125* | |
| | | | | | | | (0.1789) | |
| Demographic 2 | | | | | | | 0.0218 | |
| | | | | | | | (0.1733) | |
| Advanced Math | | 0.6793*** | | | | | 0.5373*** | |
| | | (0.2009) | | | | | (0.1692) | |
| Committee Score | | | | | | | 0.0494 | |
| | | | | | | | (0.0373) | |
| Year Transition More | 0.0548 | 0.0997 | -0.0233 | 0.6966 | 0.5674* | 63.75% | 0.0577 | |
| | (0.1261) | (0.2067) | (0.0937) | (0.5914, 0.8204) | (0.3211) | (52.78%, 73.45%) | (0.2664) | |
| Program Rank | | | | | | | -0.0283*** | |
| | | | | | | | (0.0027) | |
| $\sqrt{\text{MSE}}$ | | 1.9515*** | | | | | | |
| | | (0.0674) | | | | | | |
| Sigma | | | | | | | | 0.1347*** |
| | | | | | | | | (0.0247) |
| Obs \| Param \| LL \| AIC | 1000 | 20 | -2447.1 | 4934.2 | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

# A5: Simulated Restricted Myopic Estimates (Suboptimal)

Table 23b: Simulated Restricted Myopic Estimates (Suboptimal)

| | Advanced Math | Committee Score | Cutoff 1 | $L(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success | Sigma |
|---|---|---|---|---|---|---|---|---|
| Intercept | 0.0122 | 4.8150*** | -0.3290*** | 0.7196 | -0.0983 | 47.55% | -8.1397*** | |
| | (0.0891) | (0.1883) | (0.1039) | (0.5870, 0.8822) | (0.2266) | (36.76%, 58.56%) | (2.3601) | |
| GRE Quantitative Percentile | | | | | | | 0.0922*** | |
| | | | | | | | (0.0253) | |
| Gender | | | | | | | 0.1605 | |
| | | | | | | | (0.1277) | |
| Demographic 1 | | | | | | | -0.5394** | |
| | | | | | | | (0.2132) | |
| Demographic 2 | | | | | | | -0.0312 | |
| | | | | | | | (0.1887) | |
| Advanced Math | | 0.7051*** | | | | | 0.3544** | |
| | | (0.1963) | | | | | (0.1753) | |
| Committee Score | | | | | | | 0.1744*** | |
| | | | | | | | (0.0373) | |
| Year Transition More | 0.0489 | 0.0008 | -0.0616 | 0.6767 | 0.5779* | 61.77% | 0.1304 | |
| | (0.1263) | (0.2020) | (0.0977) | (0.5578, 0.8209) | (0.3484) | (49.80%, 72.45%) | (0.2799) | |
| Program Rank | | | | | | | -0.0279*** | |
| | | | | | | | (0.0028) | |
| $\sqrt{\text{MSE}}$ | | 1.9300*** | | | | | | |
| | | (0.0686) | | | | | | |
| Sigma | | | | | | | | 0.1544*** |
| | | | | | | | | (0.0406) |
| Obs \| Param \| LL \| AIC | 1000 | 20 | -2510.0 | 5059.9 | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

# A5: Simulated Unrestricted Mat-Dynamic Estimates (Optimal)

Table 24a: Simulated Unrestricted Matriculation-Dynamic Estimates (Optimal)

| | Advanced Math | | Committee Score | Transition | | Accept | | Success | | Matriculate | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coefficient | AME | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -1.8946** | -0.4643** | 4.8638*** | -11.5888*** | -2.2558*** | -13.1535*** | -2.3733*** | -9.5541*** | -1.4869*** | -4.5425 | -0.6043 |
| | (0.7562) | (0.1830) | (1.3238) | (1.0233) | (0.1454) | (2.0077) | (0.2886) | (2.0721) | (0.2688) | (4.6438) | (0.6185) |
| GRE Quantitative Percentile | 0.0207** | 0.0051** | 0.0008 | 0.1236*** | 0.0241*** | 0.1372*** | 0.0248*** | 0.1161*** | 0.0181*** | 0.0587 | 0.0078 |
| | (0.0090) | (0.0022) | (0.0150) | (0.0118) | (0.0018) | (0.0223) | (0.0033) | (0.0227) | (0.0029) | (0.0458) | (0.0061) |
| Gender | -0.2335* | -0.0572* | -0.0783 | 0.2266 | 0.0441 | 0.1290 | 0.0233 | 0.1263 | 0.0197 | 0.4869 | 0.0648 |
| | (0.1309) | (0.0319) | (0.2113) | (0.1464) | (0.0283) | (0.2502) | (0.0451) | (0.2699) | (0.0419) | (0.5061) | (0.0657) |
| Demographic 1 | 0.3693** | 0.0905** | -0.3431 | -0.3014 | -0.0587 | -0.2639 | -0.0476 | -0.4605 | -0.0717 | 0.3568 | 0.0475 |
| | (0.1732) | (0.0421) | (0.2885) | (0.1988) | (0.0386) | (0.3743) | (0.0672) | (0.3933) | (0.0608) | (0.7460) | (0.1001) |
| Demographic 2 | 0.0924 | 0.0226 | -0.1806 | 0.1537 | 0.0299 | 0.1706 | 0.0308 | -0.2340 | -0.0364 | 0.1812 | 0.0241 |
| | (0.1911) | (0.0468) | (0.3001) | (0.2215) | (0.0431) | (0.3844) | (0.0693) | (0.4322) | (0.0672) | (0.7963) | (0.1069) |
| Advanced Math | | | 0.6869*** | | | 0.7740*** | 0.1397*** | 0.6274** | 0.0976** | 0.8103 | 0.1078 |
| | | | (0.2024) | | | (0.2535) | (0.0443) | (0.2749) | (0.0417) | (0.5770) | (0.0733) |
| Committee Score | | | | | | 0.0783 | 0.0141 | 0.0465 | 0.0072 | -0.7409*** | -0.0986*** |
| | | | | | | (0.0602) | (0.0108) | (0.0655) | (0.0102) | (0.2066) | (0.0235) |
| Year Transition/Matriculate More | 0.0662 | 0.0162 | 0.0931 | 0.9841*** | 0.1916*** | -0.9252*** | -0.1669*** | -0.0234 | -0.0036 | 3.3262*** | 0.4425*** |
| | (0.1281) | (0.0314) | (0.2069) | (0.1472) | (0.0263) | (0.2513) | (0.0427) | (0.2763) | (0.0430) | (0.5667) | (0.0330) |
| Program Rank | | | | | | | | -0.0282*** | -0.0044*** | | |
| | | | | | | | | (0.0027) | (0.0003) | | |
| $\sqrt{\text{MSE}}$ | | | 1.9465*** | | | | | | | | |
| | | | (0.0673) | | | | | | | | |
| Obs \| Param \| LL \| AIC | 1000 | 45 | -2461.1 | 5012.2 | | | | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

# A5: Simulated Unrestricted Mat-Dynamic Estimates (Suboptimal)

Table 24b: Simulated Unrestricted Matriculation-Dynamic Estimates (Suboptimal)

| | Advanced Math | | Committee Score | Transition | | Accept | | Success | | Matriculate | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coefficient | AME | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -1.9060** | -0.4671** | 5.2717*** | -11.2295*** | -2.1801*** | -6.7297*** | -1.3296*** | -9.5961*** | -1.4972*** | -1.0400 | -0.1351 |
| | (0.7564) | (0.1830) | (1.2701) | (1.0144) | (0.1474) | (1.7376) | (0.3202) | (1.9602) | (0.2508) | (3.6082) | (0.4721) |
| GRE Quantitative Percentile | 0.0208** | 0.0051** | -0.0016 | 0.1226*** | 0.0238*** | 0.0550*** | 0.0109*** | 0.1162*** | 0.0181*** | 0.0129 | 0.0017 |
| | (0.0090) | (0.0022) | (0.0144) | (0.0117) | (0.0018) | (0.0191) | (0.0036) | (0.0216) | (0.0027) | (0.0391) | (0.0051) |
| Gender | -0.2334* | -0.0572* | -0.1023 | 0.2575* | 0.0500* | 0.0098 | 0.0019 | 0.0583 | 0.0091 | 0.1297 | 0.0168 |
| | (0.1309) | (0.0319) | (0.2051) | (0.1469) | (0.0283) | (0.2368) | (0.0468) | (0.2669) | (0.0416) | (0.4717) | (0.0608) |
| Demographic 1 | 0.3690** | 0.0904** | -0.4893* | -0.8292*** | -0.1610*** | -0.3340 | -0.0660 | -0.5154 | -0.0804 | 0.6324 | 0.0821 |
| | (0.1732) | (0.0421) | (0.2701) | (0.1966) | (0.0371) | (0.3394) | (0.0669) | (0.3647) | (0.0563) | (0.6281) | (0.0823) |
| Demographic 2 | 0.0925 | 0.0227 | -0.2744 | 0.0799 | 0.0155 | 0.1763 | 0.0348 | -0.4742 | -0.0740 | -0.2225 | -0.0289 |
| | (0.1911) | (0.0468) | (0.2740) | (0.2135) | (0.0414) | (0.3346) | (0.0659) | (0.3856) | (0.0598) | (0.6864) | (0.0880) |
| Advanced Math | | | 0.7219*** | | | 0.3580 | 0.0707 | 0.5366** | 0.0837** | 0.5877 | 0.0763 |
| | | | (0.1966) | | | (0.2375) | (0.0465) | (0.2703) | (0.0416) | (0.5321) | (0.0674) |
| Committee Score | | | | | | 0.2962*** | 0.0585*** | 0.0616 | 0.0096 | -0.6092*** | -0.0791*** |
| | | | | | | (0.0592) | (0.0106) | (0.0642) | (0.0100) | (0.1637) | (0.0176) |
| Year Transition/Matriculate More | 0.0665 | 0.0163 | 0.0031 | 1.1107*** | 0.2156*** | -0.7185*** | -0.1420*** | 0.1314 | 0.0205 | 3.4846*** | 0.4526*** |
| | (0.1281) | (0.0314) | (0.2010) | (0.1479) | (0.0257) | (0.2397) | (0.0455) | (0.2794) | (0.0434) | (0.6046) | (0.0349) |
| Program Rank | | | | | | | | -0.0279*** | -0.0044*** | | |
| | | | | | | | | (0.0028) | (0.0003) | | |
| $\sqrt{\text{MSE}}$ | | | 1.9186*** | | | | | | | | |
| | | | (0.0682) | | | | | | | | |
| Obs \| Param \| LL \| AIC | 1000 | 45 | -2506.5 | 5102.9 | | | | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

# A5: Simulated Restricted Mat-Dynamic Estimates (Optimal)

Table 25a: Simulated Restricted Matriculation-Dynamic Estimates (Optimal)

| | Advanced Math | Committee Score | Cutoff 1 | $L(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success | Matriculate | Sigma |
|---|---|---|---|---|---|---|---|---|---|
| Intercept | -2.1882*** | 5.5080*** | -1.8385*** | 0.1591 | -0.0309 | 49.23% | -9.5700*** | -9.6624** | |
| | (0.7585) | (1.3416) | (0.2138) | (0.1046, 0.2419) | (0.2298) | (38.19%, 60.33%) | (1.9922) | (4.0874) | |
| GRE Quantitative Percentile | 0.0243*** | -0.0073 | | | | | 0.1149*** | 0.1166*** | |
| | (0.0090) | (0.0153) | | | | | (0.0219) | (0.0397) | |
| Gender | -0.2211* | -0.0966 | | | | | 0.1535 | 0.7399 | |
| | (0.1308) | (0.2097) | | | | | (0.1421) | (0.4538) | |
| Demographic 1 | 0.3601** | -0.3138 | | | | | -0.3141 | 0.1062 | |
| | (0.1729) | (0.2879) | | | | | (0.2124) | (0.7215) | |
| Demographic 2 | 0.0941 | -0.1854 | | | | | 0.0161 | 0.0717 | |
| | (0.1907) | (0.2982) | | | | | (0.2065) | (0.7746) | |
| Advanced Math | | 0.6491*** | | | | | 0.6118*** | 1.0516* | |
| | | (0.2033) | | | | | (0.1753) | (0.5858) | |
| Committee Score | | | | | | | 0.0257 | -0.6841*** | |
| | | | | | | | (0.0395) | (0.2011) | |
| Year Transition/Matriculate More | 0.0704 | 0.1588 | -1.1141*** | 0.0522 | 0.6267* | 64.47% | 0.0841 | 3.0506*** | |
| | (0.1280) | (0.2057) | (0.2796) | (0.0280, 0.0972) | (0.3398) | (53.11%, 74.40%) | (0.2724) | (0.5403) | |
| Program Rank | | | | | | | -0.0281*** | | |
| | | | | | | | (0.0027) | | |
| $\sqrt{\text{MSE}}$ | | 1.9458*** | | | | | | | |
| | | (0.0672) | | | | | | | |
| Sigma | | | | | | | | | 0.1438*** |
| | | | | | | | | | (0.0288) |
| Obs \| Param \| LL \| AIC | 1000 | 36 | -2508.0 | 5088.1 | | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

# A5: Simulated Restricted Mat-Dynamic Estimates (Suboptimal)

## Table 25b: Simulated Restricted Matriculation-Dynamic Estimates (Suboptimal)

| | Advanced Math | Committee Score | Cutoff 1 | $L(\text{Cutoff})_1$ | Cutoff 2 | $P(\text{Cutoff})_2$ | Success | Matriculate | Sigma |
|---|---|---|---|---|---|---|---|---|---|
| Intercept | -2.0969*** | 5.5146*** | -1.5785*** | 0.2063 | 0.0521 | 51.30% | -8.0820*** | -6.4822* | |
| | (0.7580) | (1.2768) | (0.3118) | (0.1120, 0.3801) | (0.2631) | (38.62%, 63.82%) | (2.4280) | (3.3848) | |
| GRE Quantitative Percentile | 0.0232** | -0.0048 | | | | | 0.0924*** | 0.0774** | |
| | (0.0090) | (0.0145) | | | | | (0.0264) | (0.0364) | |
| Gender | -0.2275* | -0.1097 | | | | | 0.0806 | 0.3888 | |
| | (0.1310) | (0.2034) | | | | | (0.1641) | (0.4605) | |
| Demographic 1 | 0.3529** | -0.4632* | | | | | -0.5186** | 0.0455 | |
| | (0.1730) | (0.2684) | | | | | (0.2595) | (0.6524) | |
| Demographic 2 | 0.0929 | -0.2774 | | | | | -0.1266 | -0.4370 | |
| | (0.1909) | (0.2721) | | | | | (0.2575) | (0.6992) | |
| Advanced Math | | 0.7098*** | | | | | 0.4414** | 0.7348 | |
| | | (0.1960) | | | | | (0.1976) | (0.5514) | |
| Committee Score | | | | | | | 0.1698*** | -0.5422*** | |
| | | | | | | | (0.0393) | (0.1604) | |
| Year Transition/Matriculate More | 0.0723 | 0.0346 | -1.3772*** | 0.0520 | 0.5873 | 65.46% | 0.0893 | 3.1408*** | |
| | (0.1281) | (0.2001) | (0.3779) | (0.0208, 0.1304) | (0.4121) | (50.68%, 77.76%) | (0.2917) | (0.5889) | |
| Program Rank | | | | | | | -0.0281*** | | |
| | | | | | | | (0.0027) | | |
| $\sqrt{\text{MSE}}$ | | 1.9142*** | | | | | | | |
| | | (0.0677) | | | | | | | |
| Sigma | | | | | | | | | 0.1997*** |
| | | | | | | | | | (0.0597) |
| Obs \| Param \| LL \| AIC | 1000 | 36 | -2574.1 | 5220.1 | | | | | |

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

# A5: Simulated Unrestricted Wait-Dynamic Estimates (Optimal)

## Table 26a: Simulated Unrestricted Waitlist-Dynamic Estimates (Optimal)

| | Advanced Math | | Committee Score | $\overline{m_{2A}}$ | Transition | | Accept 2 | | Accept 3 | | Success | | Matriculate | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coefficient | AME | Coefficient | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -1.9049** | -0.4668** | 4.8636*** | -0.0252 | -11.5884*** | -2.2558*** | -11.9628*** | -1.6166*** | -13.2707*** | -1.6993*** | -9.5534*** | -1.4869*** | -5.8313 | -0.6741 |
| | (0.7564) | (0.1830) | (1.3238) | (0.0518) | (1.0232) | (0.1454) | (2.5504) | (0.3154) | (2.4113) | (0.2845) | (2.0719) | (0.2688) | (6.4970) | (0.7706) |
| GRE Quantative Percentile | 0.0208** | 0.0051** | 0.0008 | | 0.1236*** | 0.0241*** | 0.1096*** | 0.0148*** | 0.1387*** | 0.0178*** | 0.1161*** | 0.0181*** | 0.0834 | 0.0096 |
| | (0.0090) | (0.0022) | (0.0150) | | (0.0118) | (0.0018) | (0.0271) | (0.0034) | (0.0264) | (0.0032) | (0.0227) | (0.0029) | (0.0644) | (0.0075) |
| Gender | -0.2335* | -0.0572* | -0.0783 | | 0.2266 | 0.0441 | 0.0233 | 0.0032 | 0.3039 | 0.0389 | 0.1263 | 0.0197 | 0.4650 | 0.0538 |
| | (0.1309) | (0.0319) | (0.2113) | | (0.1464) | (0.0283) | (0.2899) | (0.0392) | (0.3287) | (0.0421) | (0.2699) | (0.0419) | (0.9003) | (0.1052) |
| Demographic 1 | 0.3690** | 0.0904** | -0.3430 | | -0.3014 | -0.0587 | 0.5534 | 0.0748 | -0.8831* | -0.1131* | -0.4604 | -0.0717 | -0.2627 | -0.0304 |
| | (0.1732) | (0.0421) | (0.2885) | | (0.1988) | (0.0386) | (0.4941) | (0.0665) | (0.4622) | (0.0578) | (0.3932) | (0.0608) | (0.7744) | (0.0880) |
| Demographic 2 | 0.0924 | 0.0226 | -0.1805 | | 0.1538 | 0.0299 | 0.5136 | 0.0694 | -0.0525 | -0.0067 | -0.2341 | -0.0364 | -0.3966 | -0.0459 |
| | (0.1911) | (0.0468) | (0.3001) | | (0.2215) | (0.0431) | (0.5204) | (0.0699) | (0.4528) | (0.0580) | (0.4322) | (0.0672) | (1.2105) | (0.1393) |
| Advanced Math | | | 0.6870*** | | | | 0.5066* | 0.0685* | 0.8622*** | 0.1104*** | 0.6275** | 0.0977** | 1.1399 | 0.1318 |
| | | | (0.2024) | | | | (0.2907) | (0.0390) | (0.3306) | (0.0421) | (0.2749) | (0.0417) | (0.9839) | (0.1080) |
| Committee Score | | | | | | | 0.1061 | 0.0143 | 0.0378 | 0.0048 | 0.0464 | 0.0072 | -0.9123** | -0.1055*** |
| | | | | | | | (0.0689) | (0.0093) | (0.0672) | (0.0086) | (0.0655) | (0.0102) | (0.3787) | (0.0317) |
| Year Transition/Matriculate More | 0.0664 | 0.0163 | 0.0932 | | 0.9839*** | 0.1915*** | | | | | -0.0235 | -0.0037 | 3.3933*** | 0.3922*** |
| | (0.1281) | (0.0314) | (0.2069) | | (0.1472) | (0.0263) | | | | | (0.2763) | (0.0430) | (0.7682) | (0.0697) |
| Program Rank | | | | | | | | | | | -0.0282*** | -0.0044*** | | |
| | | | | | | | | | | | (0.0027) | (0.0003) | | |
| $\ddot{m}_2$ | | | | 1.0892*** | | | -1.5983** | -0.2160** | | | | | | |
| | | | | (0.0851) | | | (0.7869) | (0.1049) | | | | | | |
| $\overline{m_{2A}}$ | | | | | | | | | -2.0175*** | -0.2583*** | | | | |
| | | | | | | | | | (0.7697) | (0.0964) | | | | |
| $\sqrt{\text{MSE}}$ | | | 1.9465*** | 0.0015*** | | | | | | | | | | |
| | | | (0.0673) | (0.0005) | | | | | | | | | | |
| Obs \| Param \| LL \| AIC | 1000 | 48 | -594.3 | 1284.6 | | | | | | | | | | |

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

Table 26b: Simulated Unrestricted Waitlist-Dynamic Estimates (Suboptimal)

| | Advanced Math | | Committee Score | $\overline{m_{2A}}$ | Transition | | Accept 2 | | Accept 3 | | Success | | Matriculate | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coefficient | AME | Coefficient | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -1.9050** | -0.4669** | 5.2717*** | -0.0498* | -11.2295*** | -2.1801*** | -6.7113*** | -1.0637*** | -7.1639*** | -0.8002*** | -9.5990*** | -1.4976*** | -1.4373 | -0.2080 |
| | (0.7564) | (0.1830) | (1.2701) | (0.0260) | (1.0144) | (0.1474) | (2.0007) | (0.3043) | (2.5069) | (0.2796) | (1.9604) | (0.2508) | (3.8091) | (0.5533) |
| GRE Quantative Percentile | 0.0208** | 0.0051** | -0.0016 | | 0.1226*** | 0.0238*** | 0.0387* | 0.0061* | 0.0717*** | 0.0080** | 0.1162*** | 0.0181*** | 0.0355 | 0.0051 |
| | (0.0090) | (0.0022) | (0.0144) | | (0.0117) | (0.0018) | (0.0215) | (0.0034) | (0.0278) | (0.0031) | (0.0216) | (0.0027) | (0.0435) | (0.0063) |
| Gender | -0.2334* | -0.0572* | -0.1023 | | 0.2575* | 0.0500* | -0.2767 | -0.0439 | 0.4103 | 0.0458 | 0.0583 | 0.0091 | -0.3544 | -0.0513 |
| | (0.1309) | (0.0319) | (0.2051) | | (0.1469) | (0.0283) | (0.2746) | (0.0431) | (0.3417) | (0.0380) | (0.2669) | (0.0416) | (0.5436) | (0.0788) |
| Demographic 1 | 0.3690** | 0.0904** | -0.4893* | | -0.8293*** | -0.1610*** | 0.2577 | 0.0408 | -1.1549** | -0.1290** | -0.5155 | -0.0804 | 0.3334 | 0.0482 |
| | (0.1732) | (0.0421) | (0.2701) | | (0.1966) | (0.0371) | (0.3774) | (0.0594) | (0.5110) | (0.0563) | (0.3647) | (0.0563) | (0.7088) | (0.1030) |
| Demographic 2 | 0.0925 | 0.0227 | -0.2744 | | 0.0797 | 0.0155 | 0.5369 | 0.0851 | -0.3697 | -0.0413 | -0.4742 | -0.0740 | -0.7602 | -0.1100 |
| | (0.1911) | (0.0468) | (0.2740) | | (0.2135) | (0.0414) | (0.3886) | (0.0605) | (0.4652) | (0.0521) | (0.3856) | (0.0598) | (0.7927) | (0.1089) |
| Advanced Math | | | 0.7219*** | | | | 0.2363 | 0.0374 | 0.4865 | 0.0543 | 0.5366** | 0.0837** | 0.3982 | 0.0576 |
| | | | (0.1966) | | | | (0.2639) | (0.0417) | (0.3566) | (0.0399) | (0.2704) | (0.0416) | (0.6445) | (0.0922) |
| Committee Score | | | | | | | 0.3737*** | 0.0592*** | 0.0653 | 0.0073 | 0.0616 | 0.0096 | -0.6725*** | -0.0973*** |
| | | | | | | | (0.0674) | (0.0096) | (0.0837) | (0.0093) | (0.0643) | (0.0100) | (0.1981) | (0.0217) |
| Year Transition/Matriculate More | 0.0664 | 0.0163 | 0.0031 | | 1.1108*** | 0.2156*** | | | | | 0.1314 | 0.0205 | 2.9410*** | 0.4256*** |
| | (0.1281) | (0.0314) | (0.2010) | | (0.1479) | (0.0257) | | | | | (0.2794) | (0.0434) | (0.6722) | (0.0578) |
| Program Rank | | | | | | | | | | | -0.0279*** | -0.0044*** | | |
| | | | | | | | | | | | (0.0028) | (0.0003) | | |
| $\ddot{m}_2$ | | | | 0.8883*** | | | -0.5496 | -0.0871 | | | | | | |
| | | | | (0.0489) | | | (0.5706) | (0.0900) | | | | | | |
| $\overline{m_{2A}}$ | | | | | | | | | -2.9213*** | -0.3263*** | | | | |
| | | | | | | | | | (0.9934) | (0.1102) | | | | |
| $\sqrt{\text{MSE}}$ | | | 1.9186*** | -0.0010*** | | | | | | | | | | |
| | | | (0.0682) | (0.0002) | | | | | | | | | | |
| Obs \| Param \| LL \| AIC | 1000 | 48 | -430.6 | 957.3 | | | | | | | | | | |

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

# A5: Simulated Restricted Wait-Dynamic Estimates (Optimal)

## Table 27a: Simulated Restricted Waitlist-Dynamic Estimates (Optimal)

| | Advanced Math | Committee Score | $\overline{m_{2A}}$ | Cutoff 1 | $L(\text{Cutoff})_1$ | Cutoff 3 | $P(\text{Cutoff})_3$ | Success | Matriculate | Beta | Sigma |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Intercept | -2.1483*** | 4.7010*** | -0.0252 | -0.2209** | 0.8018 | 0.1801 | 60.02% | -9.5045*** | -5.8313 | | |
| | (0.7577) | (1.3237) | (0.0518) | (0.0865) | (0.6768, 0.9500) | (0.3070) | (49.17%, 69.97%) | (1.9667) | (6.4970) | | |
| GRE Quantative Percentile | 0.0238*** | 0.0028 | | | | | | 0.1132*** | 0.0834 | | |
| | (0.0090) | (0.0150) | | | | | | (0.0217) | (0.0644) | | |
| Gender | -0.2140 | -0.0645 | | | | | | 0.2230* | 0.4650 | | |
| | (0.1305) | (0.2113) | | | | | | (0.1231) | (0.9003) | | |
| Demographic 1 | 0.3579** | -0.3525 | | | | | | -0.3259* | -0.2627 | | |
| | (0.1725) | (0.2903) | | | | | | (0.1842) | (0.7744) | | |
| Demographic 2 | 0.0934 | -0.1816 | | | | | | 0.0202 | -0.3966 | | |
| | (0.1906) | (0.3010) | | | | | | (0.1773) | (1.2105) | | |
| Advanced Math | | 0.6923*** | | | | | | 0.5637*** | 1.1399 | | |
| | | (0.2026) | | | | | | (0.1717) | (0.9839) | | |
| Committee Score | | | | | | | | 0.0501 | -0.9123** | | |
| | | | | | | | | (0.0376) | (0.3787) | | |
| Year Transition/Matriculate More | 0.0734 | 0.0954 | | -0.0200 | 0.7859 | | 74.43% | 0.0701 | 3.3933*** | | |
| | (0.1278) | (0.2069) | | (0.0820) | (0.6583, 0.9384) | | (59.73%, 85.10%) | (0.2666) | (0.7682) | | |
| Program Rank | | | | | | | | -0.0282*** | | | |
| | | | | | | | | (0.0027) | | | |
| $\ddot{m}_2$ | | | 1.0892*** | | | | | | | | |
| | | | (0.0851) | | | | | | | | |
| $\overline{m_{2A}}$ | | | | | | 1.3960* | | | | | |
| | | | | | | (0.8072) | | | | | |
| $\sqrt{\text{MSE}}$ | | 1.9469*** | 0.0015*** | | | | | | | | |
| | | (0.0673) | (0.0005) | | | | | | | | |
| Beta | | | | | | | | | | 0.9751*** | |
| | | | | | | | | | | (0.0345) | |
| Sigma | | | | | | | | | | | 0.1363*** |
| | | | | | | | | | | | (0.0249) |
| Obs \| Param \| LL \| AIC | 1000 | 32 | -625.7 | 1315.4 | | | | | | | |

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

# A5: Simulated Restricted Wait-Dynamic Estimates (Suboptimal)

## Table 27b: Simulated Restricted Waitlist-Dynamic Estimates (Suboptimal)

| | Advanced Math | Committee Score | $\overline{m_{2A}}$ | Cutoff 1 | $L(\text{Cutoff})_1$ | Cutoff 3 | $P(\text{Cutoff})_3$ | Success | Matriculate | Beta | Sigma |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Intercept | -2.1670*** | 4.3821*** | -0.0498* | -0.2877** | 0.7500 | 0.2323 | 61.62% | -6.8847*** | -1.4373 | | |
| | (0.7596) | (1.2671) | (0.0260) | (0.1209) | (0.5918, 0.9505) | (0.3539) | (47.35%, 74.14%) | (2.2787) | (3.8091) | | |
| GRE Quantative Percentile | 0.0241*** | 0.0091 | | | | | | 0.0782*** | 0.0355 | | |
| | (0.0090) | (0.0144) | | | | | | (0.0242) | (0.0435) | | |
| Gender | -0.2191* | -0.0595 | | | | | | 0.1414 | -0.3545 | | |
| | (0.1307) | (0.2029) | | | | | | (0.1168) | (0.5436) | | |
| Demographic 1 | 0.3467** | -0.5657** | | | | | | -0.4512** | 0.3334 | | |
| | (0.1730) | (0.2717) | | | | | | (0.1991) | (0.7088) | | |
| Demographic 2 | 0.1001 | -0.2626 | | | | | | 0.0015 | -0.7602 | | |
| | (0.1908) | (0.2740) | | | | | | (0.1684) | (0.7927) | | |
| Advanced Math | | 0.7355*** | | | | | | 0.3288** | 0.3982 | | |
| | | (0.1973) | | | | | | (0.1652) | (0.6445) | | |
| Committee Score | | | | | | | | 0.1647*** | -0.6725*** | | |
| | | | | | | | | (0.0367) | (0.1981) | | |
| Year Transition/Matriculate More | 0.0686 | 0.0039 | | -0.0379 | 0.7221 | | 77.13% | 0.1685 | 2.9410*** | | |
| | (0.1278) | (0.2010) | | (0.0870) | (0.5848, 0.8915) | | (53.88%, 90.69%) | (0.2714) | (0.6722) | | |
| Program Rank | | | | | | | | -0.0281*** | | | |
| | | | | | | | | (0.0027) | | | |
| $\ddot{m}_2$ | | | 0.8883*** | | | | | | | | |
| | | | (0.0489) | | | | | | | | |
| $\overline{m_{2A}}$ | | | | | | 1.7371 | | | | | |
| | | | | | | (1.1521) | | | | | |
| $\sqrt{\text{MSE}}$ | | 1.9217*** | -0.0010*** | | | | | | | | |
| | | (0.0689) | (0.0002) | | | | | | | | |
| Beta | | | | | | | | | | 0.8711*** | |
| | | | | | | | | | | (0.0429) | |
| Sigma | | | | | | | | | | | 0.1370*** |
| | | | | | | | | | | | (0.0403) |
| Obs \| Param \| LL \| AIC | 1000 | 32 | -486.0 | 1036.0 | | | | | | | |

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

# A5: Simulated Unrestricted Wait-Hybrid Estimates (Optimal)

Table 28a: Simulated Unrestricted Waitlist-Hybrid Estimates (Optimal)

| | $\overline{m_{2A}}$ | Transition | | Accept 2 | | Accept 3 | | Success 1 | | Success 3 | | Matriculate | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -0.0252 | -11.5905*** | -2.2561*** | -11.9661*** | -1.6170*** | -13.2739*** | -1.6996*** | -8.1384*** | -1.1679*** | -9.5601*** | -1.4878*** | -5.8313 | -0.6741 |
| | (0.0518) | (1.0234) | (0.1454) | (2.5509) | (0.3154) | (2.4117) | (0.2845) | (1.1381) | (0.1344) | (2.0725) | (0.2688) | (6.4970) | (0.7706) |
| GRE Quantitative Percentile | | 0.1236*** | 0.0241*** | 0.1097*** | 0.0148*** | 0.1387*** | 0.0178*** | 0.1079*** | 0.0155*** | 0.1161*** | 0.0181*** | 0.0834 | 0.0096 |
| | | (0.0118) | (0.0018) | (0.0271) | (0.0034) | (0.0264) | (0.0032) | (0.0133) | (0.0015) | (0.0227) | (0.0029) | (0.0644) | (0.0075) |
| Gender | | 0.2266 | 0.0441 | 0.0233 | 0.0031 | 0.3039 | 0.0389 | 0.1399 | 0.0201 | 0.1264 | 0.0197 | 0.4650 | 0.0538 |
| | | (0.1464) | (0.0283) | (0.2899) | (0.0392) | (0.3287) | (0.0421) | (0.1739) | (0.0249) | (0.2699) | (0.0419) | (0.9003) | (0.1052) |
| Demographic 1 | | -0.3014 | -0.0587 | 0.5530 | 0.0747 | -0.8834* | -0.1131* | -0.2535 | -0.0364 | -0.4606 | -0.0717 | -0.2627 | -0.0304 |
| | | (0.1988) | (0.0386) | (0.4940) | (0.0665) | (0.4622) | (0.0578) | (0.2359) | (0.0338) | (0.3933) | (0.0608) | (0.7744) | (0.0880) |
| Demographic 2 | | 0.1538 | 0.0299 | 0.5128 | 0.0693 | -0.0526 | -0.0067 | -0.1180 | -0.0169 | -0.2340 | -0.0364 | -0.3966 | -0.0459 |
| | | (0.2215) | (0.0431) | (0.5204) | (0.0699) | (0.4528) | (0.0580) | (0.2581) | (0.0371) | (0.4322) | (0.0672) | (1.2105) | (0.1393) |
| Advanced Math | | | | 0.5065* | 0.0684* | 0.8621*** | 0.1104*** | | | 0.6273** | 0.0976** | 1.1399 | 0.1318 |
| | | | | (0.2907) | (0.0390) | (0.3306) | (0.0421) | | | (0.2749) | (0.0417) | (0.9839) | (0.1080) |
| Committee Score | | | | 0.1061 | 0.0143 | 0.0378 | 0.0048 | | | 0.0464 | 0.0072 | -0.9123** | -0.1055*** |
| | | | | (0.0689) | (0.0093) | (0.0672) | (0.0086) | | | (0.0655) | (0.0102) | (0.3787) | (0.0317) |
| Year Transition/Matriculate More | | 0.9842*** | 0.1916*** | | | | | -0.0586 | -0.0084 | -0.0233 | -0.0036 | 3.3933*** | 0.3922*** |
| | | (0.1472) | (0.0263) | | | | | (0.1690) | (0.0242) | (0.2763) | (0.0430) | (0.7682) | (0.0697) |
| Program Rank | | | | | | | | -0.0310*** | -0.0045*** | -0.0282*** | -0.0044*** | | |
| | | | | | | | | (0.0019) | (0.0002) | (0.0027) | (0.0003) | | |
| $\ddot{m}_2$ | 1.0892*** | | | -1.5983** | -0.2160** | | | | | | | | |
| | (0.0851) | | | (0.7869) | (0.1049) | | | | | | | | |
| $\overline{m_{2A}}$ | | | | | | -2.0170*** | -0.2583*** | | | | | | |
| | | | | | | (0.7696) | (0.0964) | | | | | | |
| $\sqrt{\text{MSE}}$ | 0.0015*** | | | | | | | | | | | | |
| | (0.0005) | | | | | | | | | | | | |
| Obs \| Param \| LL \| AIC | 1000 | 41 | 438.6 | -795.1 | | | | | | | | | |

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

## Table 28b: Simulated Unrestricted Waitlist-Hybrid Estimates (Suboptimal)

| | $\overline{m_{2A}}$ | Transition | | Accept 2 | | Accept 3 | | Success 1 | | Success 3 | | Matriculate | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coefficient | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME | Coefficient | AME |
| Intercept | -0.0498* | -11.2292*** | -2.1801*** | -6.7111*** | -1.0638*** | -7.1640*** | -0.8002*** | -8.2823*** | -1.1912*** | -9.6011*** | -1.4979*** | -1.4373 | -0.2080 |
| | (0.0260) | (1.0143) | (0.1474) | (2.0006) | (0.3043) | (2.5069) | (0.2796) | (1.1392) | (0.1341) | (1.9606) | (0.2508) | (3.8091) | (0.5533) |
| GRE Quantitative Percentile | | 0.1226*** | 0.0238*** | 0.0387* | 0.0061* | 0.0717*** | 0.0080** | 0.1097*** | 0.0158*** | 0.1162*** | 0.0181*** | 0.0355 | 0.0051 |
| | | (0.0117) | (0.0018) | (0.0215) | (0.0034) | (0.0278) | (0.0031) | (0.0133) | (0.0015) | (0.0216) | (0.0027) | (0.0435) | (0.0063) |
| Gender | | 0.2576* | 0.0500* | -0.2763 | -0.0438 | 0.4099 | 0.0458 | 0.1354 | 0.0195 | 0.0584 | 0.0091 | -0.3544 | -0.0513 |
| | | (0.1468) | (0.0283) | (0.2746) | (0.0431) | (0.3417) | (0.0380) | (0.1741) | (0.0250) | (0.2669) | (0.0416) | (0.5436) | (0.0788) |
| Demographic 1 | | -0.8291*** | -0.1610*** | 0.2570 | 0.0407 | -1.1547** | -0.1290** | -0.2774 | -0.0399 | -0.5155 | -0.0804 | 0.3334 | 0.0482 |
| | | (0.1966) | (0.0371) | (0.3773) | (0.0594) | (0.5110) | (0.0563) | (0.2344) | (0.0337) | (0.3648) | (0.0563) | (0.7088) | (0.1030) |
| Demographic 2 | | 0.0799 | 0.0155 | 0.5360 | 0.0850 | -0.3696 | -0.0413 | -0.1806 | -0.0260 | -0.4743 | -0.0740 | -0.7602 | -0.1100 |
| | | (0.2135) | (0.0414) | (0.3885) | (0.0605) | (0.4652) | (0.0521) | (0.2566) | (0.0369) | (0.3856) | (0.0598) | (0.7927) | (0.1089) |
| Advanced Math | | | | 0.2361 | 0.0374 | 0.4867 | 0.0544 | | | 0.5367** | 0.0837** | 0.3982 | 0.0576 |
| | | | | (0.2639) | (0.0417) | (0.3566) | (0.0399) | | | (0.2704) | (0.0416) | (0.6445) | (0.0922) |
| Committee Score | | | | 0.3736*** | 0.0592*** | 0.0653 | 0.0073 | | | 0.0617 | 0.0096 | -0.6725*** | -0.0973*** |
| | | | | (0.0674) | (0.0096) | (0.0837) | (0.0093) | | | (0.0643) | (0.0100) | (0.1981) | (0.0217) |
| Year Transition/Matriculate More | | 1.1108*** | 0.2157*** | | | | | -0.0767 | -0.0110 | 0.1318 | 0.0206 | 2.9410*** | 0.4256*** |
| | | (0.1479) | (0.0257) | | | | | (0.1688) | (0.0243) | (0.2794) | (0.0434) | (0.6722) | (0.0578) |
| Program Rank | | | | | | | | -0.0304*** | -0.0044*** | -0.0279*** | -0.0044*** | | |
| | | | | | | | | (0.0019) | (0.0002) | (0.0028) | (0.0003) | | |
| $\ddot{m}_2$ | 0.8883*** | | | -0.5486 | -0.0870 | | | | | | | | |
| | (0.0489) | | | (0.5705) | (0.0900) | | | | | | | | |
| $\overline{m_{2A}}$ | | | | | | -2.9216*** | -0.3263*** | | | | | | |
| | | | | | | (0.9934) | (0.1102) | | | | | | |
| $\sqrt{\text{MSE}}$ | 0.0010*** | | | | | | | | | | | | |
| | (0.0002) | | | | | | | | | | | | |
| Obs \| Param \| LL \| AIC | 1000 | 41 | 618.0 | -1153.9 | | | | | | | | | |

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Conclusion

# A5: Simulated Restricted Wait-Hybrid Estimates (Optimal)

## Table 29a: Simulated Restricted Waitlist-Hybrid Estimates (Optimal)

| | $\overline{m_{2A}}$ | Cutoff 1 | $P(\text{Cutoff})_1$ | Cutoff 3 | $P(\text{Cutoff})_3$ | Success 1 | Success 3 | Matriculate | Beta | Sigma |
|---|---|---|---|---|---|---|---|---|---|---|
| Year Transition/Matriculate Fewer \| Intercept | -0.0252 | 0.6891*** | 66.58% | 0.4185 | 69.29% | -8.0491*** | -9.6599*** | -5.8313 | | |
| | (0.0518) | (0.1830) | (58.19%, 74.03%) | (0.4184) | (54.94%, 80.68%) | (1.1651) | (1.7228) | (6.4970) | | |
| GRE Quantitative Percentile | | | | | | 0.1060*** | 0.1151*** | 0.0834 | | |
| | | | | | | (0.0135) | (0.0191) | (0.0644) | | |
| Gender | | | | | | 0.1997** | 0.1149 | 0.4650 | | |
| | | | | | | (0.1017) | (0.1664) | (0.9003) | | |
| Demographic 1 | | | | | | -0.2477* | -0.3286 | -0.2627 | | |
| | | | | | | (0.1349) | (0.2390) | (0.7744) | | |
| Demographic 2 | | | | | | 0.0405 | -0.0036 | -0.3966 | | |
| | | | | | | (0.1507) | (0.2410) | (1.2105) | | |
| Advanced Math | | | | | | | 0.6284*** | 1.1399 | | |
| | | | | | | | (0.1758) | (0.9839) | | |
| Committee Score | | | | | | | 0.0548 | -0.9123** | | |
| | | | | | | | (0.0419) | (0.3787) | | |
| Year Transition/Matriculate More | | -0.8464*** | 46.08% | | 87.77% | -0.0609 | 0.0204 | 3.3933*** | | |
| | | (0.2185) | (39.62%, 52.67%) | | (63.97%, 96.67%) | (0.1668) | (0.2685) | (0.7682) | | |
| Program Rank | | | | | | -0.0310*** | -0.0283*** | | | |
| | | | | | | (0.0019) | (0.0027) | | | |
| $\ddot{m}_2$ | 1.0892*** | | | | | | | | | |
| | (0.0851) | | | | | | | | | |
| $\overline{m_{2A}}$ | | | | 2.4392* | | | | | | |
| | | | | (1.4478) | | | | | | |
| $\sqrt{\text{MSE}}$ | 0.0015*** | | | | | | | | | |
| | (0.0005) | | | | | | | | | |
| Beta | | | | | | | | | 0.9493*** | |
| | | | | | | | | | (0.0396) | |
| Sigma | | | | | | | | | | 0.1917*** |
| | | | | | | | | | | (0.0251) |
| Obs \| Param \| LL \| AIC | 1000 | 25 | 434.1 | -818.1 | | | | | | |

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Conclusion

# A5: Simulated Restricted Wait-Hybrid Estimates (Suboptimal)

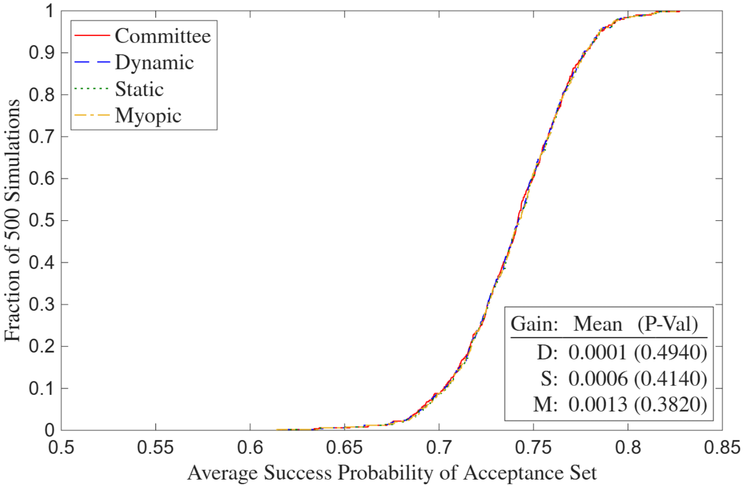
Table 29b: Simulated Restricted Waitlist-Hybrid Estimates (Suboptimal)

| | $\overline{m_{2A}}$ | Cutoff 1 | $P(\text{Cutoff})_1$ | Cutoff 3 | $P(\text{Cutoff})_3$ | Success 1 | Success 3 | Matriculate | Beta | Sigma |
|---|---|---|---|---|---|---|---|---|---|---|
| Year Transition/Matriculate Fewer \| Intercept | -0.0498* | 0.7428*** | 67.76% | 0.5736 | 74.61% | -8.0749*** | -6.3299*** | -1.4373 | | |
| | (0.0260) | (0.2082) | (58.29%, 75.97%) | (0.5852) | (55.80%, 87.24%) | (1.2691) | (1.7373) | (3.8091) | | |
| GRE Quantitative Percentile | | | | | | 0.1082*** | 0.0718*** | 0.0355 | | |
| | | | | | | (0.0147) | (0.0186) | (0.0435) | | |
| Gender | | | | | | 0.2189** | 0.0369 | -0.3545 | | |
| | | | | | | (0.1045) | (0.1594) | (0.5436) | | |
| Demographic 1 | | | | | | -0.5509*** | -0.3579 | 0.3334 | | |
| | | | | | | (0.1397) | (0.2337) | (0.7088) | | |
| Demographic 2 | | | | | | -0.0156 | -0.0774 | -0.7602 | | |
| | | | | | | (0.1510) | (0.2321) | (0.7927) | | |
| Advanced Math | | | | | | | 0.3857** | 0.3982 | | |
| | | | | | | | (0.1669) | (0.6445) | | |
| Committee Score | | | | | | | 0.1748*** | -0.6725*** | | |
| | | | | | | | (0.0374) | (0.1981) | | |
| Year Transition/Matriculate More | | -1.0055*** | 43.47% | | 93.27% | -0.0797 | 0.1064 | 2.9410*** | | |
| | | (0.2419) | (37.12%, 50.04%) | | (41.58%, 99.63%) | (0.1668) | (0.2766) | (0.6722) | | |
| Program Rank | | | | | | -0.0306*** | -0.0281*** | | | |
| | | | | | | (0.0019) | (0.0027) | | | |
| $\ddot{m}_2$ | 0.8883*** | | | | | | | | | |
| | (0.0489) | | | | | | | | | |
| $\overline{m_{2A}}$ | | | | 3.6299 | | | | | | |
| | | | | (3.2728) | | | | | | |
| $\sqrt{\text{MSE}}$ | 0.0010*** | | | | | | | | | |
| | (0.0002) | | | | | | | | | |
| Beta | | | | | | | | | 0.8280*** | |
| | | | | | | | | | (0.0422) | |
| Sigma | | | | | | | | | | 0.2012*** |
| | | | | | | | | | | (0.0325) |
| Obs \| Param \| LL \| AIC | 1000 | 25 | 596.3 | -1142.6 | | | | | | |

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

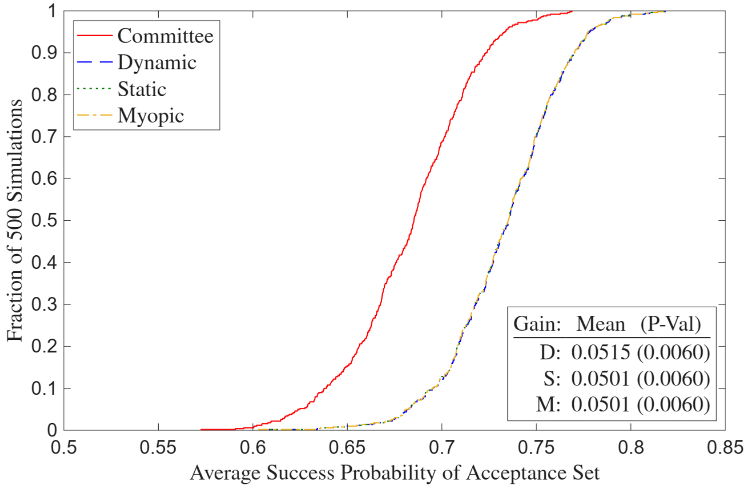
Conclusion

# A5: Simulated Deviation Gains (Optimal)



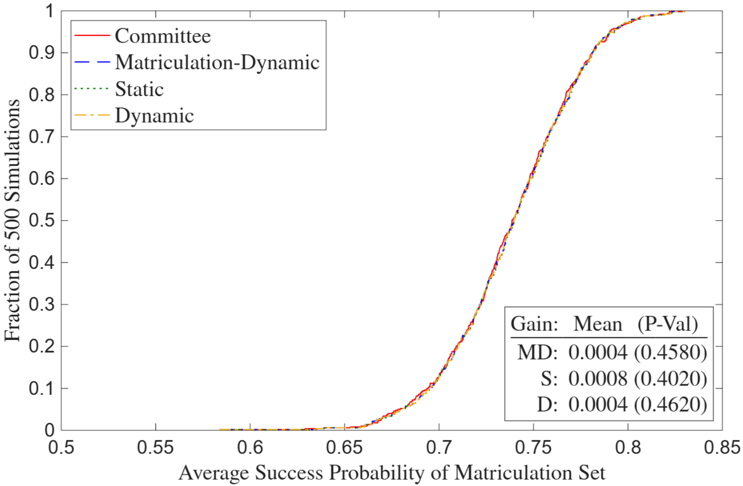

Conclusion

# A5: Simulated Deviation Gains (Suboptimal)



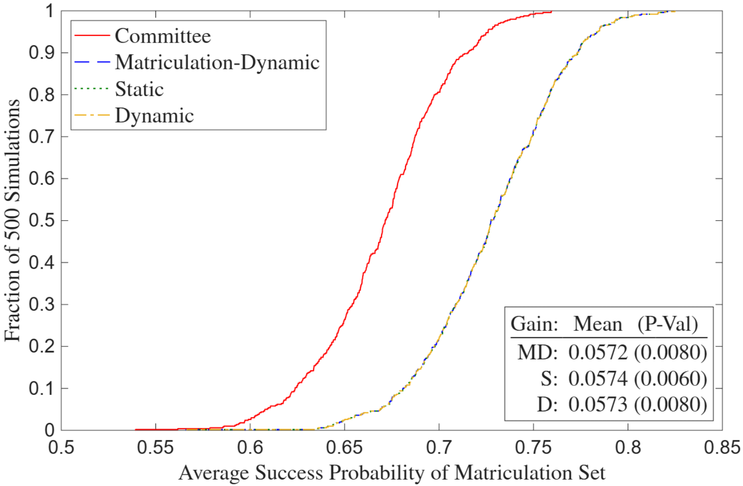

Conclusion

# A5: Simulated Matriculation Deviation Gains (Optimal)



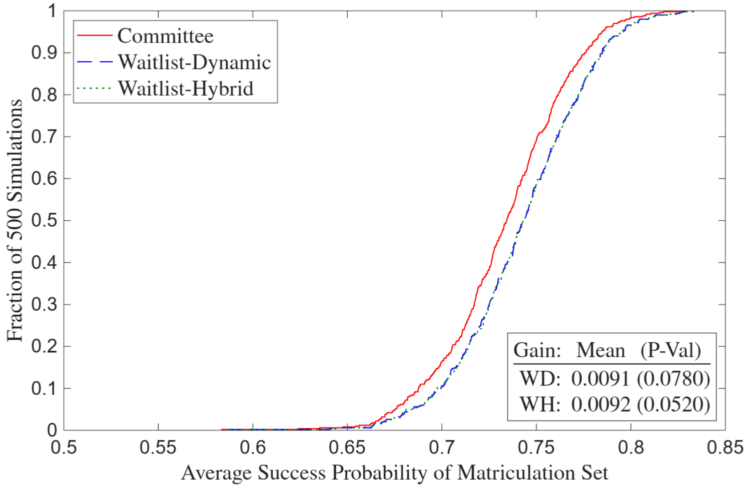

Conclusion

# A5: Simulated Matriculation Deviation Gains (Suboptimal)



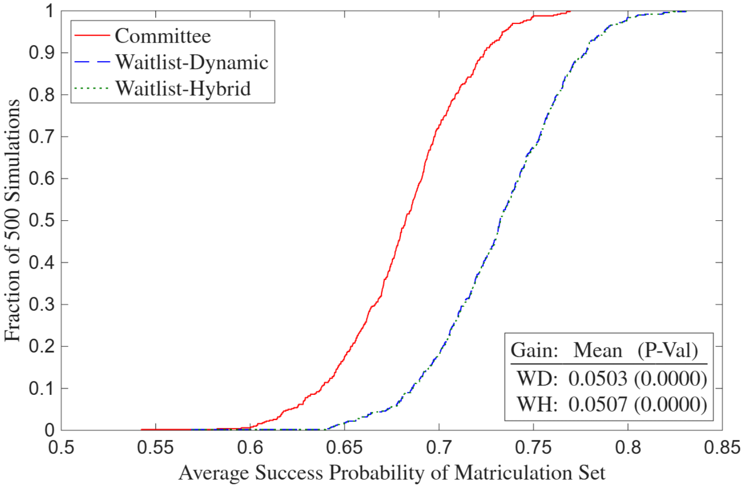

Conclusion

# A5: Simulated Waitlist Deviation Gains (Optimal)




Conclusion

# A5: Simulated Waitlist Deviation Gains (Suboptimal)




▸▸ Conclusion

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